

Reviewed Preprint

v1 • October 10, 2024

Not revised

Reviewed Preprint

v2 • January 7, 2026

Revised by authors

✉ For correspondence:

kenghui@gate.sinica.edu.tw

tktang@ibms.sinica.edu.tw

Competing interests:

No competing interests declared

Funding:

See [page 47](#)

Reviewing editor:

Heidi Hehnl,
Syracuse University, United States

© 2024, Wang et al. This article is distributed under the terms of the

[Creative Commons Attribution](#)

[License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Centrosome Migration and Apical Membrane Formation in Polarized Epithelial Cells: Insights from the MDCK Cyst Model

Po-Kai Wang^{1,5}, Keng-Hui Lin^{2,3,4}✉, Tang K Tang^{1,5}✉

¹Program in Molecular Medicine, National Yang Ming Chiao Tung University and Academia Sinica, Taipei, Taiwan •

²Institute of Physics, Academia Sinica, Taipei, Taiwan • ³Institute of Cellular and Organismic Biology, Academia

Sinica, Taipei, Taiwan • ⁴Genome and Systems Biology Degree Program, National Taiwan University, Taipei, Taiwan

• ⁵Institute of Biomedical Sciences, Academia Sinica, Taipei, Taiwan

eLife Assessment

This **important** study contributes to our understanding of how epithelial cells establish polarity by identifying a hierarchy in which Par3 acts upstream of centrosome positioning and apical membrane initiation. The evidence supporting the main conclusions is **convincing**, although several aspects of the model remain only partially supported due to unresolved questions about microtubule organization and the need for clearer integration of quantitative and conceptual points raised in review. The work will be of interest to cell and developmental biologists, but the conclusions would be strengthened by greater precision in methodology, terminology, and interpretation.

<https://doi.org/10.7554/eLife.101088.2.sa3>

Abstract

Polarization is crucial for the proper functioning of epithelial cells. Early hallmarks include the trafficking and enrichment of polarity molecules to form the apical membrane (AM) or cell-cell junctions, and the apical positioning of the centrosome. However, the dependencies among polarity molecules, AM formation, and centrosome positioning remain poorly understood. When cultured in Matrigel, *de novo* polarization of a single epithelial cell is often coupled with mitosis. During mitotic exit, centrosomes move to the future AM site, raising questions about their role in polarization. We perturbed centrosomes and polarity regulators in Matrigel-cultured cells and manipulated polarity direction using suspension culture to examine the relationships among polarization features. Surprisingly, centrosomal microtubules are not essential for centrosome positioning or AM formation, but their absence reduces the efficiency of AM formation. The polarity regulator Par3, rather than AM component trafficking, influences centrosome positioning. In suspension culture, centrosome migrate in the direction opposite to AM formation. Taken together, our findings define the hierarchical relationships among several polarization features and show that centrosome-based polarity is not universal in epithelial cells, providing new insights into the mechanisms of epithelial polarization.

Impact statement

Single-cell observations refute the expectation that centrosome positioning is a critical factor in the polarization of epithelial cells, while showing that proper centrosome positioning enhances the efficiency of cell polarization.

Introduction

Cell polarity refers to spatial asymmetry in the shape, organization, and function of a cell. The development of cell polarity is a fundamental process in developmental and physiological biology, and aberrations in cell polarization can contribute to various developmental disorders and cancer (Hakanen et al., 2019 [DOI](#); Halaoui & McCaffrey, 2015 [DOI](#); Murdoch et al., 2003 [DOI](#); Royer & Lu, 2011 [DOI](#); Schneeberger et al., 2018 [DOI](#); Wilson, 2011 [DOI](#)). Although polarizations in different cell types present in different morphologies, they share evolutionarily conserved machinery components that mediate these processes. For example, the partitioning defective (PAR) complexes, Crumbs complexes, and Scribble complexes are known to be asymmetrically occupied in different cellular compartments to regulate cell polarity (Goldstein & Macara, 2007 [DOI](#); Iden & Collard, 2008 [DOI](#); St Johnston & Ahringer, 2010 [DOI](#)). Furthermore, the centrosome is always located at a specific position, resulting in the centrosome–nucleus axis aligning with the polarization axis, which may be crucial for various types of polarized cells (Burakov & Nadezhdina, 2020 [DOI](#); Tang & Marshall, 2012 [DOI](#)).

The centrosome is a distinct organelle in cells, composed of a centriole pair surrounded by a multi-protein cloud called pericentriolar material (PCM) (Conduit et al., 2015 [DOI](#)). The centrioles in the pair exhibit structural and functional asymmetry based on their generation, with the older, fully mature mother centriole carrying characteristic subdistal appendages (SDAs) and distal appendages (DAs) (Bornens, 2012 [DOI](#); Gomes Pereira et al., 2021 [DOI](#)). The SDAs anchor microtubules to the mother centriole (Tateishi et al., 2013 [DOI](#)), while the DAs allow the mother centriole to dock to the plasma membrane (Tanos et al., 2013 [DOI](#)). The PCM comprises hundreds of proteins, including the scaffold protein pericentrin (PCNT) and the γ -tubulin ring complex, which are required for microtubule nucleation (Fong et al., 2008 [DOI](#); Zimmerman et al., 2004 [DOI](#)). Because of its strong ability to nucleate microtubules, the centrosome functions as the primary microtubule organizing center and is considered to play a crucial role in regulating cell polarity (Bornens, 2012 [DOI](#); de Anda et al., 2005 [DOI](#)). Laser ablation of the pericentrosomal region alters the polarized cell shape and affects cell migration (Wakida et al., 2010 [DOI](#)). Light inactivation or depletion of centrosomal proteins disrupts axon formation and neuronal migration (de Anda et al., 2010 [DOI](#)). Depletion of Par3 and dynein indirectly interferes with centrosome positioning and causes polarity defects in migrating cells (Schmoranzner et al., 2009 [DOI](#)). However, researchers have reported contradictory results in some cases. For instance, axons continue to grow when the centrosome in rodent hippocampal neurons is disrupted via laser ablation (Stiess et al., 2010 [DOI](#)). Moreover, the treatment of human umbilical vein endothelial cells (HUVECs) with centrinone, an inhibitor of centriole duplication (Wong et al., 2015 [DOI](#)), to deplete centrosomes does not affect the polarization of endothelial sprouting or cell migration. Notably, under conditions lacking non-centrosomal microtubules, centrosome depletion partially restored cell polarization, suggesting that under such conditions, the centrosome instead hinders cell polarization (Martin et al., 2018 [DOI](#)). Therefore, the exact role of the centrosome in cell polarity requires further clarification.

In polarized epithelial cells, the centrosome is localized at the apical region during interphase (Buckley & St Johnston, 2022 [DOI](#); Ching et al., 2022 [DOI](#); Rodriguez-Boulan & Macara, 2014 [DOI](#)), which contributes to the construction of an asymmetric microtubule network conducive to polarized vesicle trafficking (Feldman & Priess, 2012 [DOI](#); Meiring et al., 2020 [DOI](#)). Establishing apical-basal polarity in epithelial cells requires the correct intracellular transportation of different polarity molecules to distinct domains of the plasma membrane (Jewett & Prekeris, 2018 [DOI](#); Mellman & Nelson, 2008 [DOI](#); Roman-Fernandez & Bryant, 2016 [DOI](#)). These processes have been extensively studied using *in vitro* three-dimensional (3D) cultures of vertebrate epithelial cells, such as Madin-Darby Canine Kidney (MDCK) cells seeded in the extracellular matrix (ECM), which spontaneously form polarized cysts with central lumens to mimic tubular structures *in vivo*. According to this model, previous studies have found that the orientation of apical-basal polarity depends on ECM-mediated signaling. In suspended culture without ECM or inhibition of ECM protein laminin, its receptor β 1-integrin, or downstream signaling molecule small GTPase Rac1, polarity reversal occurs, leading to failure in lumen formation (O'Brien et al., 2001 [DOI](#); Yu et al., 2005 [DOI](#)). Both integrin and Rac1 are necessary for laminin accumulation around the cyst and for inhibiting the

RhoA-ROCK-mediated actomyosin on the basal side, which then enables the internalization of apical membrane proteins from the basolateral membrane (Bryant et al., 2014 [DOI](#); Yu et al., 2008 [DOI](#)). Following ECM signaling, the well-known apical membrane marker Gp135 (also known as podocalyxin) begins to be internalized from the cell membrane into Rab11a-positive recycling endosomes when cells aggregate together or when a single cell undergoes cell division. These Gp135-positive vesicles containing Crumbs3 (Crb3) and Cdc42 are then delivered and tethered to the cell-cell contact site or cytokinesis site. Accumulation of these apical determinants forms the apical membrane initiation site (AMIS). Expansion of the apical membrane from the AMIS, combined with fluid pumping, generates a single lumen at the center of MDCK cysts (Bryant et al., 2010 [DOI](#); Chou et al., 2016 [DOI](#); Galvez-Santisteban et al., 2012 [DOI](#); Klinkert et al., 2016 [DOI](#); Mangan et al., 2016 [DOI](#); Schluter et al., 2009 [DOI](#)).

Although numerous reports have been on the molecular networks involved in the *de novo* generation of the apical membrane and lumen, the sequential recruitment of apical determinants and the relative timing of centrosome positioning remain poorly characterized. The differences between cell-aggregation-derived polarization and cell-division-derived polarization are still critical in this field (Liang et al., 2022 [DOI](#); Rathbun et al., 2020 [DOI](#)). Moreover, the role of apical migration of centrosomes in epithelial polarization remains unclear. Interestingly, previous studies have demonstrated that centrosomes can move toward the cytokinetic bridge during the late stages of cell division (Jonsdottir et al., 2010 [DOI](#); Krishnan et al., 2022 [DOI](#); Piel et al., 2001 [DOI](#)). Therefore, it is worthwhile to understand the coupling between the apical migration of centrosomes and cytokinesis.

In this work, we employed three major polarity indicators: (1) apical membrane protein recruitment, defined by Gp135 accumulation (Meder et al., 2005 [DOI](#); Yu et al., 2007 [DOI](#)), (2) centrosome positioning, defined by the angles and relative distance between the nucleus and centrosome (Burute et al., 2017 [DOI](#); Rodriguez-Fraticelli et al., 2012 [DOI](#)), and (3) organization of polarity regulators such as Par3, which controls tight junction assembly to partition the apical surface from the basolateral surface (Chen & Macara, 2005 [DOI](#); Horikoshi et al., 2009 [DOI](#)). Here, we elucidate the detailed coupling between these polarity indicators during cytokinesis-induced *de novo* epithelial polarization in Matrigel and other cell culture conditions, which allow us to impose different polarity states. Our findings indicate that cytokinesis-induced centrosome positioning promotes epithelial polarization. The tight-junction-localized component Par3 is considered the upstream regulator of centrosome positioning and polarized vesicle trafficking. The trafficking of apical membrane components is downstream of centrosome positioning. The position of the centrosome is not sensitive to the disruption of individual centrosome components. Migrating centrosomes promote Gp135 accumulation during *de novo* epithelial polarization. Our study clarifies the relationship between different polarity indicators and the role of the centrosome in epithelial polarization.

Results

Apical membrane component proteins are recruited to the centrosome during *de novo* epithelial polarization

To capture the centrosome position and the distribution of Gp135 vesicles, we performed immunostaining at several stages in the transition from a single MDCK cell to a two-cell cyst. As a single non-polarized MDCK cell is embedded into Matrigel, most Gp135 proteins cluster around the interphase centrosome (Figure 1A, i [DOI](#)) and strongly co-localize with the Rab11a-labeled recycling endosomes (Figure 1—figure supplement 1A [DOI](#)). During mitosis, the Gp135 proteins are well dispersed in the cytosol at metaphase (Figure 1A, ii [DOI](#)), weakly cluster near the centrosomes in early telophase (Telo, Figure 1A, iii [DOI](#)), and gradually accumulate at the centrosomes from late telophase to the stage of cytokinetic bridge pre-abscission (pre-Abs) (Figure 1A, iii-iv [DOI](#)), when the acetyl-tubulin-labeled cytokinetic bridge condenses significantly. Cytokinesis in most animal cells begins in the middle point of anaphase and ends shortly after the completion of mitosis in telophase. Cytokinetic abscission leads to the physical cut of the cytokinetic bridge connecting the

daughter cells and concludes cell division. After cytokinetic bridge abscission (post-cytokinesis), the Gp135-positive vesicles are transported to the middle membrane region, which will form the AMIS between the two daughter cells (Figure 1A, v). Subsequently, exocytosis continues to expand the apical membrane and produces a lumen structure (Figure 1A, vi). The relative Gp135 signals near the centrosome regions of different stages show a clear maximum at the cytokinetic pre-abscission stage (Figure 1A–C). Notably, before bridge abscission, we detected a stronger signal of Gp135-positive vesicles surrounding the centrosomes compared to the cytosol and cytokinetic bridge (Figure 1A–B and D).

We quantify the degree of polarization by the centrosome positions using the following quantities. The center point between two nuclei is set as the origin (O), representing the cyst center. The polarity index is defined as the nucleus (N)–centrosome (C) distance, L_{N-C} , normalized by the average nuclear radius, r_N . In addition, the nucleus–centrosome orientation (θ_{N-C} , the angle between the N–C and N–N axes) characterizes the movement of the centrosome (Figure 1E, see Materials and methods) (Burute et al., 2017; Rodriguez-Fraticelli et al., 2012). Compared to the non-polarized 1-cell stage, the polarity index gradually increases from the cytokinetic pre-abscission to the post-cytokinesis stage, reaching the highest level during the lumen-open (LO) stage (Figure 1F). A high θ_{N-C} value indicates that the centrosomes are located at the outside of two nuclei, which also corresponds to a larger L_{O-C} (origin–centrosome distance) and L_{IC} (inter-centrosomal distance) (Figure 1—figure supplement 1B–E). From telophase to post-cytokinesis, centrosomes move from outside to the inside of two nuclei, and thus θ_{N-C} decreases (Figure 1G).

Because recycling endosomes play a crucial role in trafficking apical membrane components such as Gp135, we examined other apical membrane components such as Crb3, Cdc42, and aPKC (Bryant et al., 2010; Klinkert et al., 2016; Schluter et al., 2009). We found that aPKC appears to diffuse in the cytoplasm during the M phase and concentrates at the AMIS only after cytokinesis (Figure 1—figure supplement 1F). This result may be due to a lower aPKC concentration, contributing to a weak immunostaining signal. Nevertheless, the signals of enhanced green fluorescent protein (EGFP)-Crb3 and EGFP-Cdc42 in MDCK cells exhibited similar patterns near the centrosome during cytokinetic pre-abscission and were transported to the AMIS from the centrosome in the post-cytokinesis stage, identical to Gp135 (Figure 1—figure supplement 2A–B).

Taken together, our data suggest that the centrosome serves as a hub for the recruitment of apical recycling endosomes. Therefore, the prior migration of the centrosome toward the center of cell doublets may facilitate the trafficking of apical recycling endosomes to the AMIS (Figure 1H).

Centrosomes exhibit robust migration toward the center of two daughter cells after anaphase onset before Gp135 targeting the AMIS

To precisely define the hierarchy of centrosome migration, apical membrane formation, and apical-basal polarity establishment, we performed live cell imaging using MDCK cells that expressed fluorescently labeled centrosome protein, PACT-mKO1 (the PACT domain is a conserved centrosomal targeting motif of PCNT) (Gillingham & Munro, 2000), and apical membrane protein, EGFP-Gp135, cultured in Matrigel. The onset of anaphase is set as 0:00 (h: min). Consistent with our fixation data, live cell imaging showed that EGFP-Gp135-positive vesicles did not surround PACT-labeled centrosomes during metaphase. When cleavage furrow ingression occurred, EGFP-Gp135-positive vesicles were recruited to a region surrounding the centrosome (Figure 2A, 0:10). The movement of these EGFP-Gp135-positive vesicles followed the same path as centrosome migration (Figure 2A, 0:20–0:40), with the vesicles finally fusing to the middle membrane of the cell doublet (Figure 2A, 0:55). Using our live cell movies, we measured the polarity index and found that cell polarization was approximately begun after the onset of anaphase (Figure 2B). θ_{N-C} and L_{O-C} decreased significantly within 30 min after the anaphase onset and then plateaued (Figure 2C, D). These results suggest that centrosomes migrate toward the center of the cell doublet during the initial 30 min following the onset of anaphase, followed by a stable phase with reduced movement. Intriguingly, we observed this type of centrosome

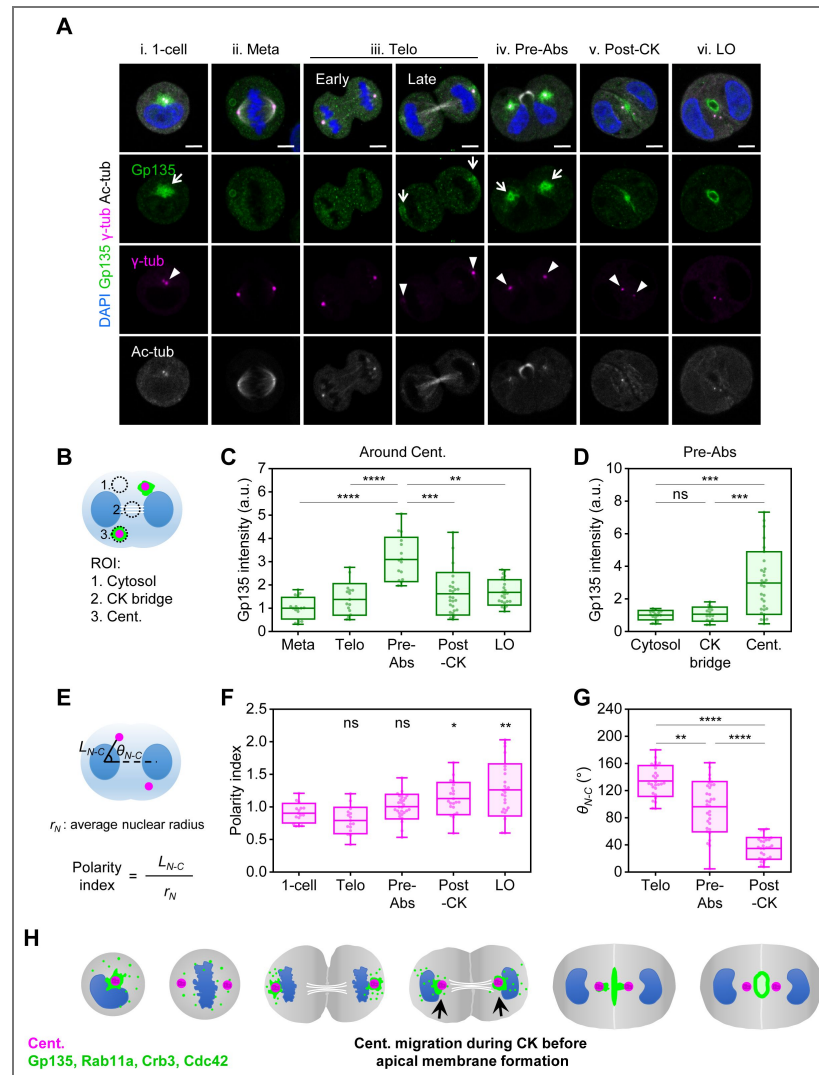


Figure 1. Localization of apical membrane protein Gp135 around the centrosome and its transport during epithelial polarization

(A) Single MDCK cells were cultured in Matrigel for 12 h. Immunostaining was performed with the following markers: apical membrane marker glycoprotein 135 (Gp135, green), centrosome marker γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). Single confocal sections through the middle of a cyst are shown. The order of polarization is arranged from single cell (1-cell), metaphase (Meta), telophase (Telo), cytokinetic pre-abscission (Pre-Abs), post-cytokinesis (Post-CK), to lumen open (LO). Arrows indicate Gp135 clusters around centrosomes, while arrowheads point to centrosome positions. Scale bar: 5 μ m. (B) Illustration showing the use of a fixed-size oval tool to select and measure Gp135 intensity in the chosen region. 1. Cytosol. 2. Cytokinesis (CK) bridge. 3. Centrosome (Cent.). (C–D) Boxplots of Gp135 intensity surrounding centrosomes at different stages normalized to the mean in metaphase (C) and at different cellular regions at the cytokinetic pre-abscission stage normalized to the mean in the cytosol (D). We analyzed >15 cells for each stage or region in three independent experiments. Statistical analyses were performed via one-way ANOVA and Dunn’s multiple comparisons (**** $p < 0.0001$, *** $p < 0.001$, ** $p < 0.01$, ns: not significant). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units). (E) Illustration showing how the polarity index (L_{N-C} [nucleus–centrosome distance] divided by r_N [average nuclear radius]) and θ_{N-C} (angle between the N–C and N–N axes) are calculated. (F–G) Boxplots depicting centrosome positions at different stages, shown by the polarity index (F) and θ_{N-C} (G). We analyzed >16 cells per measurement in three independent experiments. Statistical analyses used one-way ANOVA and Dunn’s multiple comparisons (* $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$, ns: not significant). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values. (H) Localization overview summarizing the localization of the centrosome (magenta) and recycling endosome/apical membrane components (green) during cell-division-directed polarization in Matrigel culture. White lines depict intercellular bridge microtubules, and nuclei are shown in blue. Arrows indicate apical membrane component localization around centrosomes and their transport from the centrosome to the AMIS.

movement only in 3D culture cells embedded in Matrigel. In contrast, in adherent two-dimensional (2D) culture cells grown on a cover glass, the centrosomes exhibited random migration patterns (Figure 2—figure supplement 1A–C).

To investigate the temporal process of apical membrane formation, we measured the intensity profile of EGFP-Gp135 along the line connecting the two nuclei; we took the central value as an indicator of the amount of Gp135 on the AMIS (Figure 2E, white arrows). The intensity continually increased after the onset of anaphase (Figure 2F). As shown in Figure 2A, the EGFP-Gp135 signal at the middle membrane of the cell doublet increased significantly after 30 min from the onset of anaphase, coinciding with a reduced movement of the centrosomes (Figure 2C, D). These findings are consistent with our immunostaining results, indicating that centrosome migration precedes the targeting of Gp135 to the AMIS.

Centrosomes influence the efficiency of early polarization but are not essential for late-stage lumen formation

To investigate the role of centrosomes in guiding Gp135-labeled vesicle trafficking for polarization, we depleted centrosomes in MDCK cells by applying centrinone, a reversible inhibitor of PLK4 kinase for centriole duplication in vertebrate cells. Centrinone treatment effectively blocked centriole duplication in MDCK cells and progressively depleted centriole numbers during cell cycle progression (Figure 3—figure supplement 1A, B). Three days after centrinone treatment, nearly 80% of cells lost their centrioles (Figure 3—figure supplement 1A), indicating the effective removal of centrosomes by centrinone treatment in MDCK cells.

Previous studies have reported that a loss of centrosomes in cells activates p53, followed by activation of p21, which causes cell cycle arrest in non-transformed cells (Fong et al., 2016; Lambrus et al., 2016; Meitinger et al., 2016). However, cell division is required in our experimental setup to initiate epithelial polarization. To overcome the problem of cell cycle arrest, we generated a p53 knockout (KO) line of MDCK cells using the clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9 editing system (Figure 3—figure supplement 1C, D). We found that the MDCK cells carry normal p53 activity and exhibit increased p21 protein levels after centrinone treatment (Figure 3—figure supplement 1E, F). As expected, the p53-KO MDCK cells do not induce p21 and continue proliferating after centrosome depletion (Figure 3—figure supplement 1F). Compared with wild-type (WT) cells, the p53-KO cells show no difference in polarity index, θ_{N-C} , or L_{O-C} during *de novo* polarization (Figure 3A–C and Figure 3—figure supplement 1G), indicating that the loss of p53 does not interfere with centrosome movement.

Next, we treated the p53-KO MDCK cells with centrinone for three days, seeded them into Matrigel, and observed them by live cell imaging (Figure 3—figure supplement 1H). Our results showed that, in the absence of centrosomes, EGFP-Gp135 did not cluster in the cytoplasm at any time points (Figure 3A), and the accumulation of EGFP-Gp135 at the middle membrane was significantly reduced (Figure 3A, D). Together, our time-lapse imaging and centrosome depletion studies suggest that centrosome depletion causes a reduction in Gp135 accumulation at the AMIS due to a lack of Gp135 near the centrosome during cytokinesis. We then performed immunostaining, which supported our live cell imaging observations (Figure 3—figure supplement 2A, B).

We extended the observation time to 48 and 72 hours and found that more than 60% of centrosome-depleted p53-KO cells, which overcome the proliferation arrest, form cysts with a central lumen (Figure 3E, F). For both WT and p53-KO cells, the percentage of single-lumen cysts is higher than centrosome-depleted p53-KO cells at 24 hours, but the differences decrease over time (Figure 3E, F). However, the cell heights of the centrosome-depleted cysts are less uniform, and the Gp135 levels at the apical region remained lower than those of the controls (WT and p53-KO) (Figure 3G). Although p53-KO can overcome cell cycle arrest after centrosome depletion, its cell proliferation rate is still reduced (Figure 3—figure supplement 2C). We, therefore, compared the percentage of lumen formation in two-cell stage cysts to eliminate the effect of proliferation variation. 24 hours after seeding in Matrigel, more than 50% of WT and p53-KO two-cell cysts

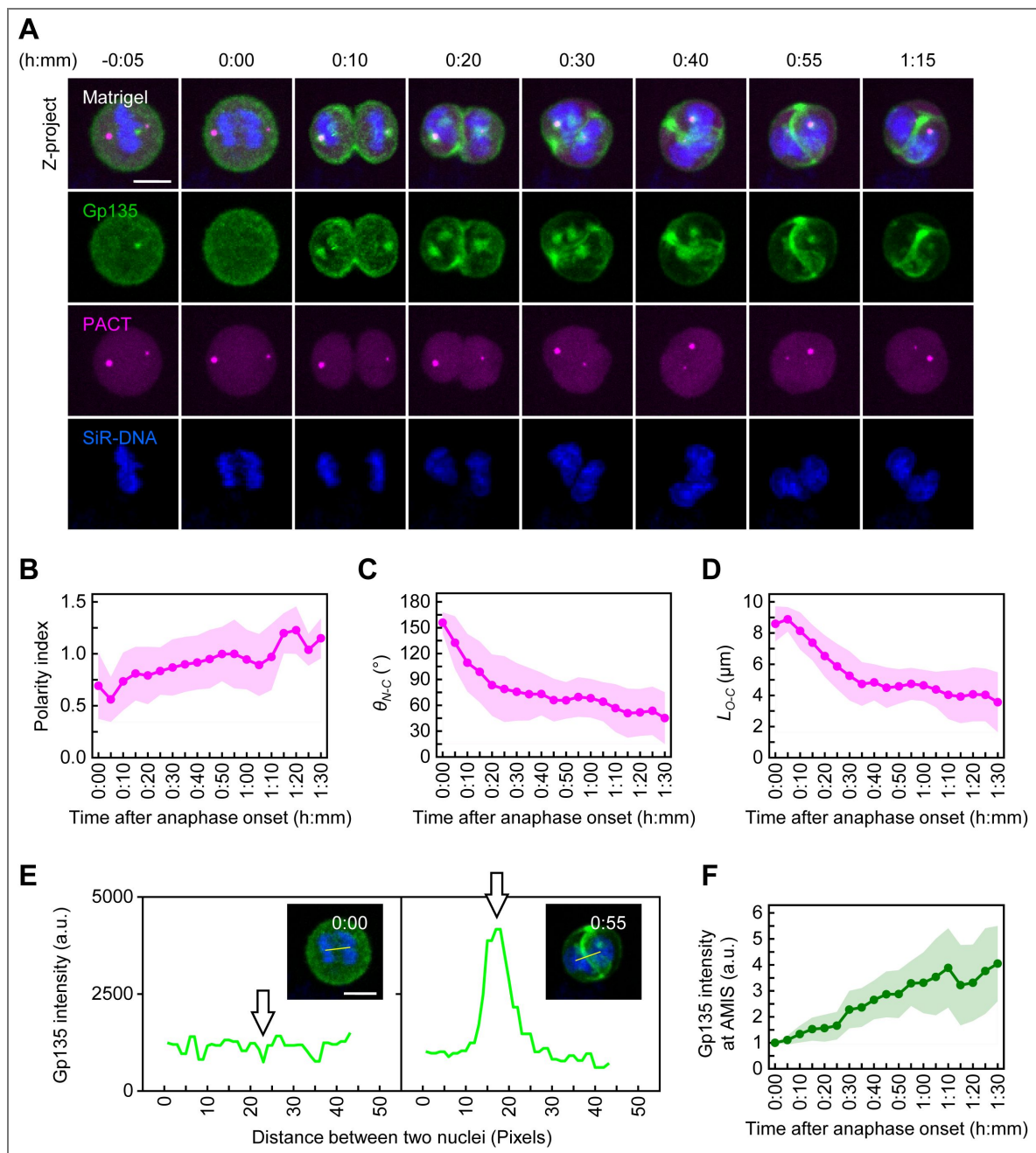


Figure 2. Timing of centrosome migration and Gp135 targeting to the AMIS

(A) Time-lapse snapshots of MDCK cells expressing EGFP-Gp135 (green) and PACT-mKO1 (magenta, centrosome marker) in Matrigel. Nuclei were labeled with SiR-DNA (blue) before live imaging. Z-projection images of a dividing cell are shown. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μm. (B–D) Change in polarity index, θ_{NC} , and L_{OC} over time. Each data point represents the average at a given time (>10 cells in Matrigel culture from three independent experiments). The lines show the means, and the shaded regions indicate SD values. (E) Fluorescent profiles of EGFP-Gp135 along the line connecting the two nuclei in a cell doublet. White arrows indicate the central value used to indicate the level of Gp135 on the AMIS (a.u., arbitrary units). Time stamps show hours and minutes, with 0:00 set to the first frame after anaphase onset. Scale bar: 10 μm. (F) Change in the central value in the EGFP-Gp135 fluorescent profile over time. The value of each time point was normalized to the value at 0:00. Each data point represents the average fluorescent intensity at a given time (10 cells in Matrigel culture from three independent experiments). The line shows the mean, and the shaded region indicates SD values (a.u., arbitrary units).

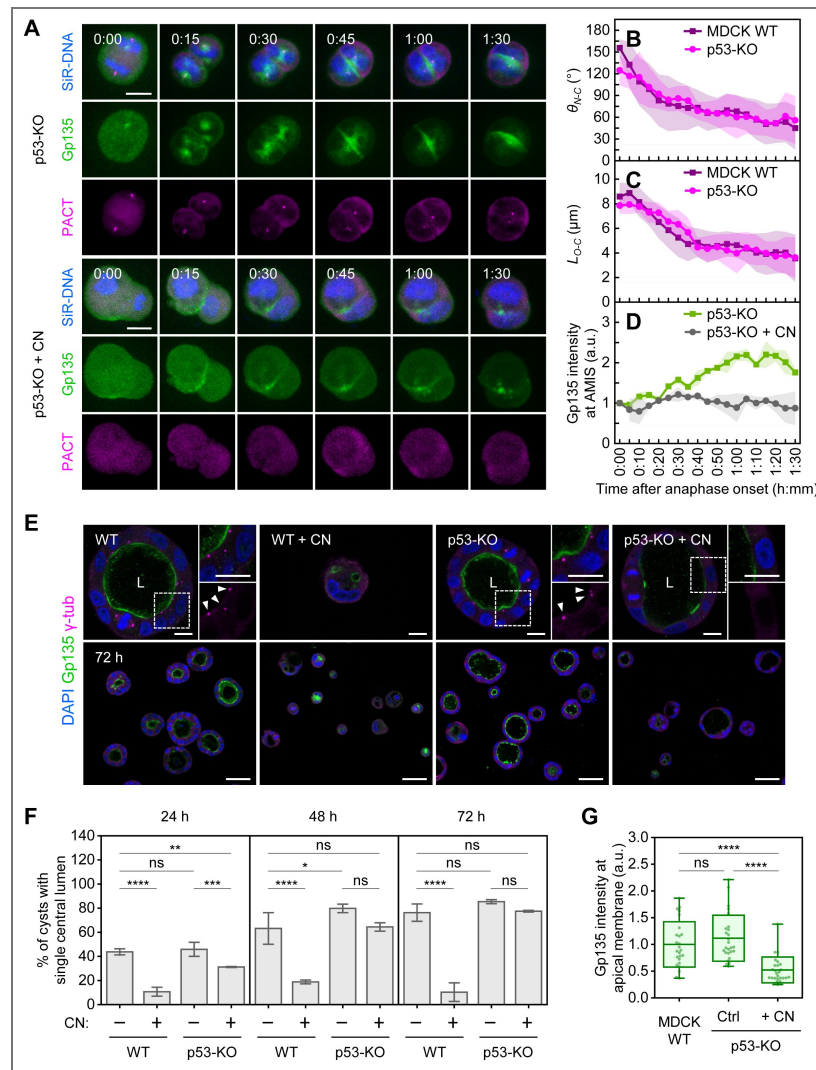


Figure 3. Impact of centriole depletion on apical lumen formation and Gp135 protein levels

(A) Time-lapse snapshots of p53-KO and centrione (CN)-treated cells expressing EGFP-Gp135 (green) and PACT-mKO1 (magenta, centrosome marker) in Matrigel. Nuclei were labeled with SiR-DNA (blue) before live imaging. Z-projection images of dividing cells are shown. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m. (B–C) Change in θ_{NC} and L_{OC} over time. Each data point represents the average at a given time (>3 p53-KO cells in Matrigel culture from three independent experiments). The lines show the means, and the shaded regions indicate SD values. (D) Change in the central value of the EGFP-Gp135 fluorescent profile over time. The value of each time point was normalized to the value at 0:00. Each data point represents the average fluorescent intensity at a given time (n = 3 [p53-KO; olive] and 3 [p53-KO + CN; gray]). The lines show the means, and the shaded regions indicate SD values (a.u., arbitrary units). (E) Single MDCK WT and p53-KO cells, with or without CN treatment, after 72 h of Matrigel culture. Single confocal sections through the middle of cysts are shown with immunofluorescent signals of indicated markers: apical membrane Gp135 (green), centrosome marker γ -tubulin (magenta), and DAPI for nuclei (blue). "L" denotes the lumen. WT cells treated with CN remain at the single-cell stage. The insets display enlarged images of the region in the yellow box. Arrowheads indicate the presence of centrosomes. Scale bar: 10 μ m. The bottom panels represent a larger view of the 3D culture. Scale bar: 50 μ m (F) Quantification of the proportion of MDCK cysts with a single central lumen after being cultured for different durations (24, 48, 72 h). The lumen structure was identified by Gp135 staining. n = 92 (24 h), 231 (48 h), 78 (72 h) cysts (MDCK WT); n = 22 (24 h), 133 (48 h), 59 (72 h) cysts (WT + CN); n = 154 (24 h), 170 (48 h), 204 (72 h) cysts (p53-KO); n = 105 (24 h), 128 (48 h), 149 (72 h) cysts (p53-KO + CN) analyzed for each measurement in three independent experiments. Statistical analyses used two-way ANOVA and Tukey multiple comparisons (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001, ns: not significant). Values represent the mean $\pm$ SD. (G) Boxplot of Gp135 intensity on the apical membrane normalized to the mean in WT cells. n = 25 (MDCK WT), 26 (p53-KO), 25 (p53-KO + CN) cysts were analyzed in three independent experiments. Statistical analyses were performed via one-way ANOVA and Dunn's multiple comparisons (ns: not significant, ****p<0.0001). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units).

formed a single central lumen. In contrast, only 30% of centrinone-treated p53-KO two-cell cysts exhibited a single central lumen (Figure 3—figure supplement 2D [↗](#)). Taken together, our data suggest that centrosome depletion primarily affects the early steps of epithelial polarization but not the later steps of lumen formation.

Disrupting centriole/centrosome structures and centrosomal microtubules does not alter centrosome positioning, but Gp135 trafficking is facilitated by centrosomal microtubules

Centrosomes are a large protein complex; thus, we sought to determine which components affect the position of centrosomes and the trafficking of Gp135 vesicles by perturbing centrosome components without completely removing the centrosome. We perturbed molecules associated with centriole/centrosome structures, such as CEP164 (an outer-layer protein of the DA), ODF2 (the base component of the SDA), and PCNT (the scaffold protein of the PCM) (Figure 4A [↗](#)). Each of these molecules has been reported to affect centrosome migration in different experimental systems (Hannaford et al., 2022 [↗](#); Hung et al., 2016 [↗](#); Krishnan et al., 2022 [↗](#); Pitaval et al., 2017 [↗](#)).

We generated KO lines of MDCK cells for CEP164, ODF2, and PCNT (Figure 4—figure supplement 1 [↗](#) and 2 [↗](#)), as well as a KO line of CEP120 in the $p53^{-/-}$ background, since we were unable to generate the CEP120 KO line in p53-WT cells for unknown reasons (Figure 4—figure supplement 2E–H [↗](#)). CEP120 is an important regulator of centriole elongation (Comartin et al., 2013 [↗](#); Lin et al., 2013 [↗](#)), and its KO enables the production of centrioles lacking both DA and SDA simultaneously (Figure 4A [↗](#)) (Tsai et al., 2019 [↗](#)). Our results showed no significant differences in θ_{N-C} or L_{O-C} between the WT and KO cells in the post-cytokinesis stage despite the presence of structural defects on the centriole/centrosome (Figure 4B–D [↗](#)). Gp135-labeled vesicles are still targeted to the center of cell doublets in the Matrigel culture (Figure 4E [↗](#)). Depletion of these centriole/centrosome structures does not affect polarized vesicle trafficking or centrosome positioning.

To determine whether defective centrosomes affect the timing of centrosome migration, we examined the centrosome position at the cytokinetic pre-abscission stage, as indicated by θ_{N-C} and L_{O-C} (Figure 4F–H [↗](#)); at this stage, centrosomes have initiated migration but have not yet reached their final position (Figure 1A, G [↗](#)). We found defective centrosomes also initiated migration at the cytokinetic pre-abscission stage in various KO lines (Figure 4F [↗](#)). The θ_{N-C} and L_{O-C} values showed no significant differences between the KO and WT cells (Figure 4G, H [↗](#)). These findings suggest that the timing of centrosome migration is not delayed in KO cells following the individual disruption of SDA, DA, or PCM components. However, during the pre-abscission stage, the intensity of Gp135 surrounding the centrosome is reduced in the PCNT-KO cells compared with the WT and other KO lines (Figure 4I [↗](#)), suggesting that centrosomal microtubules from PCM may play a role in recruiting Gp135 to the centrosome.

Microtubules serve as cables that generate mechanical forces facilitating centrosome migration (Laan et al., 2012 [↗](#); Okumura et al., 2018 [↗](#)). These highlight the critical importance of the microtubule–centrosome connection. Furthermore, SDA has been implicated in anchoring microtubules to the centrosome (Tateishi et al., 2013 [↗](#)). However, disruption of SDA through ODF2 knockout did not significantly affect centrosome positioning (Figure 4B–D, F–H [↗](#)), suggesting that SDA is not the sole determinant for centrosome migration. Additionally, although PCNT knockout cells show reduced microtubule nucleation ability, they still recruit a small amount of γ -tubulin (Figure 4F [↗](#)). This observation suggests that the connection between microtubules and centrosome is not completely abolished.

To further investigate the role of centrosomal microtubules in centrosome positioning and polarized trafficking, we used a known method to disrupt centrosomal microtubule nucleation from the centrosome by expressing dominant-negative fragments of the γ -tubulin activator CDK5RAP2 and its binding protein NEDD1 (Vinopal et al., 2023 [↗](#)). It was reported that exogenous expression of two DNA constructs encoding the centrosome-targeting carboxy-terminal domain (C-

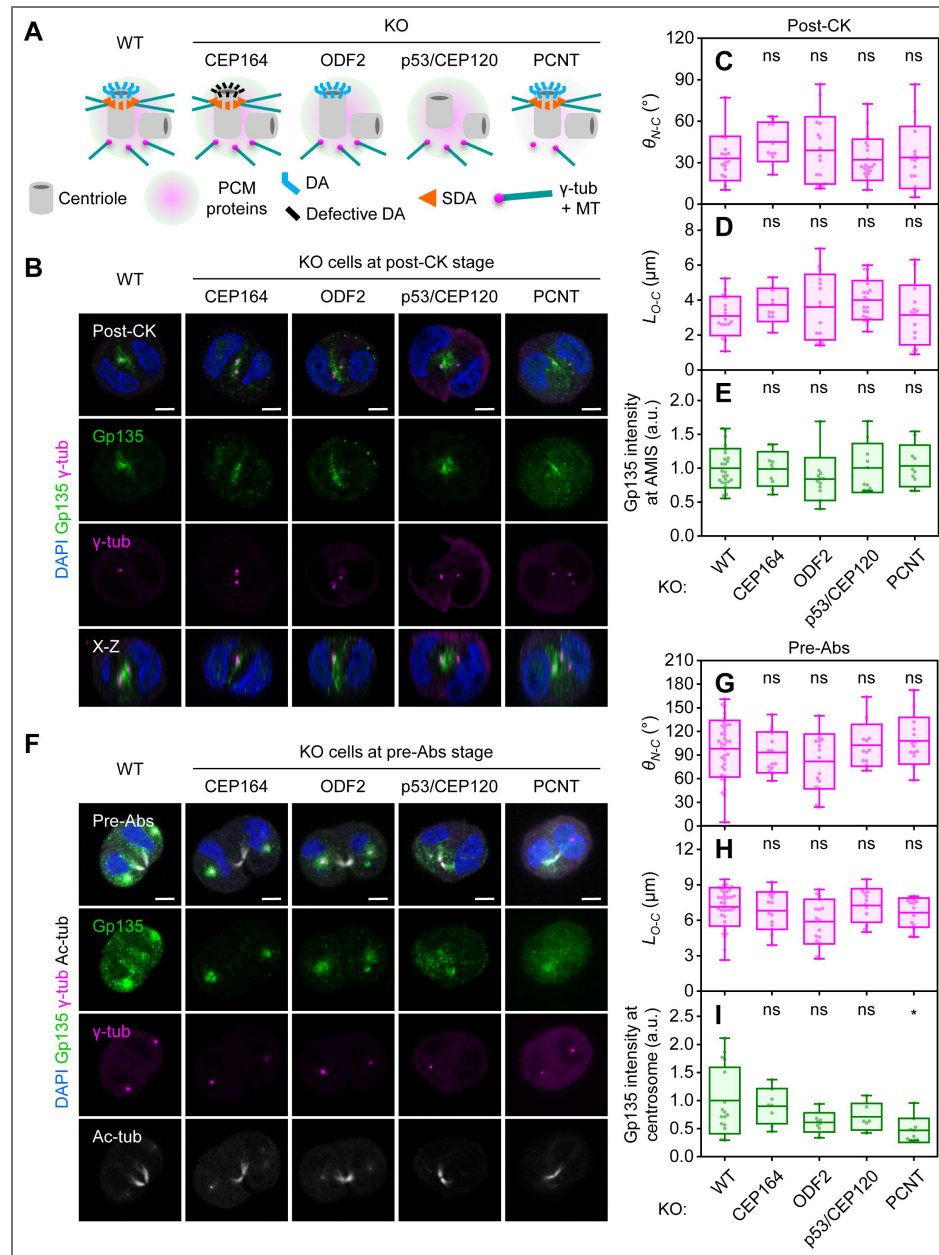


Figure 4. Centrosome migration in CEP164, ODF2, CEP120, and PCNT KO cells during epithelial polarization

(A) Illustration depicting individual centrosomal structures (DA, SDA, and PCM proteins) and the state of γ -tubulin and microtubules (γ -tub + MT) on the centrosome in cells with knockout (KO) of *CEP164*, *ODF2*, *p53/CEP120*, or *PCNT* genes. (B, F) Single MDCK cells of different genotypes were cultured in Matrigel for 12 h. Images show post-cytokinesis (CK) cells (B) or cells during pre-abscission (pre-Abs, F) with labeled markers: Gp135 (green), γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). Images shown are single confocal sections through the middle of cells and side-view x-z cross-sections. Scale bar: 5 μ m. (C–D, G–H) Boxplots of centrosome positions, represented by θ_{N-C} and L_{O-C} , for post-CK cells (C–D, n = 16 [WT], 10 [CEP164-KO], 15 [ODF2-KO], 20 [CEP120-KO], 14 [PCNT-KO]) and pre-Abs cells (G–H, n = 36 [WT], 14 [CEP164-KO], 16 [ODF2-KO], 12 [CEP120-KO], 14 [PCNT-KO]). Cells were analyzed in three independent experiments. Statistical analyses used one-way ANOVA and Dunn's multiple comparisons (ns: not significant). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values. (E, I) Boxplots of Gp135 intensity at the AMIS and centrosomes normalized to the mean in WT cells. Post-CK cell doublets: n = 25 (WT), 10 (CEP164-KO), 12 (ODF2-KO), 10 (CEP120-KO), and 10 (PCNT-KO). Pre-Abs cells: n = 16 (WT), 8 (CEP164-KO), 8 (ODF2-KO), 6 (CEP120-KO), and 10 (PCNT-KO). Data from three independent experiments. Statistical analyses were performed via one-way ANOVA and Dunn's multiple comparisons (ns: not significant, *p<0.05). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units).

CTD) of CDK5RAP2 (Figure 4—figure supplement 3A [↗](#)) and the γ -tubulin-binding domain (gTBD) of NEDD1 (N-gTBD) (Figure 4—figure supplement 3B [↗](#)) in cells resulted in almost complete depletion of γ -tubulin and a marked reduction in centrosomal microtubule nucleation at the centrosome. Here, we co-expressed N-gTBD-mCherry and C-CTD-mCherry in MDCK cells to selectively inhibit centrosomal microtubules. Cells were then embedded in Matrigel and cultured for 24 hours to assess centrosome positioning. As expected, γ -tubulin dispersed from the centrioles, whereas acetylated tubulin and CEP120 (marking the centriole) could still be detected as distinct dots (Figure 4—figure supplement 3C, D [↗](#), yellow arrowheads). Interestingly, the centriole dots were still remained positioned on the inner side of the two nuclei in cell doublets. Both the θ_{N-C} and L_{O-C} values showed no significant differences compared to those in control cells (Figure 4—figure supplement 3E, F [↗](#)), indicating that centrosomal microtubules anchored to the centrosome are not required for centrosome positioning. However, the intensity of Gp135 at the apical membrane were reduced at the two-cell stage (Figure 4—figure supplement 3G, H [↗](#)), suggesting that although centrosomal microtubules do not influence centrosome positioning, they enhance the efficiency of polarized trafficking and contribute to epithelial polarization.

The polarity regulator Par3 first emerges at the cytokinesis site to regulate centrosome positioning and polarized vesicle trafficking during *de novo* epithelial polarization

We next examined molecules that have been previously reported to be involved in MDCK polarization (Bryant et al., 2010 [↗](#); Horikoshi et al., 2009 [↗](#); Martin-Belmonte et al., 2007 [↗](#); Roland et al., 2011 [↗](#); Schluter et al., 2009 [↗](#)) to assess whether these molecules affect centrosome positioning. We first applied short hairpin RNA (shRNA) interference to deplete the molecules involved in vesicle trafficking, such as Rab11a, Sec15a, and myosin Vb (MyoVb) in WT cells (Figure 5—figure supplement 1A–C [↗](#)). Although depletion of the regulator of vesicle trafficking leads to a dispersion of Gp135, the centrosomes showed normal localization (Figure 5A–D [↗](#)). These results agree with our findings above, indicating that centrosome positioning governs the efficiency of polarized trafficking (Figure 3 [↗](#)). Inhibition of vesicle transport mediated by Rab11a, Sec15a, or MyoVb does not affect centrosome positioning.

Next, we applied shRNA to deplete polarity regulators, such as Par3, Cdc42, and Crb3, in WT cells (Figure 5—figure supplement 1D–F [↗](#)). We found that Par3 depletion, but not Cdc42 or Crb3, significantly affects the centrosome positioning, as shown by a large deviation in θ_{N-C} , L_{O-C} , and L_{IC} compared with the control (Figure 5A–D [↗](#)). The centrosomes do not move toward the AMIS after cytokinesis. Interestingly, we sometimes observed that in Par3-depleted cysts, Gp135- and Rab11a-positive vesicles are concentrated near mislocalized centrosomes in the post-cytokinesis stage (Figure 5A [↗](#) and Figure 5—figure supplement 1G [↗](#)). These data suggest that Par3 acts upstream of both centrosome positioning and polarized vesicle trafficking.

We next examined Par3 localization during *de novo* epithelial polarization through cytokinesis (Figure 5—figure supplement 1H [↗](#)). For MDCK cells in Matrigel culture, our results show that Par3 first emerges at the cell-cell interface of the mitotic cleavage site at the cytokinetic pre-abscission stage (Figure 5—figure supplement 1H [↗](#)), in contrast to the apical membrane components, which concentrate around the centrosome at this stage (Figure 1A, B [↗](#) and Figure 1—figure supplement 2A, B [↗](#)). Indeed, we found that the cell-cell junction components Par3, actin, ZO-1, and E-cadherin did not surround migrating centrosomes; rather, they localized to the cytokinesis site and cell-cell contact interface at the pre-abscission stage (Figure 5—figure supplement 2A, B [↗](#)). The cell-cell junction components, especially Par3, are recruited to the center of the cell doublets before positioning the centrosomes and targeting apical membrane components to the AMIS. The spatial-temporal localization of proteins involved in epithelial polarization is summarized in Table 1 [↗](#). The sequential order of these events and our knockdown experiments (Figure 5A–D [↗](#)) suggest that Par3 acts upstream of centrosome positioning and polarized vesicle trafficking.

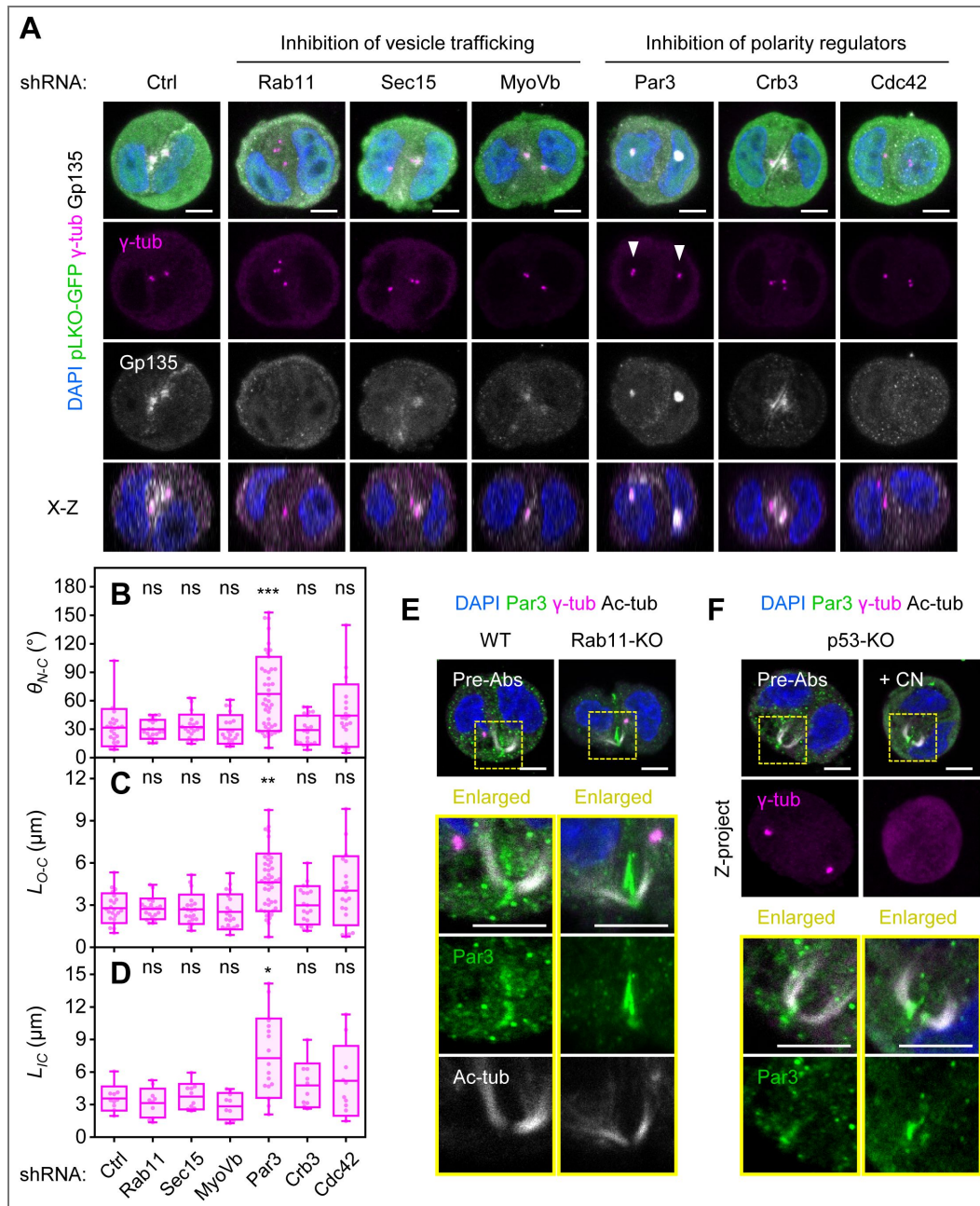


Figure 5. Par3 as the upstream regulator of centrosome migration and polarized vesicle transport

(A) Single MDCK cells, stably expressing pLKO-GFP-shRNA to knock down specified genes, cultured in Matrigel for 12 h. Images show post-cytokinesis cells with labeled markers: γ -tubulin (magenta), Gp135 (white), pLKO-GFP (green), and DAPI for nuclei (blue). Z-projection images between two centrosomes and the side-view x-z cross-section (bottom) are shown. "Ctrl" indicates the expression of scrambled-sequence shRNA. Arrowheads point to mislocalized centrosomes. Scale bar: 5 μ m. (B–D) Boxplots of centrosome positions, represented by θ_{N-C} , L_{O-C} (B–C, $n = 22$ [sh-Ctrl], 20 [sh-Rab11a], 20 [sh-Sec15a], 20 [sh-MyoVb], 44 [sh-Par3], 20 [sh-Crb3], 20 [sh-Cdc42]), and L_{I-C} (D, $n = 10$ [sh-Ctrl], 10 [sh-Rab11a], 10 [sh-Sec15a], 10 [sh-MyoVb], 16 [sh-Par3], 10 [sh-Crb3], 10 [sh-Cdc42]), in post-cytokinesis cells. Cells were analyzed in three independent experiments. Statistical analyses used one-way ANOVA and Dunn's multiple comparisons (ns: not significant, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values. (E) Single WT and Rab11a-KO MDCK cells were cultured in Matrigel for 12 h with labeled markers: Par3 (green), γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). Single confocal sections of pre-abscission cells are shown. The bridge region is enlarged from the yellow box. Scale bar: 5 μ m. (F) Single confocal sections of p53-KO cells, with or without centrinone (CN) treatment, cultured in Matrigel for 12 h with labeled markers: Par3 (green), γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). The bridge region shown in the yellow box is enlarged. The bottom panels display Z-projected images to demonstrate the complete depletion of centrosomes. Scale bar: 5 μ m.

Protein	Stage of cytokinesis directed epithelial polarization				Category
	Telophase	Pre-abscission	After cytokinesis	Lumen open	
Gp135	Vesicles/centrosome	Vesicles/ Migrating centrosome	AMIS	Apical membrane	Apical membrane components: from centrosomes
Rab11a	Vesicles/centrosome	Vesicles/ Migrating centrosome	Subapical vesicles	Subapical vesicles	
Crb3	Vesicles/centrosome	Vesicles/ Migrating centrosome	AMIS	Apical membrane	
Cdc42	Vesicles/centrosome	Vesicles/ Migrating centrosome	AMIS	Apical membrane	
aPKC	No signal	No signal	AMIS	Apical membrane Tight junction	Cell-cell junction components: before centrosome positioning
Par3	No signal	Cytokinesis site	AMIS	Tight junction	
ZO-1	No signal	Cytokinesis site	AMIS	Tight junction	
E-cad	No signal	Cell-cell contact	Cell-cell contact	Adherens junction	

Table 1. Distribution of trafficking and polarity proteins during cytokinesis-directed *de novo* epithelial polarization

To establish causality between polarity marker Par3 and polarized transport, we generated Rab11a-KO cells (Figure 5—figure supplement 2C–E). In Matrigel culture, we found that the Par3 signal still appears first at the cytokinesis site and that the centrosome begins to migrate toward the center of the cell doublet in Rab11a-KO cells (Figure 5E). Similarly, the complete depletion of centrosomes in centrinone-treated p53-KO cells did not affect Par3 localization (Figure 5F). These data provide evidence in the reverse direction, suggesting that centrosome positioning and polarized vesicle trafficking are downstream of Par3 recruitment.

Cytokinesis promotes the central migration of centrosomes during *de novo* epithelial polarization

Researchers have demonstrated an association between cytokinesis and the *de novo* polarization of epithelial cells (Klinkert et al., 2016; Li et al., 2014; Mangan et al., 2016; Rathbun et al., 2020; Schluter et al., 2009; Wang et al., 2014). Based on our results, Par3, which first localizes to the cytokinetic bridge, affects the centrosome positioning in Matrigel culture of MDCK cells. Here, we prevented cell division or changed culture conditions, which may affect the Par3 distribution in cells, to shed light on the relationship among the centrosome position, polarity regulator, and apical membrane components.

We observed cell doublets in Matrigel formed through cell-cell aggregation, not cell division (Figure 6—figure supplement 1A, see Materials and methods). Live-cell imaging showed that Gp135 is enriched on the membrane facing outward and that centrosomes remain stationary for several hours (Figure 6A). In early aggregated cell doublets, Gp135 polarity is inverted, consistent with previous studies (Bisi et al., 2020; Bryant et al., 2010; Bryant et al., 2014; Chou et al., 2016; Martin-Belmonte et al., 2007; Roman-Fernandez et al., 2018). We further performed immunostaining to examine polarity molecules. We found that Par3 was recruited to the cell-cell interface in the aggregated doublets (Figure 6B). Rab11a-positive recycling endosomes were concentrated around the centrosome (Figure 6C). Quantitative measurement θ_{NC} revealed that, although centrosomes are located between the two nuclei in the doublet generated by cytokinesis (Ctrl, Figure 6D, E), centrosomes are widely distributed across both nuclei in the aggregated doublets (+Aphi, Figure 6D, E). The results of the aggregated doublets indicate that the relationships among Gp135-associated apical membrane formation, Par3 location, and centrosome positioning are distinct from that of cytokinesis-induced *de novo* epithelial polarization.

We next create inverted cysts by culturing suspended single cells in low-adhesion microwells (Figure 6—figure supplement 1B) (Huang et al., 2020; Yu et al., 2005); in these conditions, the normal cell cycle progresses, but cells exhibit an inverted Gp135 polarity from cells in Matrigel culture. Immunostaining shows that Par3 proteins form a ring at the edge of the cell-cell interface (Figure 6F), distinct from the Par3 patch observed at the center of the cell-cell interface for cells in Matrigel (Figure 5—figure supplement 1H).

To observe the establishment of inverted polarity, cells were synchronized at prometaphase using nocodazole, then released and cultured in low-adhesion microwells, followed by live-cell imaging (Figure 6—figure supplement 1C). Before and at the onset of anaphase (Figure 6G, −0:15–0:00), the centrosomes form two spindle poles at two sides of the nuclei. At this time point, there is a dispersed Gp135 signal inside the cytoplasm. When cleavage furrow ingression occurs (Figure 6G, 0:05–0:20), Gp135 is concentrated at the furrow membrane between the two daughter cells. As cytokinesis progresses, Par3 first appears at the cell-cell interface at the mitotic cleavage site during the cytokinetic pre-abscission stage, similar to what is observed in Matrigel culture (Figure 6—figure supplement 1D). Gp135 dissociates from the middle membrane and becomes concentrated around the migrating centrosomes and external membranes (Figure 6G, 0:30, white and yellow arrows, Figure 6—figure supplement 1E). While Gp135 spreads toward the outside membrane, the centrosomes migrate toward the center of the cell doublets (Figure 6G, 0:40–1:05, and H–J). Finally, Gp135 is concentrated at the membranes, corresponding to locations where cells are not in contact, and forms a polarization opposite to that of cysts in Matrigel. Taken together, our results identify an interesting condition in which the polarity indicated by the

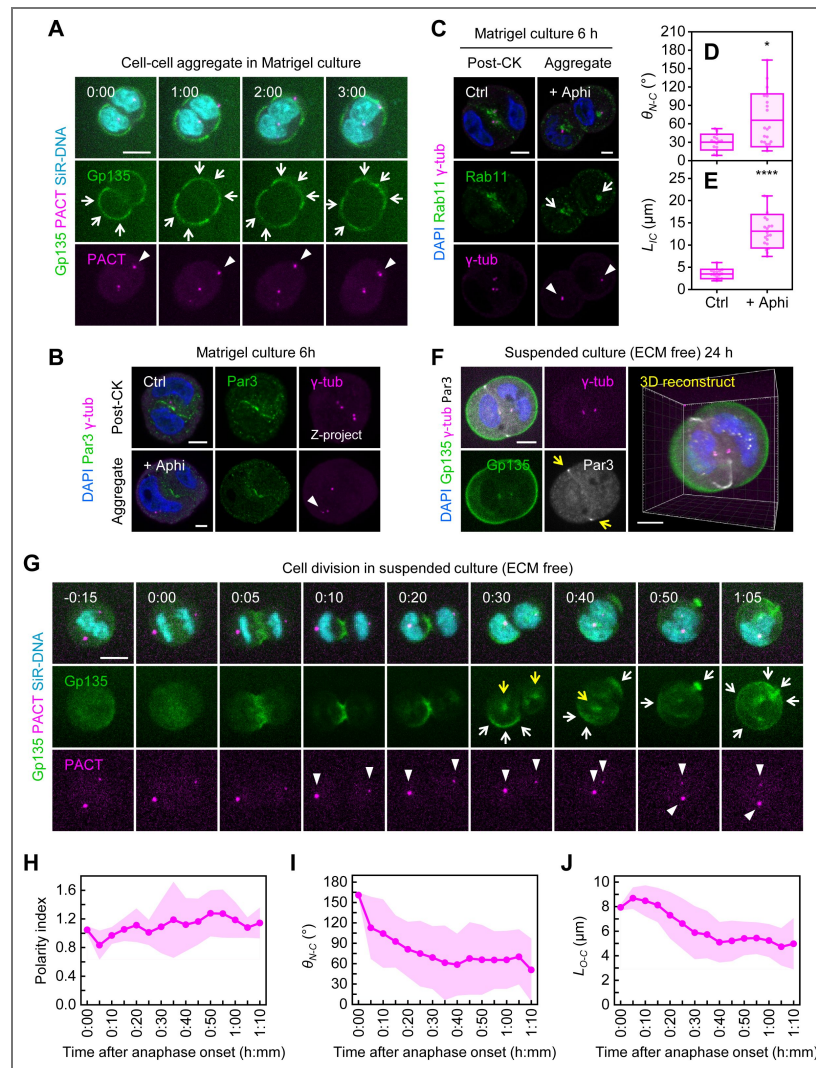


Figure 6. Centrosome migration in suspended 3D culture during cytokinetic pre-abscission

(A) Time-lapse snapshots of MDCK cells expressing EGFP-Gp135 (green) and PACT-mKO1 (magenta, centrosome marker) in Matrigel. Nuclei were labeled with SiR-DNA (cyan) before live imaging. Z-projection images of an aggregated cell doublet are shown. Arrows indicate Gp135 on the outer surface of the cell aggregate. Arrowheads point to mislocalized centrosomes that did not move over 3 h. Time stamps show hours and minutes, with 0:00 set at the beginning of live cell imaging. Scale bar: 10 μ m. (B, E) MDCK cells, with or without aphidicolin (Aphi) synchronization, were cultured in Matrigel for 6 h with labeled markers: Rab11a (B, green), Par3 (E, green), γ -tubulin (magenta), and DAPI for nuclei (blue). Single confocal sections are shown. Two types of cell doublets, formed by cell division (post cytokinesis [CK]) or aggregation, were compared. Arrows indicate Gp135 clusters around centrosomes. Arrowheads point to mislocalized centrosomes. Scale bar: 5 μ m. (C–D) Boxplots of centrosome positions, represented by θ_{N-C} (C) and L_{IC} (D), in cell doublets ($n = 15$ [unsynchronized control], 20 [Aphi-synchronized cells]). Cells were analyzed in three independent experiments. Statistical analyses used an unpaired two-tailed Mann–Whitney U test (* $p < 0.05$, **** $p < 0.0001$). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values. (F) MDCK cells were cultured in low-attachment microwells for 24 h with labeled markers: Gp135 (green), γ -tubulin (magenta), Par3 (white), and DAPI for nuclei (blue). A single confocal section through the middle of a cell doublet is shown. Yellow arrows indicate the Par3 signal at the edge of cell-cell contacts. The 3D reconstruction reveals Par3 forming a ring around the edge between cell doublets. Scale bar: 10 μ m. (G) Time-lapse snapshots of MDCK cells expressing EGFP-Gp135 (green) and PACT-mKO1 (magenta, centrosome marker) in low-attachment microwells. Nuclei were labeled with SiR-DNA (cyan) before live imaging. Z-projection images of a dividing cell are shown. White and yellow arrows indicate Gp135 on the outer surface of the cell doublet and around the centrosome, respectively. Arrowheads point to the centrosome position. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m. (H–J) Change in polarity index (H), θ_{N-C} (I), and L_{O-C} (J) over time. Each data point represents the average at a given time (three cells in low-attachment microwells from three independent experiments). The lines show the means, and the shaded regions indicate SD values.

centrosome and nucleus is opposite to the apical-basal polarity. In this case, the apical side is marked by Gp135, whereas the opposite pole is basal and in contact with adjacent cells. Furthermore, cytokinesis drives centrosome migration toward the center of the cell doublets, even in the absence of ECM and under conditions of inverted polarity.

Finally, we investigated the movement of centrosomes at the cytokinetic pre-abscission stage in highly polarized cysts (Figure 6—figure supplement 2A [↗](#)) and in an epithelial sheet (Figure 6—figure supplement 2B, C [↗](#)), of which both have apical-localized cytokinetic bridges. Both live cell imaging and immunostaining reveal that, in telophase cells, centrosomes are initially localized laterally at the mid-plane of the apical-basal axis and subsequently migrate apically to the site of the cytokinetic bridge during the pre-abscission stage (Figure 6—figure supplement 2A–C [↗](#)). Consistent with *de novo* epithelial polarization, we found that, when the apical membrane has not yet formed, the centrosomes move toward the center of cell doublets where the cytokinetic bridge is located (Figure 6—figure supplement 2A–D [↗](#)). Taken together, these findings indicate that the centrosome position is strongly associated with the cytokinesis site but not with the apical membrane.

Discussion

Because the vector from the nucleus to the centrosome is often observed aligning with the axis of other polarization markers in various cells, the centrosome has long been hypothesized to play a crucial role in regulating cell polarization (Bornens, 2012 [↗](#); Burakov & Nadezhdina, 2020 [↗](#); Tang & Marshall, 2012 [↗](#)). Based on the significant movement of centrosomes during polarization derived from cell division, as well as the dependence of polarity molecules and apical membrane location on culture conditions, we designed a series of experiments to test the role of the centrosome. Our findings, which highlight the correlation between centrosome positioning and cell polarization, as well as the impact of altered centrosome function on the distribution of polarity markers, are summarized in Figure 7 [↗](#).

In Matrigel culture, where *de novo* epithelial polarity develops, our findings show that apical membrane components, including Gp135, Crb3, and Cdc42, accumulate around the centrosomes and follow the rapidly moving centrosomes to the AMIS during cytokinesis (Figure 1 [↗](#) and 2 [↗](#)). Par3 first appears at the cytokinetic bridge, earlier than the positioning of the centrosome and the targeting of apical membrane components (Figure 5 — figure supplement 1H [↗](#)). The aligned direction of centrosome positioning, apical membrane protein trafficking, and Par3 localization significantly enhance the efficiency of polarization (Figure 7A [↗](#)). The centrosome depletion experiments indicate that loss of centrosomes still allows the formation of AMIS and lumen structures (Figure 3 [↗](#) and 5F [↗](#)), similar to their dispensable roles in cell division or spindle microtubule formation (Meitinger et al., 2016 [↗](#); Watanabe et al., 2020 [↗](#); Wong et al., 2015 [↗](#)). Under p53-KO conditions, centrosome-depleted MDCK cells can form cysts with a single central lumen at a slower rate. However, due to decreased trafficking efficiency, the protein levels on the apical membrane are reduced (Figure 7B [↗](#)). Previous studies have debated the role of centrosomes in various types of cell polarization (de Anda et al., 2010 [↗](#); Martin et al., 2018 [↗](#); Schmoranz et al., 2009 [↗](#); Stiess et al., 2010 [↗](#); Wakida et al., 2010 [↗](#)). We determine that while the centrosome is not essential, it has a synergistic effect on apical membrane establishment.

In addition to complete removal of the centrosome, we also disrupted specific substructures within the centrosome to investigate which components are responsible for regulating centrosome positioning, vesicle trafficking, and cell polarization. However, we did not identify a single centrosomal molecule whose disruption alone significantly affected centrosome positioning (Figure 4 [↗](#)). These findings led us to further investigate whether centrosome-nucleating microtubules are essential for proper centrosome positioning and polarized vesicle trafficking. By competitively displacing γ -tubulin from centrosomes, we were able to produce centrioles devoid of associated γ -tubulin. Due to the displacement of γ -tubulin, these centrioles were unable to nucleate microtubules, demonstrating the effective elimination of centrosomal microtubules. Interestingly, despite the loss of centrosomal microtubules, centrioles were still localized to the

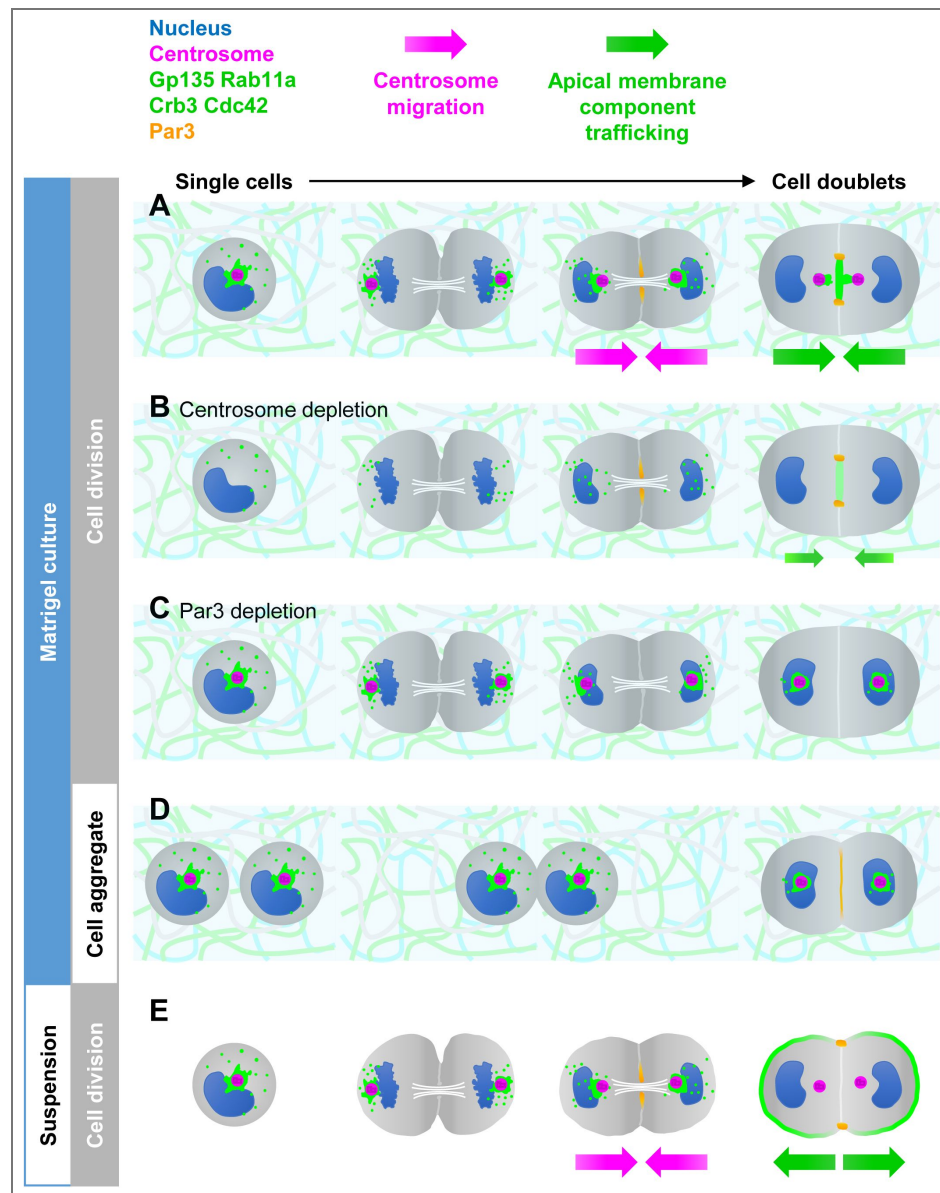


Figure 7. Summary of centrosome migration, apical membrane component trafficking, and Par3 recruitment in different experiments and culture conditions

(A) In conventional Matrigel culture, centrosomes move directionally toward AMIS following cell division (magenta arrow). Apical membrane components like Gp135, Crb3, and Cdc42 follow the centrosomes (green arrow). Par3 first emerges at the cytokinesis site, which regulates centrosome positioning and polarized vesicle trafficking during polarization. (B) Loss of centrosomes diminishes the effectiveness of apical membrane component trafficking (smaller green arrow), yet Par3 still localizes to the cytokinetic bridge. Additionally, centrosome loss mainly impacts the initial stages of epithelial polarization rather than the later stages of lumen formation. (C) Loss of Par3 results in randomized centrosome positioning, obstructs polarized trafficking, and traps apical membrane components around mislocalized centrosomes. (D) Two-cell aggregates without cell division cannot effectively guide centrosome migration and polarized vesicle trafficking. Despite Par3 being recruited to the cell-cell interface, apical membrane components remain trapped around mislocalized centrosomes. (E) MDCK cells suspended in the ECM-free condition. Centrosomes migrate to the center of cell doublets during cytokinesis (magenta arrow), while Apical membrane components transport in the opposite direction (green arrow). After mitosis, Par3 exhibits a pattern distinct from that observed in Matrigel culture.

apical region of Matrigel-cultured MDCK cells (Figure 4—figure supplement 3 [3](#)). These findings challenge the previously proposed model that centrosomal microtubules are required to actively pull centrosomes to a specific cellular position.

Among all our experiments involving molecular perturbations to elucidate relationships between different polarity features, we have discovered that Par3 plays a pivotal role in upstream signaling during cytokinesis-induced polarization, which affects centrosome positioning and polarized trafficking (Figure 7C [3](#)). Par3 has been reported to regulate centrosome position in micropattern-cultured MCF10A cells and in the intestine of *C. elegans* (Burute et al., 2017 [3](#); Feldman & Priess, 2012 [3](#)) and has also been shown to affect centrosome position through interaction with dynein in migrating NIH 3T3 cells (Schmoranz et al., 2009 [3](#)). However, Par3 is not directly localized to the centrosome, but rather to the cell cortex or cell-cell junctions. Therefore, Par3 is likely to regulate centrosome positioning through some intermediate molecules or mechanisms, but its specific mechanism is still unclear and requires further investigation. Previous study have also suggested that Par3 functions as an exocyst receptor in epithelial cells, mediating the targeting of delivered proteins to specific domains of the plasma membrane through interaction with Sec8 and Exo70, two exocyst components (Ahmed & Macara, 2017 [3](#)). Depletion of Par3 leads to membrane proteins and Rab11a residing in the cytoplasm (Ahmed & Macara, 2017 [3](#); Horikoshi et al., 2009 [3](#)). Consistent with these findings, we found that Rab11a-positive vesicles sometimes pause during delivery and become trapped around mislocalized centrosomes in Par3-depleted MDCK cells (Figure 5A [3](#) and Figure 5—figure supplement 1G [3](#)). Taken together, our findings support the concept that Par3 participates in centrosome positioning and mediates the transportation of proteins on the Rab11a-positive endosomes to AMIS during cytokinesis-induced polarization. However, cultured conditions influence cellular polarization preferences. For instance, when cells aggregate in Matrigel, we observe recruitment of Par3 to the cell-cell interface, despite the absence of a division site to specify its location, but both centrosome positioning and Gp135 transport are disrupted (Figure 6A-E [3](#) and 7D [3](#)). It is necessary to determine the differences in the function of Par3 between cell aggregates and during cytokinesis, such as whether it also interacts with dynein and exocysts in these cell aggregates.

When MDCK cells are suspended in ECM-free conditions, they form inverted cysts after division, with Gp135 facing outward, while Par3 forms a ring at the edge of the cell-cell contact surface (Figure 6F-J [3](#)). Intriguingly, the vector from the nucleus to the centrosome is opposite from the direction of the apical membrane (Figure 7E [3](#)). Notably, we found that centrosomes do not constantly localize near the apical surface. These different cultured conditions help us dissect the process of epithelial polarization and the relationships between different polarity indicators. During epithelial polarization, centrosome positioning, apical membrane component trafficking, and the recruitment of the polarity regulator Par3 appear to be executed through distinct mechanisms. It is known that apical membrane components undergo endocytosis and are transported by the Rab11a cascade to target in the opposite direction of the ECM contact side (Jewett & Prekeris, 2018 [3](#); Yu et al., 2005 [3](#)). However, the mechanisms for centrosome positioning and Par3 recruitment remain unclear. Recent studies have shown that the cortical actin flows in *C. elegans* blastomeres attracts the enrichment of Par3 (Ng et al., 2023 [3](#)), and that the acto-myosin network plays a role in the centrosome off-centering process in enucleated cells (Jimenez et al., 2021 [3](#); Yamamoto et al., 2022 [3](#)). Therefore, the dynamics of the actomyosin network may be crucial for both Par3 recruitment and centrosome migration, as it undergoes significant rearrangements during cytokinesis. During the pre-abscission stage, the actomyosin network loses its symmetric distribution and a contractile ring forms at the center of the cell doublet, while the amount of actin decreases at the cell periphery (Figure 5—figure supplement 2A [3](#)). These events during cytokinesis may promote Par3 recruitment and trigger centrosome migration. Subsequently, a dense actin network remains at the center of the cell doublet, contributing to the formation of the apical actin network during the post-cytokinesis stage. In conventional Matrigel cultures, ECM signals and the actin network are aligned in the same direction, synergistically enhancing the efficiency of epithelial polarization. In cell-aggregated

conditions, cells have not yet divided and the actin network does not show dynamic changes. This lack of remodeling may prevent centrosome movement, resulting in a reduced efficiency of epithelial cell polarization.

We also explored the possibility of affecting the apical positioning of centrosomes by disrupting centrosomal components, such as the appendage structures and PCM proteins that link centrosomes to microtubules (Figure 4 [↗](#)), as microtubules are believed to generate pulling or pushing forces to aid in centrosome positioning (Hooikaas et al., 2020 [↗](#); Kapitein et al., 2005 [↗](#); Yi et al., 2013 [↗](#)). We found that individually disrupting centrosomal components does not affect centrosome positioning. Therefore, we hypothesize that either the connections between microtubules and centrosomes are very robust or additional mechanisms may play a role in centrosome positioning. For instance, recent studies have proposed that motor proteins transport centrosomes along non-centrosomal microtubules as cargo (Hannaford et al., 2022 [↗](#); Hannaford & Rusan, 2024 [↗](#)), and other studies suggest that the actin network contributes to centrosome positioning (Jimenez et al., 2021 [↗](#); Yamamoto et al., 2022 [↗](#)).

Although our investigation primarily focused on the 3D culture of epithelial cells *in vitro*, the insights gleaned from this study may have implications for understanding polarity establishment in various epithelial-like tissues *in vivo*.

Materials and methods

3D cell culture in Matrigel and drug treatment

MDCK (NBL-2) cells (Bioresource Collection and Research Center, Taiwan) were tested for mycoplasma, cultured in minimum essential medium (MEM) with glucose, 2 mM L-glutamine, and 10% fetal bovine serum, and supplemented with 0.1 mM non-essential amino acids and 1 mM sodium pyruvate, penicillin, and streptomycin. We conducted 3D cell culture assays for cyst formation using a previously described method (Lee et al., 2007 [↗](#); Martin-Belmonte et al., 2007 [↗](#)). We trypsinized 70% of confluent cells to a single cell suspension of 4×10^4 cells/mL in complete MEM containing 2% growth factor reduced Matrigel (356231, Corning, Glendale, AZ). We plated cells in 2% Matrigel medium (300 μ L) in 8-well cover-glass chamber slides (80827, ibidi, Gräfelfing, Germany) pre-coated with 8 μ L of 100% Matrigel. The cells were incubated for the indicated period, and the medium was changed every 2 days.

For the centrosome depletion assay, we pre-treated MDCK WT and p53-KO cells with 300 nM centrinone (LCR-263, HY-18682, MedChemExpress, Monmouth Junction, NJ) (Wong et al., 2015 [↗](#)) for 3 days on a petri dish to achieve >80% of cells without centrosomes. The centrosome-depleted cells were then trypsinized to a single-cell suspension and plated into a 3D culture in the continuous presence of 300 nM centrinone.

To assess cell aggregates, we treated MDCK cells with aphidicolin (Aphi, 2.5 μ g/mL) for 15 hours, arresting almost all cells in the S phase. Subsequently, we released these synchronized cells from the S phase and simultaneously cultured them in Matrigel. Based on past experience, MDCK cells require approximately 8 hours to progress from the S phase to the M phase. Therefore, we observed cell doublets at 6 hours after release to ensure that these cell doublets were formed through aggregation rather than cell division.

To enrich mitotic cells in suspension culture, MDCK cells were treated with nocodazole (100 ng/mL) for 4 hours to arrest cells predominantly at the prophase stage. Cells were then trypsinized and released from the nocodazole block, and immediately transferred to low-attachment culture dishes to maintain suspension conditions. Based on previous observations, cell doublets typically appeared ~1 hour after release, corresponding to the telophase to pre-abscission stages.

Antibodies

We used rabbit polyclonal antibodies against Gp135 (IF: 1:400, a kind gift from Dr. Jou Tzue-Shuh, NTU, Taiwan) (Lim et al., 2017 [↗](#)), Rab11a (IF: 1:400, WB: 1:2000, 20229-1-AP, Proteintech, Rosemont, IL), aPKC (IF: 1:200, 610175, BD Biosciences, Franklin Lakes, NJ), CEP120 (IF: 1:1000, WB:

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (Canis lupus familiaris)	MDCK (NBL-2)	BCRC, Taiwan	BCRC No. 60004	
Cell line (Canis lupus familiaris)	MDCK GFP-Gp135/PACT-mKO1 stable line	This work		
Cell line (Canis lupus familiaris)	MDCK p53-KO line	This work		
Cell line (Canis lupus familiaris)	MDCK CEP164-KO line	This work		

Key reagent or resource table.

Cell line (Canis lupus familiaris)	MDCK ODF2-KO line	This work		
Cell line (Canis lupus familiaris)	MDCK p53/CEP120-KO line	This work		
Cell line (Canis lupus familiaris)	MDCK PCNT-KO line	This work		
Cell line (Canis lupus familiaris)	MDCK Rab11a-KO line	This work		
Antibody	Gp135 (Rabbit polyclonal)	a kind gift from Dr. Jou Tzuu-Shuh, NTU, Taiwan		IF: 1:400
Antibody	Rab11a (Rabbit polyclonal)	Proteintech	20229-1-AP	IF: 1:400, WB: 1:2000
Antibody	CEP120 (Rabbit polyclonal)	Lin et al., 2013		IF: 1:1000, WB: 1:2000
Antibody	p53 (Rabbit polyclonal)	Cell Signaling	92825	IF: 1:100
Antibody	p21 (Rabbit polyclonal)	Proteintech	10355-1-AP	IF: 1:200, WB: 1:2000
Antibody	CEP164 (Rabbit polyclonal)	NOVUS	NBP1-81445	IF: 1:400
Antibody	Pericentrin (Rabbit polyclonal)	Abcam	ab4448	IF: 1:200
Antibody	Par3 (Rabbit polyclonal)	EMD Millipore	07-330	IF: 1:200, WB: 1:1000
Antibody	β -actin (Rabbit polyclonal)	Sigma-Aldrich	A2066	IF: 1:200
Antibody	E-cadherin (Rabbit monoclonal)	Cell Signaling	24E10, 3195	IF: 1:200

Key reagent or resource table. (continued)

Antibody	γ -tubulin (Mouse monoclonal IgG1)	Sigma-Aldrich	GTU-88, T6557	IF: 1:400
Antibody	Acetylated tubulin (Mouse monoclonal IgG2b)	Sigma-Aldrich	T6793	IF: 1:400
Antibody	Gp135 (Mouse monoclonal IgG1)	a kind gift from Dr. Jou Tzuu-Shuh, NTU, Taiwan	3F2D8	IF: 1:10
Antibody	aPKC (Mouse monoclonal IgG2b)	BD Biosciences	610175	IF: 1:200
Antibody	ODF2 (Mouse monoclonal IgG2a)	NOVUS	H00004957-M01	IF: 1:400
Antibody	Glutamylated tubulin (Mouse monoclonal IgG1)	AdipoGen	GT335, AG-20B-0020	IF: 1:100
Antibody	α -tubulin (Mouse monoclonal IgG1)	Sigma-Aldrich	DM1 α , T9026	WB: 1:5000
Antibody	Cdc42 (Mouse monoclonal IgG1)	BD Biosciences	610928	WB: 1:1000
Antibody	ZO-1 (Mouse monoclonal IgG1)	Thermo Fisher Scientific	33-9100	IF: 1:400
Antibody	Alexa Fluor 488-, 561-, 633-, and 647-secondaries	Thermo Fisher Scientific	A-11029; A-11031; A-21236; A-11034; A-11036; A-21245; A-21124; A-21146	IF: 1:400
Antibody	HRP-labeled secondaries	Perkin Elmer	NEF822; NEF812	WB: 1:5000

Key reagent or resource table. (continued)

Recombinant DNA reagent	pLAS2w.Ppuro-EGFP-Crb3	This work		
Recombinant DNA reagent	pLAS2w.Ppuro-EGFP-Cdc42	This work		
Recombinant DNA reagent	pLAS2w.Ppuro-EGFP-Gp135	This work		
Recombinant DNA reagent	pLAS2w.Ppuro-PACT-mKO1	This work		
Sequence-based reagent	sgRNA target canine <i>p53</i> exon 4 #1	This study	5'-TCCCAGAGAGCGTCGTGAAC-3'	fw
Sequence-based reagent	sgRNA target canine <i>p53</i> exon 4 #2	This study	5'-ACTTGGCTGTCCCGGAATGC-3'	rev
Sequence-based reagent	sgRNA target canine <i>CEP164</i> exon 4 #1	This study	5'-GTCCCACATGGACTGCCC GT-3'	rev
Sequence-based reagent	sgRNA target canine <i>CEP164</i> exon 4 #2	This study	5'-AACTTGGTGATTCAAGAGCG-3'	fw
Sequence-based reagent	sgRNA target canine <i>ODF2</i> exon 7 #1	This study	5'-CCTCATGTCCAAGCTGGTAG-3'	fw
Sequence-based reagent	sgRNA target canine <i>ODF2</i> exon 7 #2	This study	5'-ATCGGGAAGCTGAAGACGGT-3'	fw
Sequence-based reagent	sgRNA target canine <i>CEP120</i> exon 3 #1	This study	5'-ACAGTTGGCTACCGACCC TG-3'	fw
Sequence-based reagent	sgRNA target canine <i>PCNT</i> exon 4 #1	This study	5'-TCGCCAGGGAGCAGCATGCG-3'	fw
Sequence-based reagent	sgRNA target canine <i>Rab11a</i> exon 1 #1	This study	5'-ATAGTCGTACTCGTCGTCGC-3'	rev
Sequence-based reagent	sgRNA target canine <i>Rab11a</i> exon 2 #2	This study	5'-CTTACGCTGATGTTATAGCT-3'	rev

Key reagent or resource table. (continued)

Sequence-based reagent	p53-KO sequencing primer F	This study	5'-TGTTCTGgtaaggactgggtg-3'	
Sequence-based reagent	p53-KO sequencing primer R	This study	5'-tgaggccaaaggtcacagaa-3'	
Sequence-based reagent	CEP164-KO sequencing primer F	This study	5'-cagggactaatgcctgacctct-3'	
Sequence-based reagent	CEP164-KO sequencing primer R	This study	5'-tcaaaggaaaattgccagaaga-3'	
Sequence-based reagent	ODF2-KO sequencing primer F	This study	5'-gagcctcgaaggcattgt-3'	
Sequence-based reagent	ODF2-KO sequencing primer R	This study	5'-ctggcccagagatgtcataaag-3'	
Sequence-based reagent	CEP120-KO sequencing primer F	This study	5'-tcctgtgcatgttgatgttga-3'	
Sequence-based reagent	CEP120-KO sequencing primer R	This study	5'-ggaactgagaacagcagtcac-3'	
Sequence-based reagent	PCNT-KO sequencing primer F	This study	5'-gttgctgctggtgccgtga-3'	
Sequence-based reagent	PCNT-KO sequencing primer R	This study	5'-ggcagggactctgtgtgaatc-3'	
Sequence-based reagent	Rab11a-KO EX1 sequencing primer F	This study	5'-ggccggaagtgcgtagag-3'	
Sequence-based reagent	Rab11a-KO EX1 sequencing primer R	This study	5'-cgacccgctgaggaaagt-3'	

Key reagent or resource table. (continued)

Sequence-based reagent	Rab11a-KO EX2 sequencing primer F	This study	5'-ATTGGAGATTCTGGTGTTGGAA-3'	
Sequence-based reagent	Rab11a-KO EX2 sequencing primer R	This study	5'-caatggtaagatggaagggaat-3'	
Sequence-based reagent	shRNA against Rab11a #1	Bryant et al., 2010	5'-AAGGCACAGATATGGGACACG-3'	
Sequence-based reagent	shRNA against Rab11a #2	Bryant et al., 2010	5'-ATGGTTTGTTCATTATTGA-3'	
Sequence-based reagent	shRNA against Sec15a #1	Bryant et al., 2010	5'-AAGAGGATGAGAATGAAGAGG-3'	
Sequence-based reagent	shRNA against Sec15a #2	Bryant et al., 2010	5'-AAGCACGGGTCATGATAGTTT-3'	
Sequence-based reagent	shRNA against MyoVb #1	Roland et al., 2011	5'-AAGGTGGAGTATCTTTCA-3'	
Sequence-based reagent	shRNA against MyoVb #2	Roland et al., 2011	5'-AACGGGTAAGTGTGGCCTTTA-3'	
Sequence-based reagent	shRNA against Par3 #1	Horikoshi et al., 2009	5'-GACAGACTGGTAGCAGTGT-3'	
Sequence-based reagent	shRNA against Par3 #2	Horikoshi et al., 2009	5'-AGGATAAAGCTGGCAAG-3'	
Sequence-based reagent	shRNA against Cdc42 #1	Bryant et al., 2010	5'-GCGATGGTGCCGTTGGTA-3'	
Sequence-based reagent	shRNA against Cdc42 #2	Bryant et al., 2010	5'-GATTACGACCGCTGAGTTA-3'	

Key reagent or resource table. (continued)

Sequence-based reagent	shRNA against Crumbs3	Whiteman et al., 2008	5'-GCTTAAGAGTAGAAGGG AA-3'	
Sequence-based reagent	Sec15a qPCR primer F	Bryant et al., 2010	5'-GTCAGCCTGCCAGCATCT GT-3'	
Sequence-based reagent	Sec15a qPCR primer R	Bryant et al., 2010	5'-CTGCTGAACAGCTCCCAT GC-3'	
Sequence-based reagent	MyoVb qPCR primer F	Roland et al., 2011	5'-CAAGTCCCACTGGTGGC TG-3'	
Sequence-based reagent	MyoVb qPCR primer R	Roland et al., 2011	5'-GAAGACGAGCCCTCCCA GA-3'	
Sequence-based reagent	Crumbs3 qPCR primer F	Whiteman et al., 2008	5'-GAGAACGGCACCATTACA CC-3'	
Sequence-based reagent	Crumbs3 qPCR primer R	Whiteman et al., 2008	5'-GAGGGAGAAGACCACAA TGAT-3'	
Sequence-based reagent	GAPDH qPCR primer F	Bryant et al., 2010	5'-AGTCAAGGCTGAGAACG GGAAACT-3'	
Sequence-based reagent	GAPDH qPCR primer R	Bryant et al., 2010	5'-CATGGTTCACGCCCATCA CAAACA-3'	
Chemical compound	Matrigel, growth factor reduced, phenol red-free	Corning	356231	
Chemical compound	4',6-diamidino-2-phenylindole (DAPI)	Sigma-Aldrich	D8417	1:10000
Chemical compound	SiR-DNA	Spirochrome	SC007	500 nM
Chemical compound	Centrinone	MedChemExpress	LCR-263, HY-18682	300 nM

Key reagent or resource table. (continued)

Chemical compound	Aphidicolin	Sigma-Aldrich	A0781	2.5 µg/ml
Chemical compound	Nocodazole	Sigma-Aldrich	M1404	100 ng/ml
Chemical compound	Phalloidin-iFluor 647	Abcam	ab176759	1:1000
Chemical compound	SiR-tubulin	Spirochrome	SC002	750 nM
Software, algorithm	Imaris	Bitplane		version 9.8.0

Key reagent or resource table. (continued)

1:2000, human residues 639–986, described in our previous paper) (Lin et al., 2013 [DOI](#)), p53 (IF: 1:100, 9282, Cell Signaling Technology, Danvers, MA), p21 (WB: 1:2000, 10355-1-AP, Proteintech), CEP164 (IF: 1:400, NBP1-81445, NOVUS, Centennial, CO), PCNT (IF: 1:200, ab4448, Abcam, Cambridge, UK), Par3 (IF: 1:200, WB: 1:1000, 07-330, EMD Millipore, Burlington, MA), and beta-actin (IF: 1:200, A2066, Sigma-Aldrich, St. Louis, MO); rabbit monoclonal antibodies against E-cadherin (IF: 1:200, 24E10, #3195, Cell Signaling Technology); and mouse monoclonal antibodies against γ -tubulin (IF: 1:400, GTU-88, T6557, Sigma-Aldrich), acetylated tubulin (IF: 1:400, T6793, Sigma-Aldrich), Gp135 (IF: 1:10, the culture medium of hybridoma clone 3F2D8, a kind gift from Dr. Jou Tzuu-Shuh, NTU, Taiwan) (Lim et al., 2017 [DOI](#)), polyglutamylated tubulin (IF: 1:100, GT335, AG-20B-0020, Adipogen, San Diego, CA), ODF2 (IF: 1:400, H00004957-M01, NOVUS), α -tubulin (WB: 1:5000, DM1 α , T9026, Sigma-Aldrich), CDC42 (WB: 1:1000, 610928, BD Biosciences), and ZO-1 (IF: 1:400, 33-9100, ThermoFisher Scientific, Waltham, MA).

For western blots, we used the following secondary antibodies: HRP-labeled goat anti-rabbit and mouse IgG (1:10000, NEF812 and NEF822, PerkinElmer, Waltham, MA). For immunofluorescence, we used Alexa Fluor 488, 561, and 647 conjugated goat antibodies against rabbit, mouse IgG1, and mouse IgG2b (1:400, all purchased from ThermoFisher Scientific) as secondary antibodies, together with Phalloidin-iFluor 647 reagent (1:1000, ab176759, Abcam) and 4',6-diamidino-2-phenylindole (DAPI) (1:10000, D8417, Sigma-Aldrich).

3D immunofluorescence staining, image acquisition, and analysis

We stained 3D cell cultures according to a previously published protocol, with some modifications (Lee et al., 2007 [DOI](#)). We pre-fixed whole cultures in each well with 4% paraformaldehyde at room temperature for 10 min. If the staining included centrosome markers, we post-fixed the samples with pure cold methanol at -20°C for another 10 min. After fixation, cells were permeabilized with 0.25% Triton X-100 in phosphate-buffered saline (PBS). We performed primary antibody staining in 100 μL PBS plus 10% normal goat serum at 4°C for 48 h. After four 15 min washes in 250 μL PBS with 0.1% Tween 20 at room temperature, we added secondary antibodies with DAPI in 100 μL PBS plus 10% normal goat serum for another 48 hours at 4°C . Finally, we performed another four 15 min washes, followed by adding 100 μL mounting medium (VectaShield, Vector Laboratories, Newark, CA) to each well. We analyzed the samples on the LSM 880 confocal (Zeiss, Jena, Germany) using a Plan-Apochromat 63x oil immersion objective with a numerical aperture (NA) 1.4.

For fluorescence intensity measurements, confocal images were acquired with constant parameters (pinhole size, laser power, master gain, and z-stack interval) and introduced to Fiji. We applied the maximum intensity projection to project the z-stack. The oval selection tool was fixed on size to select the region of interest (ROI) (e.g., pericentrosomal region, cytokinetic bridge, cytosol, AMIS, or apical lumen) and to measure the mean fluorescence intensity of the ROI.

Quantification of centrosome position

We considered the polarity index, θ_{N-C} , L_{O-C} , and L_{IC} of cell doublets cultured in 3D conditions to assess centrosome positions. We obtained the centroid of each cell nucleus and centrosome, as well as the volume of the cell nucleus, using Imaris software (version 9.8.0, Bitplane, Belfast, UK). The cell nucleus radius, r_N , was determined by dividing its volume by $4/3\pi$ and then taking the cube root. We calculated the polarity index from the nucleus–centrosome distance, L_{N-C} , divided by the r_N .

The nucleus–centrosome orientation (θ_{N-C}) is represented by the angle between the line from the nucleus to the centrosome and the line from the nucleus to the other nucleus. The central points of each nucleus and centrosome were marked as three points (nuclei: n1 and n2; centrosome: c) in 3D

$$\cos\theta = \frac{\overrightarrow{n_1n_2} \cdot \overrightarrow{n_1c}}{|\overrightarrow{n_1n_2}| |\overrightarrow{n_1c}|} \quad \text{Angle } \theta = \arccos(\cos\theta) * \frac{180}{\pi}$$

space to give three x , y , z coordinates. We computed the angle between the vector of the nucleus to the centrosome and the vector of the nucleus to the other nucleus using the vector dot product formula. Centrosomes oriented within 90° from the nucleus–nucleus axis were considered to be correctly oriented toward the apical region.

The distance between the centrosome and apical membrane was represented by the distance from the origin point (the center point of the two nuclei) to the centrosome, L_{O-C} , because the AMIS and apical lumen will form at the center of the cell doublet in Matrigel culture. We also calculated the distance using x , y , z coordinates in 3D space. A smaller distance indicates that the centrosome and apical membrane are closer. The distance between the centrosomes in each cell of a cell doublet is referred to as the inter-centrosomal distance, L_{IC} .

Lentiviral transductions and stable cell lines

We generated MDCK cells expressing EGFP-rabbit podocalyxin (Gp135, complementary DNA [cDNA] was a kind gift from K. Simons), EGFP-Crb3 (cDNA was obtained by reverse transcription polymerase chain reaction [RT-PCR] following previously described procedures) (Schluter et al., 2009), EGFP-Cdc42, PACT-mKO1, mCherry–C-CTD, and mCherry–N-gTBD via lentiviral transduction using the following protocol: MDCK cells were transduced with lentiviruses that package with the pLAS2w.Ppuro vector (RNAi Core, Taiwan) containing the relevant cDNA. The transduction processes were performed in complete MEM with $8 \mu\text{g/mL}$ polybrene. The transduced cells were sterile-sorted by a cell sorter (FACSJazz, BD Biosciences) for EGFP, mKO1, or mCherry. Fluorescence-positive cells were selected and expanded.

3D live cell imaging in Matrigel and analysis

We resuspended stable MDCK lines in a culture medium containing 2% Matrigel and plated the cells in a rose chamber with the cover-glass bottom pre-coated by 100% Matrigel. We allowed the cells to remain in the incubator for 2 hours to ensure that the cells fell and were fully adherent on the Matrigel. We labeled the nucleus in live cells with a far-red fluorescence probe, SiR-DNA (500 nM, SC007 Spirochrome, Cytoskeleton Inc., Denver, CO), in phenol red-free medium containing 2% Matrigel.

After a 1 hour incubation, we performed live cell imaging on a 40x NA1.1 water immersion objective (Leica, Wetzlar, Germany) on an inverted microscope (DMI 6000, Leica) with a confocal scan head (CSU22 Spinning Disk, Yokogawa Electric, Tokyo, Japan), a laser merge module containing 405-, 491-, 561-, and 642-nm laser lines (ILE-400, Andor, Belfast, UK), a DV885 EMCCD (Andor), an MS-2000-XYZ motorized stage with a piezo top plate (ASI, Eugene, OR), and a stage-top incubator set to 37°C under 5% CO_2 . The microscope apparatus was controlled by MetaMorph 7.7.2 (Molecular Devices, San Jose, CA). We set the time interval to 5 min and z steps to $0.5 \mu\text{m}$.

We collected images of mitotic cells to track centrosome migration during *de novo* epithelial polarization. These images were imported to Imaris software (version 9.8.0, Bitplane) for 3D reconstruction and analysis. We marked the centroid of each nucleus and centrosome as x , y , z coordinates in 3D space for each time point. We calculated the polarity index, θ_{N-C} , and L_{O-C} from the first frame of anaphase onset.

To measure the fluorescence intensity of EGFP-Gp135 on the middle membrane of the 3D cell doublet, we measured the intensity profile of EGFP-Gp135 along the line connecting the two nucleus spots. Then, we recorded the average value of the five center pixels representing the region of the middle membrane from the first frame of anaphase onset.

CRISPR/Cas9 knockouts

To generate MDCK lines lacking either p53, CEP164, ODF2, CEP120, PCNT, or Rab11a, we used CRISPR/Cas9 technology. Because CEP120-null cells cannot survive in the presence of WT p53 (Tsai et al., 2019), we performed a knock-out of CEP120 in MDCK p53-null cells. We transfected MDCK

cells using a Lipofectamine 2000 transfection reagent (11668019, ThermoFisher Scientific) with the pSpCas9(BB)-2A-GFP (PX458) vector (Ran et al., 2013) (648138, Addgene, Watertown, MA) bearing the appropriate targeting sequence:

p53 #1 (+): 5'-TCCCAGAGAGCGTCGTGAAC-3'; *p53* #2 (-): 5'-ACTTGCTGTCCCGGAATGC-3';
CEP164 #1 (-): 5'-GTCCCACATGGACTGCCCGT-3'; *CEP164* #2 (+): 5'-
 AACTTGGTGATTCAAGAGCG-3'; *ODF2* #1 (+): 5'-CCTCATGTCCAAGCTGGTAG-3'; *ODF2* #2 (+): 5'-
 ATCGGGAAGCTGAAGACGGT-3'; *CEP120* #1 (+): 5'-ACAGTTGGCTACCGACCTG-3'; *PCNT* #1 (+):
 5'-TCGCCAGGGAGCAGCATGCG-3'; *Rab11a* #1 (-): 5'-GCGACGACGAGTACGACTAT-3'; *Rab11a*
 #2 (-): 5'-AGCTATAACATCAGCGTAAG-3'

Cells were sterile-sorted by a cell sorter (FACSJazz, BD Biosciences) for GFP. Fluorescence-positive cells were serially diluted to pick single colonies and were expanded to pure clones. Genomic DNAs isolated from each clone were subjected to PCR and sequencing for verification. We examined protein expression levels in each clone via immunofluorescence microscopy or western blotting.

RNA interference

We achieved RNA interference (RNAi) using pLKO.1-EGFP (TRC011, RNAi Core, Taiwan) lentiviral vector. RNAi target sequences were based on previously published reports (Bryant et al., 2010; Horikoshi et al., 2009; Roland et al., 2011; Schluter et al., 2009) for canines, as follows:

Rab11a #1: 5'-AAGGCACAGATATGGGACACG-3'; *Rab11a* #2: 5'-ATGGTTTGTTCATTGAGA-3';
Sec15a #1: 5'-AAGAGGATGAGAATGAAGAGG-3'; *Sec15a* #2: 5'-AAGCACGGGTCATGATAGTTT-3';
Par3 #1: 5'-GACAGACTGGTAGCAGTGT-3'; *Par3* #2: 5'-AGGATAAGCTGGCAAAGA-3'; *MyoVb*
 #1: 5'-AAGGTGGAGTATCTTTCAGAT-3'; *MyoVb* #2: 5'-AACGGGTAAGTGGCCTTTA-3'; *Cdc42*
 #1: 5'-GCGATGGTGCCGTTGGTAA-3'; *Cdc42* #2: 5'-GATTACGACCGCTGAGTTA-3'; *Crb3*: 5'-
 GCTTAAGAGTAGAAGGGAA-3'

We verified knockdown by western blot or quantitative real-time PCR procedures normalized to GAPDH expression. The designed primer sequences were as follows:

Sec15a Fwd: 5'-GTCAGCTGCCAGCATCTGT-3'; *Sec15a* Rev: 5'-CTGCTGAACAGCTCCCATGC-3';
MyoVb Fwd: 5'-CAAGTCCCACTGGTGGCTG-3'; *MyoVb* Rev: 5'-GAAGACGAGCCCTTCCAGA-3';
Crb3 Fwd: 5'-GAGAACGGCACCATTACAC-3'; *Crb3* Rev: 5'-GAGGGAGAAGACCACAATGAT-
 3'; *GAPDH* Fwd: 5'-AGTCAAGGCTGAGAACGGGAACT-3'; *GAPDH* Rev: 5'-
 TGTTTGTGATGGGCGTGAACCATG-3'

3D live cell imaging in ECM-free conditions

Following a previously published protocol (Huang et al., 2020), we generated multiarray spherical microwells for an ECM-free suspension culture. In brief, 100 μ L polyacrylamide (PA) solution containing 7% acrylamide monomer (Bio-Rad, Hercules, CA), 0.35% bisacrylamide crosslinker (Bio-Rad), 0.15% N, N, N', N'-tetramethylethylenediamine catalyst (Sigma-Aldrich), and 0.5% ammonium persulfate (Sigma-Aldrich) was sandwiched between an epoxy mold and 3-(trimethoxysilyl) propyl methacrylate silanized cover glass (22 $\times$ 22 mm). Waiting for the complete gelation of PA, we carefully separated the cover-glass-bound microwell substrate from the epoxy template. To avoid cell adhesion, we conjugated poly-L-lysine/polyethylene glycol to the PA substrate using the bi-functional crosslinker sulfo-SANPAH (ThermoFisher Scientific). We sterilized low-attachment microwell substrates with antibiotics and then seeded a single-cell suspension of MDCK stable lines onto the substrate surface. After cells fell into the microwells, we rinsed the PA substrate with the culture medium to remove excess cells outside the microwells. The samples were assembled into a rose chamber and imaged by a CSU22 spinning disk confocal microscope, as described above, for 3D live cell imaging in Matrigel.

Transwell-cultured 2D epithelial sheet

To grow polarized epithelial cell monolayers suitable for observation under an inverted microscope, we cultured MDCK cells on the outer surface of Transwell inserts (3470, Corning). Typically, 50 μL of a suspension containing 4.25×10^5 cells/mL was seeded into each 6.5-mm Transwell insert. Cells were cultured for 3 days before immunofluorescence staining or live cell imaging. We mounted Transwell inserts in a custom-made glass-bottom chamber for confocal imaging. For live cell imaging, we stained epithelial cell monolayers with SiR-tubulin (500 nM, SC002, Spirochrome) and monitored the cells using LSM 980 confocal microscopy (Zeiss).

Statistical analysis

Statistical analyses utilized GraphPad Prism 9 and OriginPro 2021, employing the Mann-Whitney U test and one-way and two-way analysis of variance (ANOVA) with multiple comparisons. Details on statistical tests and sample sizes are provided in the figure legends. n denotes the sample size, and the boxplot boxes indicate the standard deviation (SD).

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

Supplemental Figures

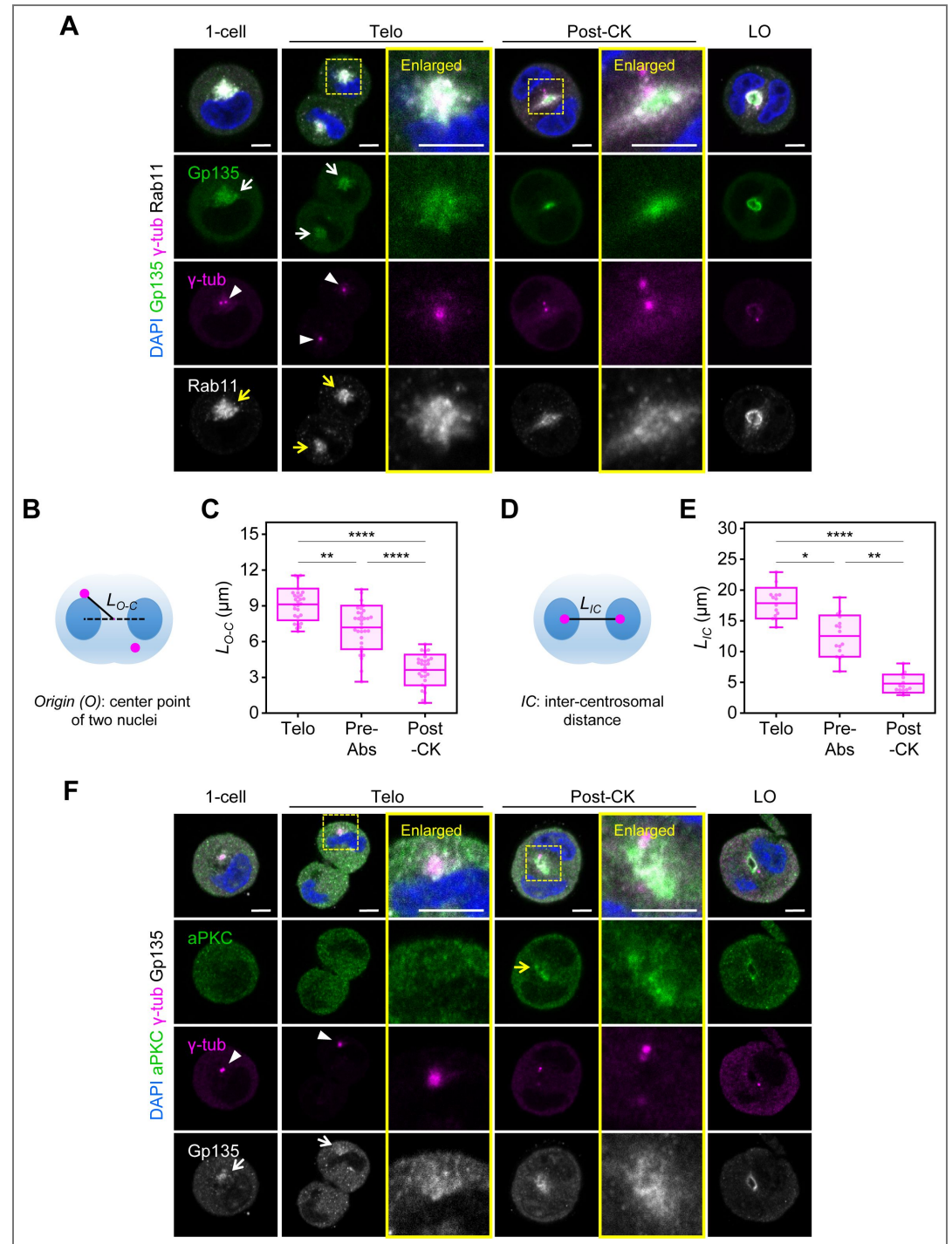


Figure 1—figure supplement 1. Distribution of Rab11a, Gp135, and aPKC proteins and centrosome position during *de novo* epithelial polarization (A, F) Single MDCK cells were cultured in Matrigel for 12 h with labeled markers: Gp135 (green), γ -tubulin (magenta), and Rab11a (white) in (A), and aPKC (green), γ -tubulin (magenta), and Gp135 (white) in (F). DAPI staining was applied for nuclei (blue). Single confocal sections are shown through the middle of the cells. The centrosome region during telophase (Telo) and the AMIS region in post-cytokinesis (Post-CK) cells, as shown in the yellow box, are enlarged. Yellow arrows indicate Rab11a and aPKC around centrosomes or at the AMIS. White arrows indicate Gp135. Arrowheads point to the centrosome position. LO: lumen open. Scale bar: 5 μm . (B, D) Illustration showing L_{O-C} (origin to centrosome distance) and L_{IC} (inter-centrosomal distance) for cell doublets. The center point between the two nuclei is taken as the origin (O). (C, E)

Boxplots depicting centrosome positions at different stages, shown by L_{O-C} (C) and L_{IC} (E). We analyzed >15 cells per measurement in three independent experiments. Statistical analyses used one-way ANOVA and Dunn's multiple comparisons (* $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values.

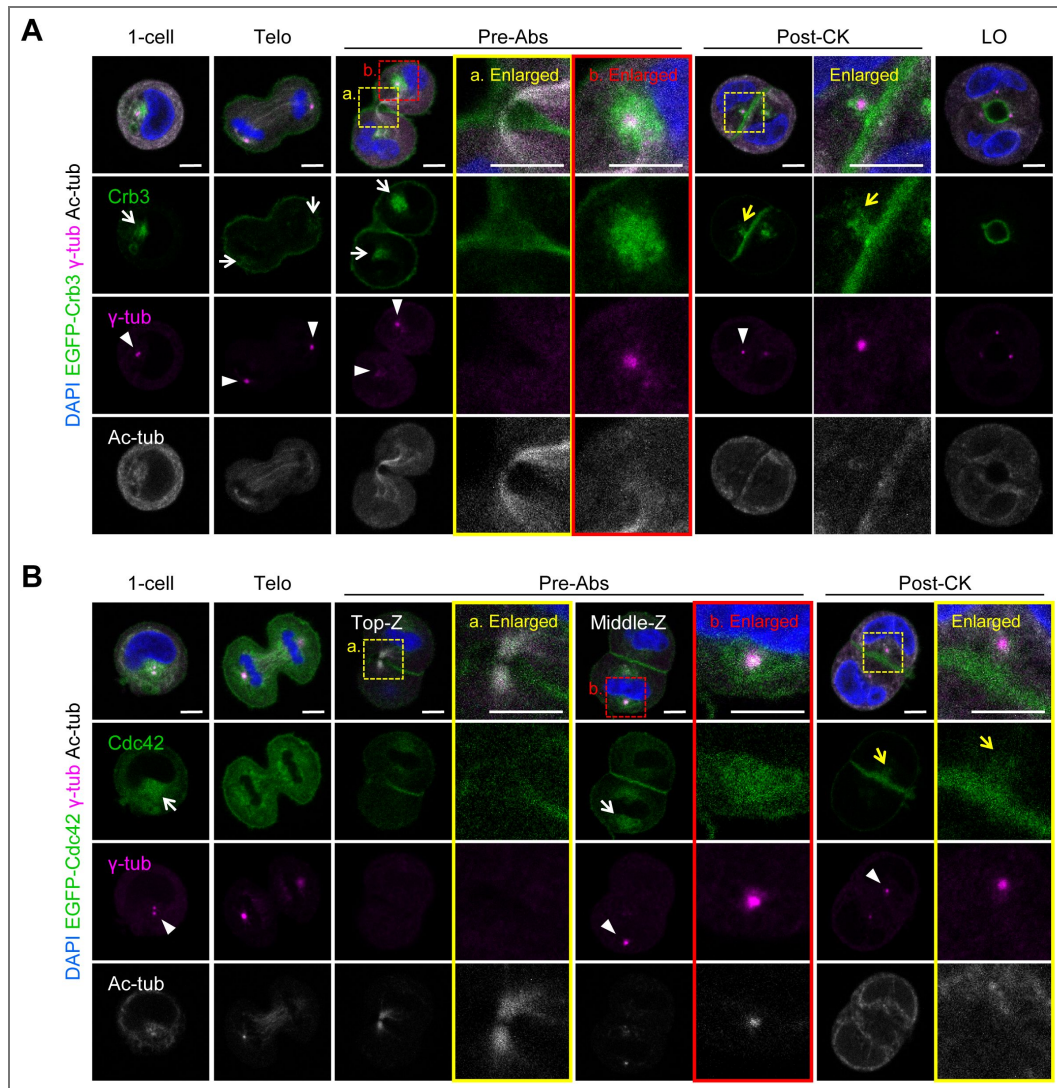


Figure 1—figure supplement 2. Clustering of Crb3 and Cdc42 around centrosomes during *de novo* epithelial polarization

(A–B) Single MDCK cells stably expressing EGFP-Crb3 (A, green) or EGFP-Cdc42 (B, green) were cultured in Matrigel for 12 h with labeled markers: γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). Single confocal sections through the middle of cells are shown. The centrosome, cytokinesis (CK) bridge, and AMIS region are enlarged from the boxed area in pre-abscission (Pre-Abs) and post-CK cells. White arrows indicate EGFP-Crb3 (A) or EGFP-Cdc42 (B) around centrosomes. Arrowheads point to the centrosome position. Yellow arrows indicate the transport of Crb3 and Cdc42 from the centrosome to the mid-membrane. Scale bar: 5 μ m.

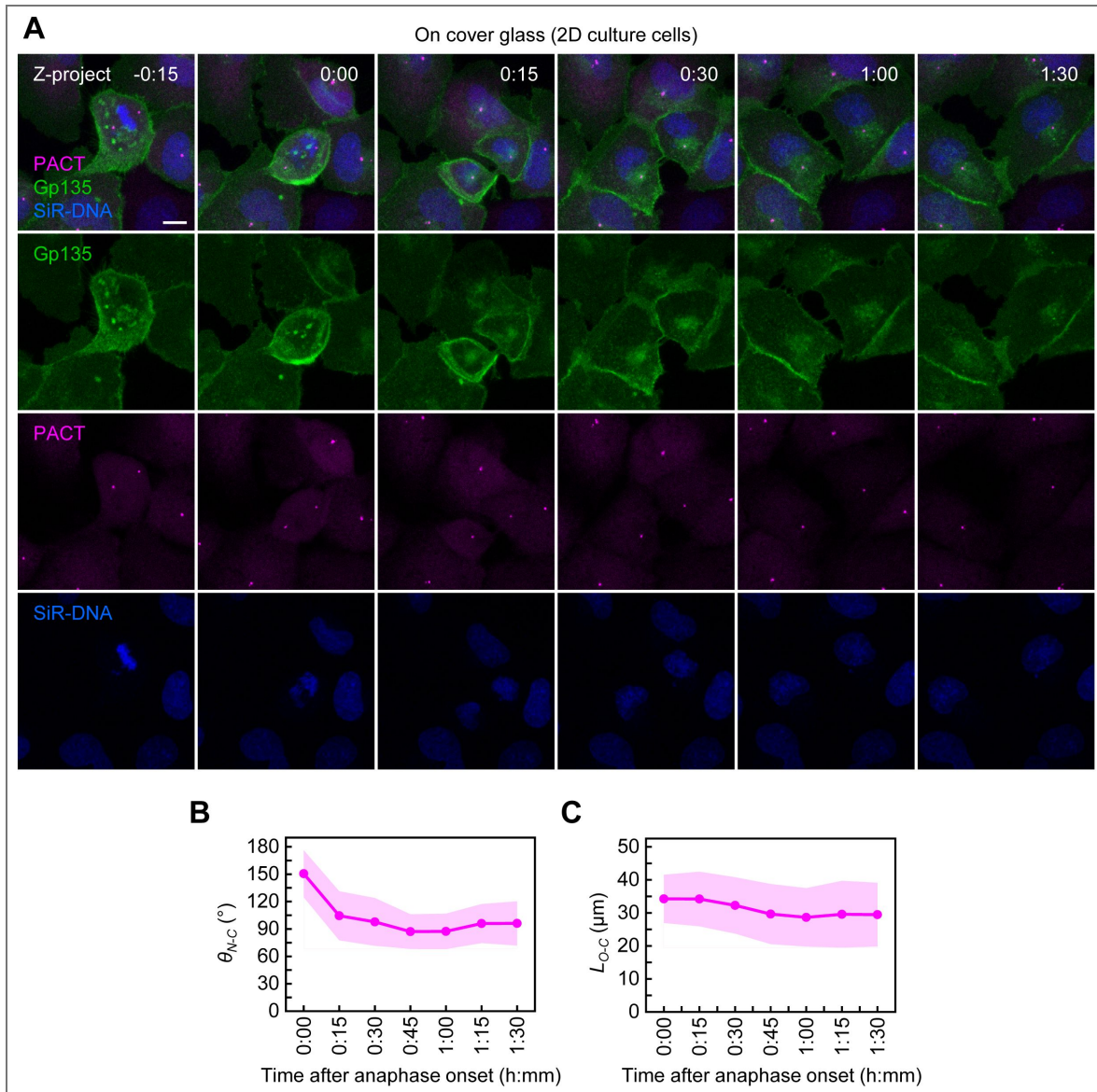


Figure 2—figure supplement 1. Differential centrosome migration of MDCK cells in adherent 2D culture versus 3D culture conditions during cytokinetic pre-abscission

(A) Time-lapse snapshots of MDCK cells expressing EGFP-Gp135 (green) and PACT-mKO1 (magenta, centrosome marker) on a cover glass. Nuclei were labeled with SiR-DNA (blue) before live imaging. Z-projection images of a dividing cell are shown. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m. (B–C) Change in θ_{NC} and L_{OC} over time. Each data point represents the average at a given time (6 cells on cover glass from three independent experiments). The lines show the means, and the shaded regions indicate SD values.

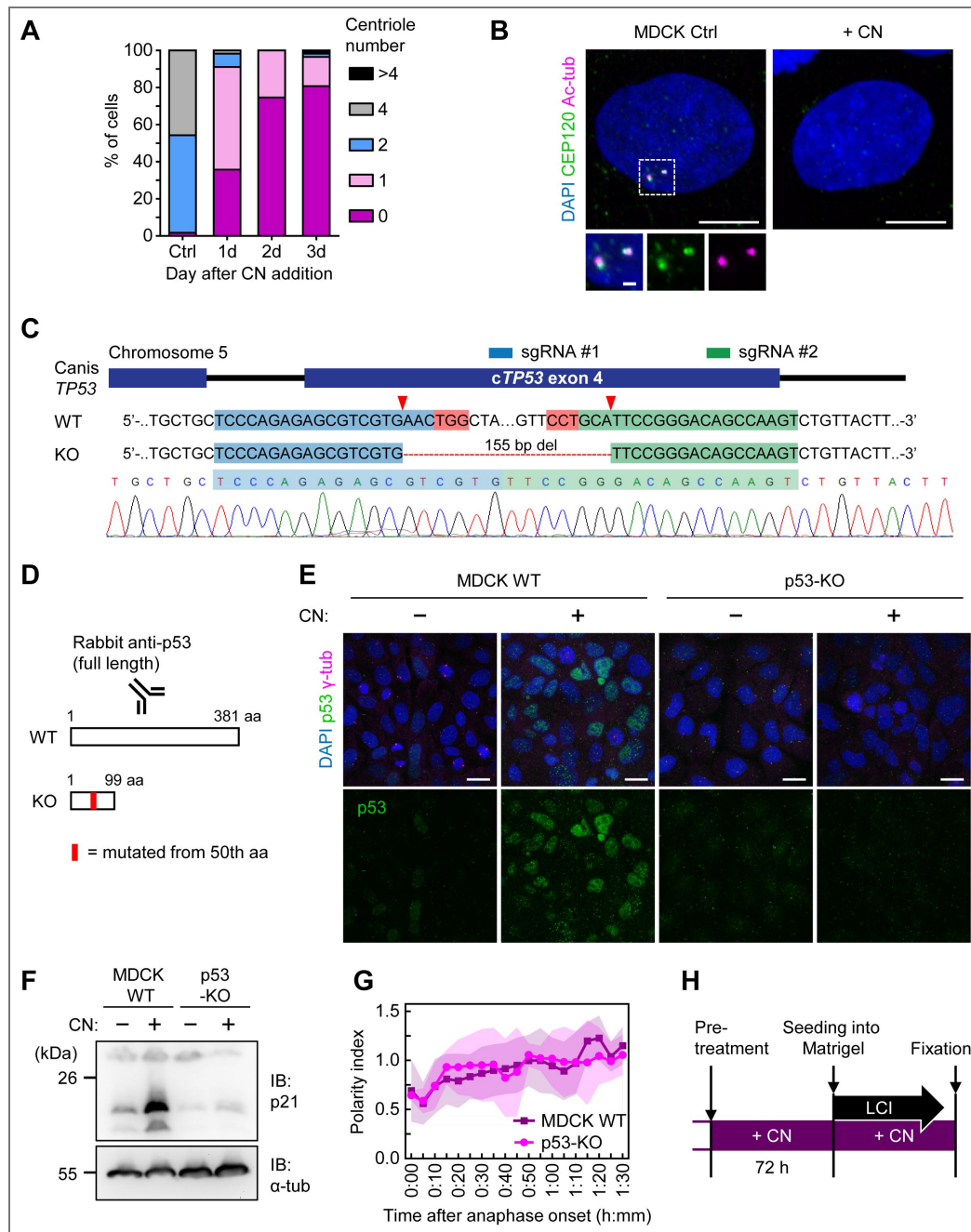


Figure 3—figure supplement 1. Assessment of centrinone (CN) treatment and generation of p53-KO MDCK cells

(A) Bar graph showing progressive centrosome depletion after CN treatment in MDCK cells. The distribution of centrosome numbers over time after the addition of CN is shown. (B) MDCK cells are present three days after CN addition and an untreated control. Scale bar: 10 μm . Insets show enlargements of the image in the boxed area. Scale bar: 1 μm . (C) Generation of the p53-KO MDCK cell line. The diagram illustrates two single-guide RNA (sgRNA) sequences and their target sites in p53 exon 4 (blue and green boxes). The red boxes indicate the position of the protospacer-adjacent motif (PAM) sites and the predicted Cas9 cut sites are indicated by red arrowheads. Genotyping results show a genomic mutation in the p53-KO cell line with a deletion (del) within exon 4 of the p53-encoding gene. (D) Binding of p53 antibody and mutation induced in the p53-encoding gene. (E) Immunofluorescence images showing p53 expression in MDCK WT and p53-KO cells with or without CN treatment. Scale bar: 10 μm . (F) Immunoblot results for p21 protein in MDCK WT and p53-KO cells with or without CN treatment. (G) Change in polarity index over time. Each data point represents the average at a given time (four p53-KO cells in Matrigel culture from three independent experiments). The lines show the means, and the shaded regions indicate SD values. (H) Protocol for CN pre-treatment and the time point of Matrigel culture in the presence of CN.

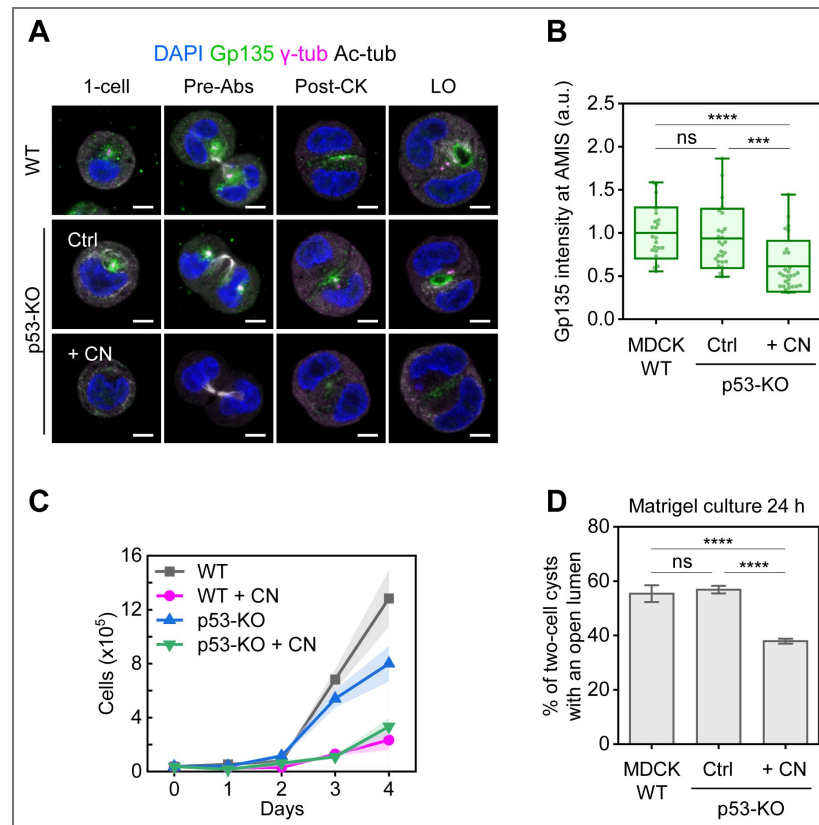


Figure 3—figure supplement 2. Impact of centriole depletion on MDCK cell proliferation and lumen formation

(A) Single MDCK WT and p53-KO cells, with or without centrinone (CN) treatment, after 8–12 h of Matrigel culture. Single confocal sections through the middle of cysts are shown with immunofluorescent signals of indicated markers: Gp135 (green), γ -tubulin (magenta), acetyl-tubulin (white), and DAPI for nuclei (blue). Acentrosomal cells fail to cluster Gp135-positive vesicles and affect lumen formation in the two-cell stage (Pre-Abs, pre-abscission; LO, lumen open). Scale bar: 5 μ m. (B) Boxplot of Gp135 intensity at the AMIS normalized to the mean in WT cells. $n = 22$ (MDCK WT), 26 (p53-KO), 28 (p53-KO + CN) post-CK cells were analyzed in three independent experiments. Statistical analyses were performed via one-way ANOVA and Dunn's multiple comparisons (ns: not significant, *** $p < 0.001$, **** $p < 0.0001$). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units). (C) Proliferation curves of MDCK WT and p53-KO cells, with or without CN treatment. The cells with CN treatment underwent a 3-day pre-treatment before examination. Values are presented as the mean $\pm$ SD. (D) Proportion of lumen opening in MDCK cell doublets after 24 h of Matrigel culture based on Gp135 staining. $n = 17$ (MDCK WT), 30 (p53-KO), and 45 (p53-KO + CN) cell doublets were analyzed in three independent experiments. Statistical analyses used two-way ANOVA and Tukey multiple comparisons (ns: not significant, **** $p < 0.0001$). Values represent the mean $\pm$ SD.

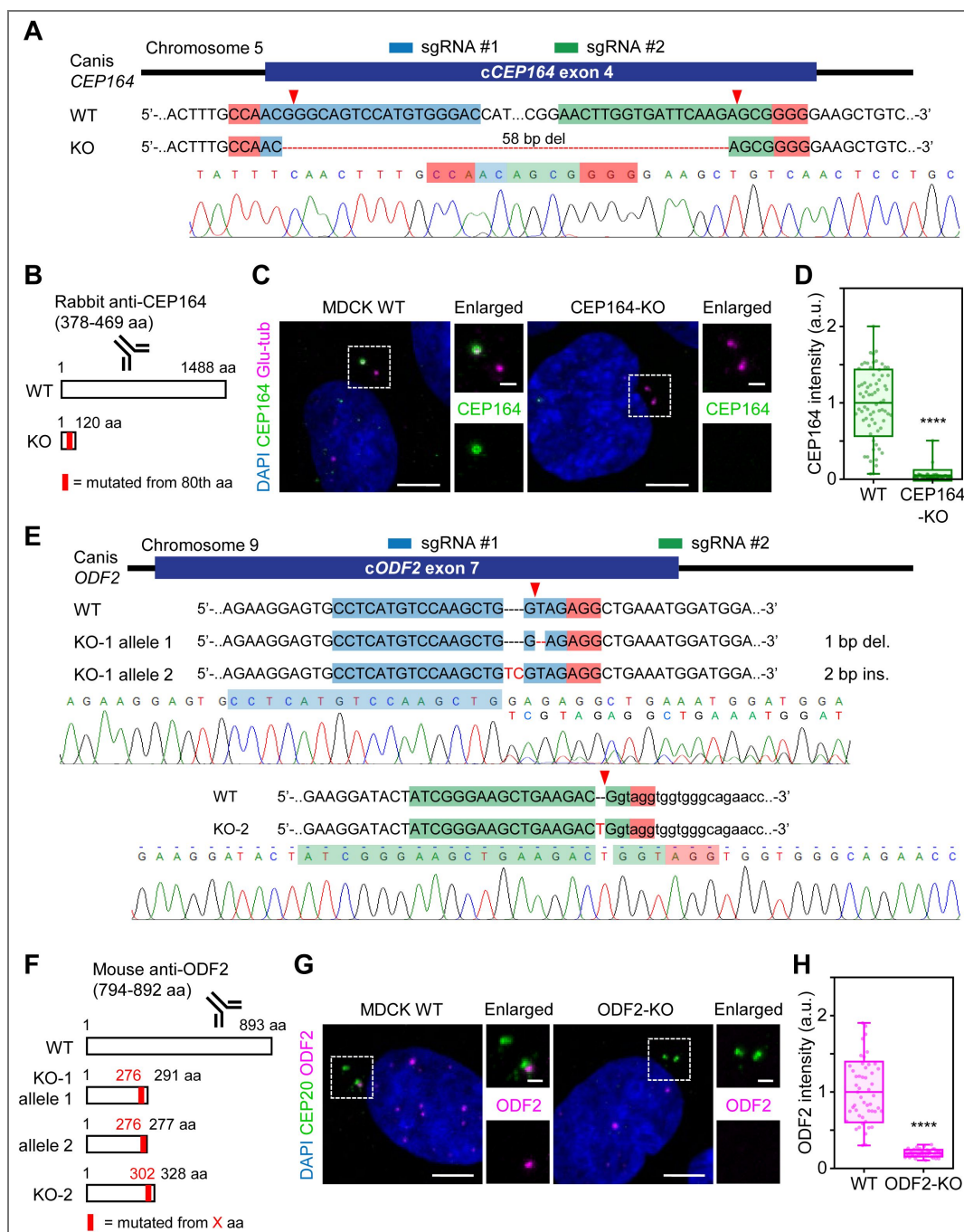


Figure 4—figure supplement 1. Generation of CEP164 and ODF2-KO MDCK cell lines

(A, E) Diagrams illustrating two single-guide RNA (sgRNA) sequences and their targeting sites within *CEP164* exon 4 and *ODF2* exon 7 (blue and green boxes). The red boxes indicate the position of the protospacer-adjacent motif (PAM) sites and the predicted Cas9 cut sites are indicated by red arrowheads. Genotyping results of the genomic mutations are shown. The predicted genomic mutations (del: deletion, ins: insertion) are illustrated. (B, F) Diagrams illustrating the binding sites of the antibody and the position of the mutation induced in the CEP164- and ODF2-encoding genes. (C, G) Immunofluorescence images depicting CEP164 and ODF2 expression in MDCK WT control and KO cells with glutamylated tubulin and CEP120, respectively, as the centrosome marker. Scale bar: 5 μ m. Insets present enlargements of the images within the boxed areas. Scale bar for insets: 1 μ m. (D, H) Boxplots of the CEP164 and ODF2 signal at the mother centriole. Fluorescent intensity values of KO cells were normalized to the mean of WT cells. $n = 70$ (MDCK WT), 51 (CEP164-KO) in (D), and $n = 50$ (MDCK WT), 50 (ODF2-KO) in (H) were analyzed in three independent experiments. Statistical significance was determined by an unpaired two-tailed Mann-Whitney U test (**** $p < 0.0001$). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units).

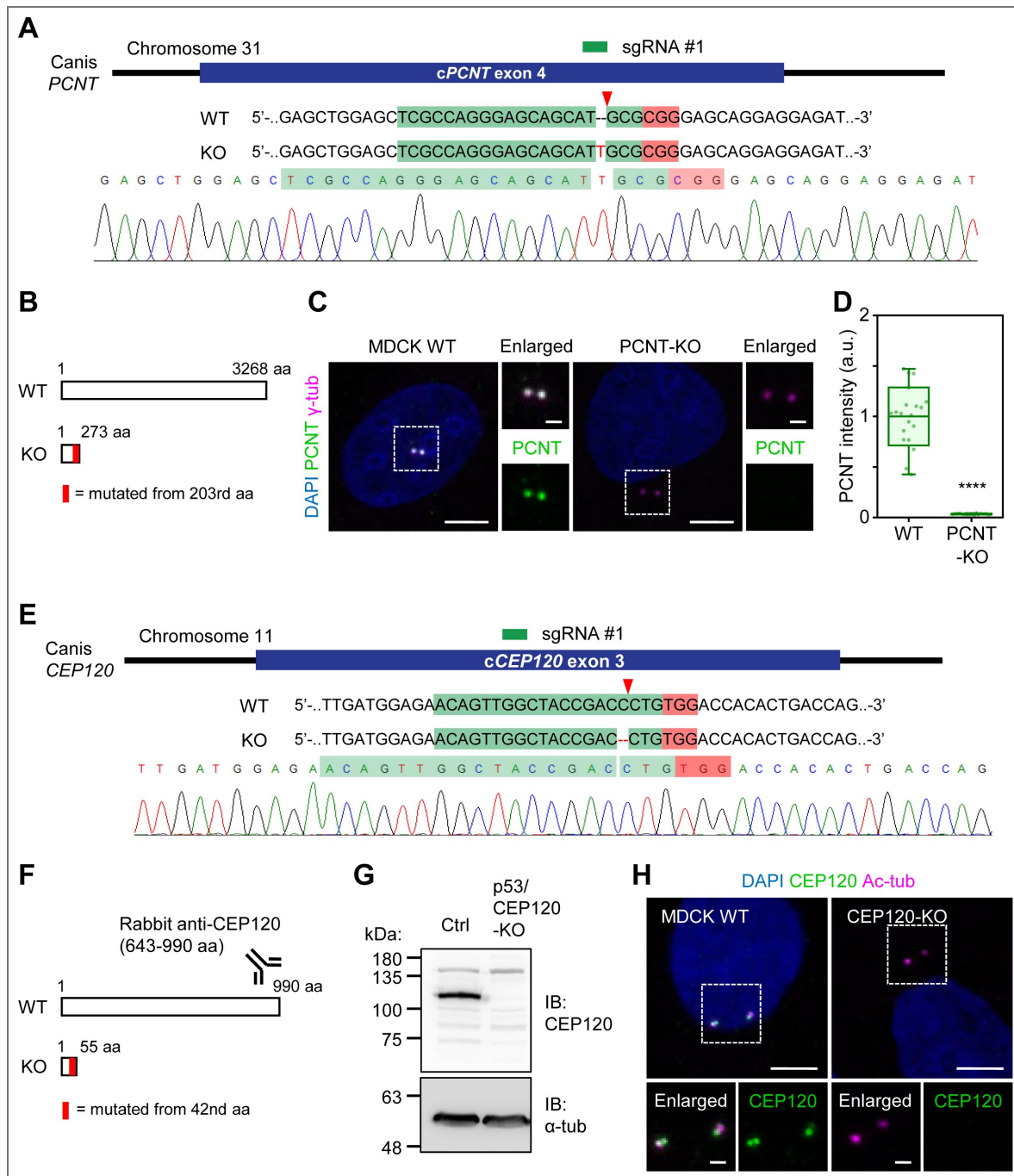


Figure 4—figure supplement 2. Generation of p53/CEP120 and PCNT-KO MDCK cell lines

(A, E) Diagrams illustrating two single-guide RNA (sgRNA) sequences and their targeting sites within *PCNT* exon 4 and *CEP120* exon 3 (blue and green boxes). The red boxes indicate the position of the protospacer-adjacent motif (PAM) sites, and the predicted Cas9 cut sites are indicated by red arrowheads. Genotyping results of the genomic mutations are shown. The predicted genomic mutations (del: deletion, ins: insertion) are illustrated. (B, F) Diagrams illustrating the binding sites of the antibody and the position of the mutation induced in the *PCNT*- and *CEP120*-encoding genes. (C, H) Immunofluorescence images depicting *PCNT* and *CEP120* expression in MDCK WT control and KO cells with acetyl-tubulin and γ -tubulin, respectively, as the centrosome marker. Scale bar: 5 μ m. Insets present enlargements of the images within the boxed areas. Scale bar for insets: 1 μ m. (D) Boxplots of the *PCNT* signal at the centrosome. Fluorescent intensity values of KO cells were normalized to the mean of WT cells. n = 20 (MDCK WT), 20 (PCNT-KO) were analyzed in three independent experiments. Statistical significance was determined by an unpaired two-tailed Mann-Whitney U test (****p<0.0001). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units). (G) Western blot confirming the loss of *CEP120* in p53/*CEP120*-KO cells.

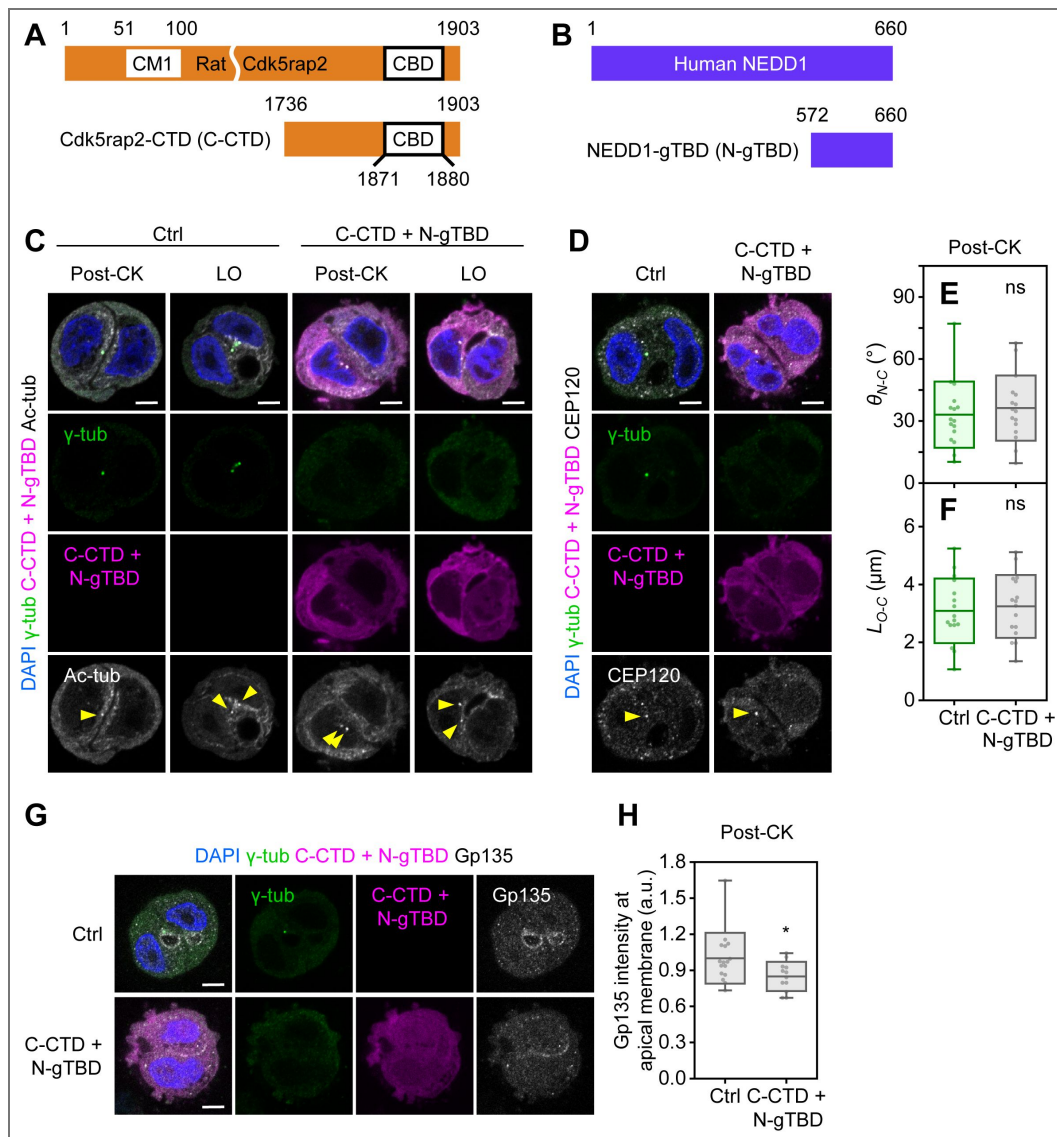


Figure 4—figure supplement 3. Disruption of centrosomal microtubules impairs epithelial polarization without affecting centrosome positioning in MDCK cells

(A) Schematic representation of rat Cdk5rap2 protein and its truncation mutant. CM1 is the γ -TuRC-binding and activating domain. CBD is the centrosome-binding domain. Numbers indicate amino acid (AA) positions. (B) Schematic representation of human NEDD1 protein and its truncation mutant. N-gTBD is the γ -tubulin-binding domain of NEDD1. (C, D, G) Single MDCK cells not expressing (control) or co-expressing mCherry-C-CTD and mCherry-N-gTBD were cultured in Matrigel for 24 h. Images show cells at the post-cytokinesis (CK) and lumen open (LO) stages, with markers including γ -tubulin (green), mCherry (magenta), acetyl-tubulin (white in C), CEP120 (white in D), Gp135 (white in G), and nuclear DAPI (blue). Images are from a single confocal section through the center of the cells. Yellow arrowheads point to the location of the centrosome. Scale bar: 5 μ m. (E, F) Boxplots of centrosome positions in control cells (Ctrl; n = 16) and transfected cells (C-CTD + N-gTBD; n = 18) at the post-CK stage, expressed as θ_{N-C} and L_{0-C} . Cells were analyzed in three independent experiments. Statistical significance was determined by an unpaired two-tailed Mann-Whitney U test (ns: not significant). Midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values. (H) Boxplot of Gp135 intensity on the apical membrane normalized to the mean of control cells. n = 16 (control) and 12 (C-CTD + N-gTBD) cysts were analyzed in three independent experiments. Statistical significance was determined by an unpaired two-tailed Mann-Whitney U test (*p<0.05). The midlines and boxes show the mean $\pm$ SD, with whiskers indicating minimum and maximum values (a.u., arbitrary units).

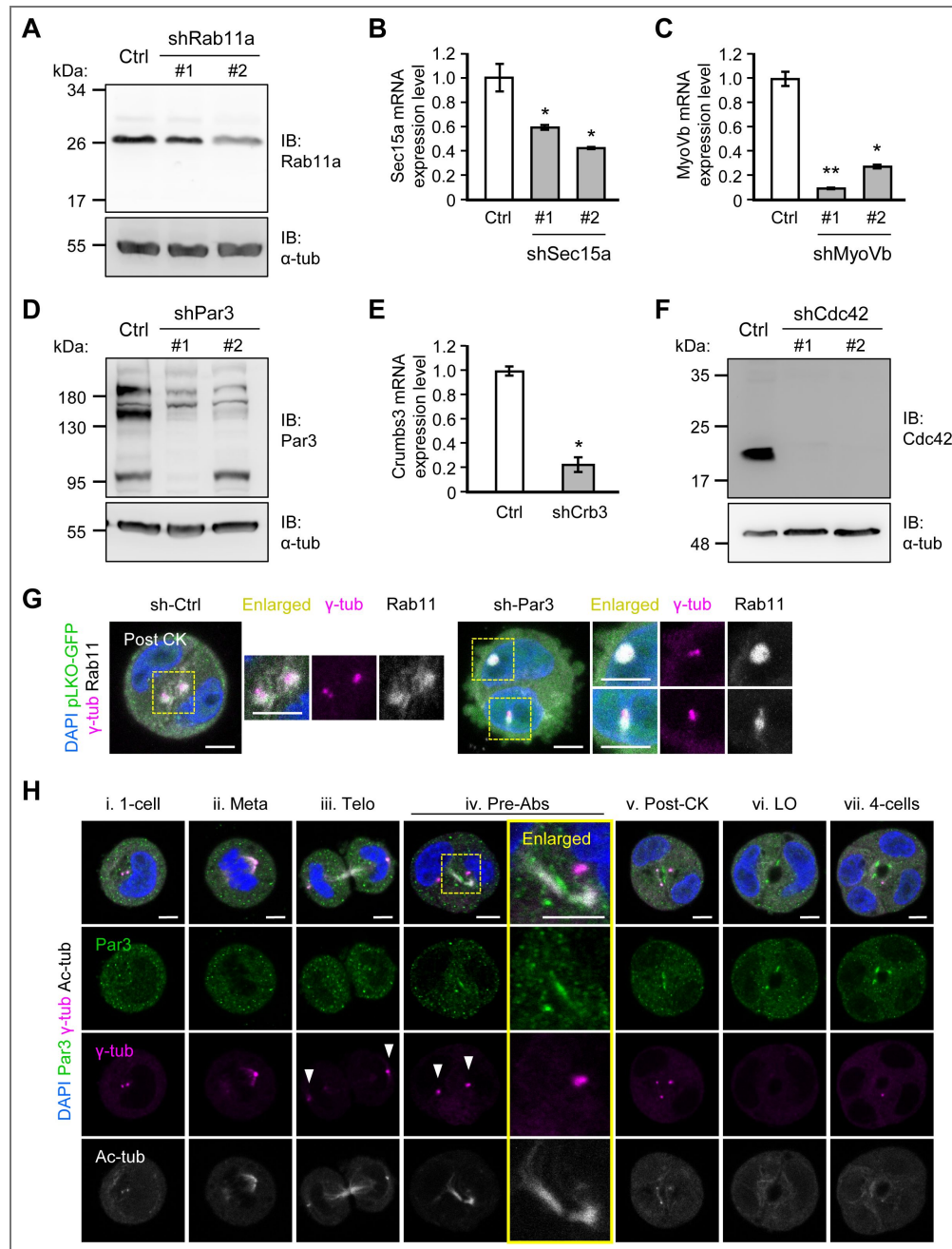


Figure 5—figure supplement 1. Characterization of MDCK RNAi and localization of Par3 during *de novo* epithelial polarization (A–F) MDCK cells stably expressing different pLKO-GFP-shRNA constructs targeting Rab11a (A), Sec15a (B), MyoVb (C), Par3 (D), Crb3 (E), and Cdc42 (F).

The knockdown efficiency was verified by blotting with appropriate antibodies (A, D, F) or quantitative RT-PCR (B, C, E), with GAPDH used as a control. For quantitative RT-PCR, statistical significance was determined by an unpaired two-tailed Mann-Whitney U test (* $p < 0.05$, ** $p < 0.01$). Values represent the mean $\pm$ SD. (G) Single MDCK cells stably express pLKO-GFP-shRNA to knock down Par3 after 12 h of Matrigel culture. Images show post-cytokinesis (CK) cells with labeled markers: γ -tubulin (magenta), Rab11a (white), pLKO-GFP (green), and DAPI for nuclei (blue). Z-projection images between two centrosomes are shown. "sh-Ctrl" indicates the expression of scrambled-sequence shRNA. The enlarged images show the centrosome position in the yellow-boxed area. Scale bar: 5 μ m. (H) Single MDCK cells after 12 h of culture in Matrigel. Immunostaining signals of the indicated markers are shown: centrosome marker γ -tubulin, polarity protein Par3, acetyl-tubulin, and DAPI. Single confocal sections through the middle of cysts are displayed. The enlarged images display the Par3 signals as they first emerge near the cytokinetic bridge, as indicated in the yellow boxed area. Arrowheads point to the centrosomes (LO, lumen open). Scale bar: 5 μ m.

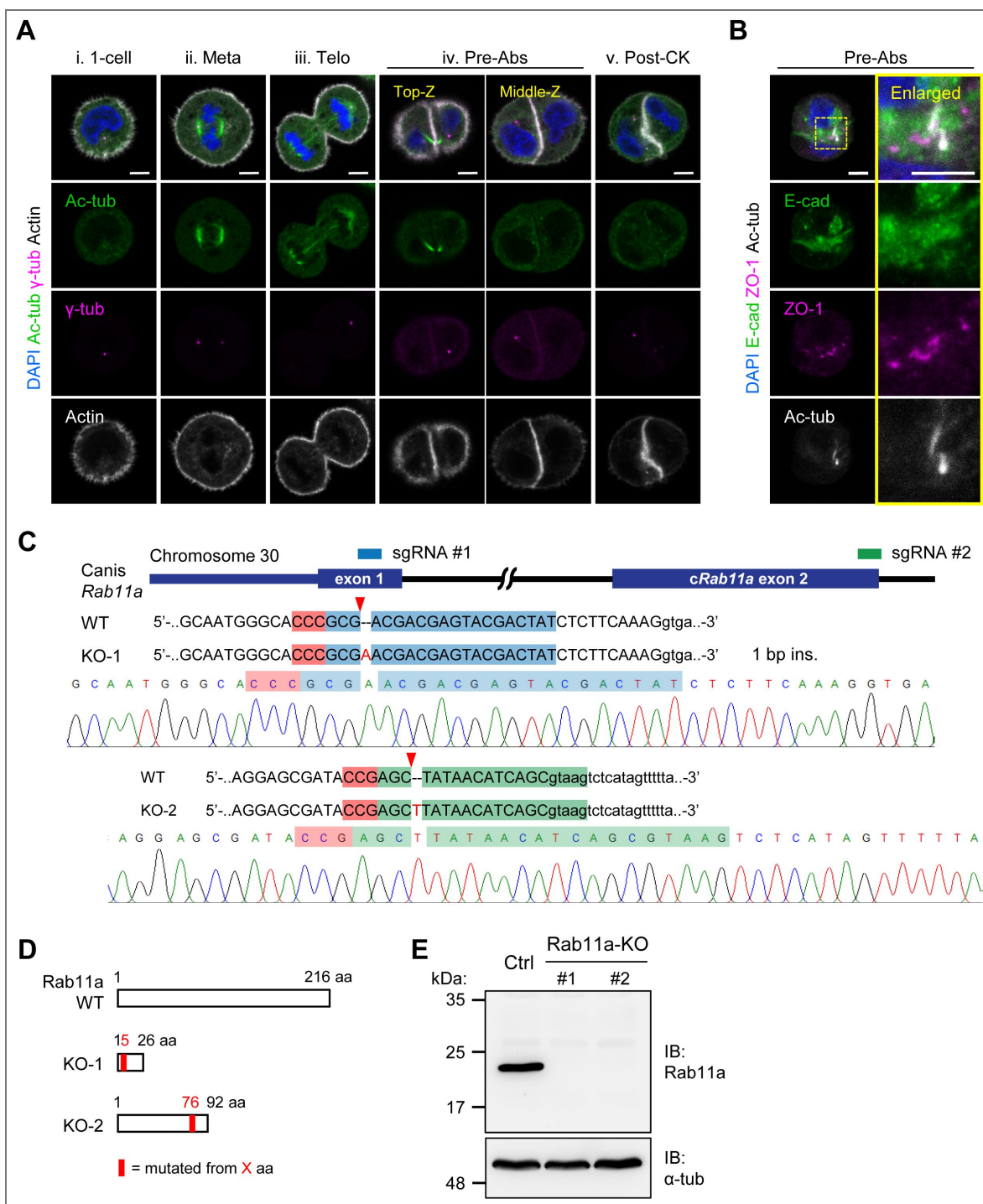


Figure 5—figure supplement 2. Localization of cell-cell junction components during *de novo* epithelial polarization and generation of Rab11a-KO MDCK cell lines

(A–B) Single MDCK cells after 12 h of culture in Matrigel. Immunostaining signals of the indicated markers are shown: centrosome marker γ-tubulin, polarity protein Par3, acetyl-tubulin, actin, E-cadherin (E-cad), ZO-1, and DAPI. Single confocal sections through the middle of cysts are displayed. The enlarged images display the Par3 and ZO-1 signals as they first emerge near the cytokinetic bridge, as indicated in the yellow boxed area. Arrowheads point to the centrosomes. Scale bar: 5 μm. (C) Diagram illustrating two single-guide RNA (sgRNA) sequences and their targeting sites within *Rab11a* exon 1 and exon 2 (blue and green boxes). The red boxes indicate the position of the protospacer-adjacent motif (PAM) sites and the predicted Cas9 cut sites are indicated by red arrowheads. Genotyping results of the genomic mutations, obtained via gel-purified PCR products covering exon 1 or 2 of the *Rab11a*-encoding gene in the Rab11a-KO cell line, are shown. The predicted genomic insertion (ins) within exons 1 and 2 of the *Rab11a*-encoding gene is illustrated. (D) Diagram illustrating the mutation position predicted in the *Rab11a*-encoding gene. (E) Western blot confirming the loss of Rab11a in two Rab11a-KO lines.

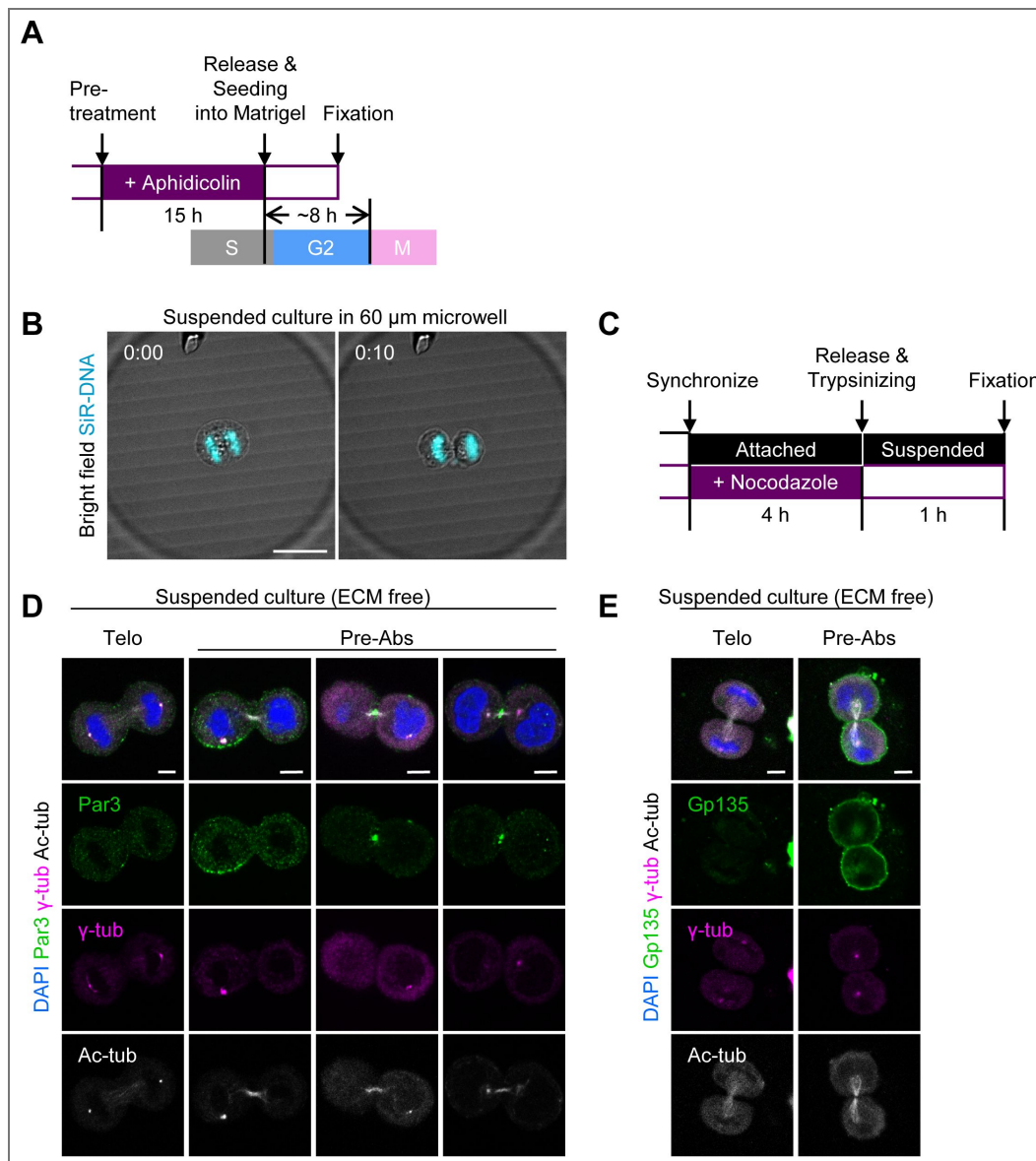


Figure 6—figure supplement 1. Experimental design and polarity marker dynamics from telophase to pre-abscission in suspension culture

(A) Experimental design: Aphidicolin (Aphi)-treated cells (synchronized at the S phase) were seeded into Matrigel and observed within 8 h to exclude the possibility of cell doublet formation through cell division. (B) Bright-field images of MDCK cells cultured in low-attachment microwells. The images are merged with the DAPI (cyan) signal to show cell division occurring over time. (C) Experimental design: Nocodazole-treated cells (synchronized at prophase) were resuspended in low-attachment dishes and observed within 1 hour to enrich for mitotic cells in suspension culture. (D, E) MDCK cells were treated as indicated in (C), and were labeled with Par3 (green in D), Gp135 (green in E), γ -tubulin (magenta), acetyl-tubulin (white), and DAPI (blue). Images show telophase (Telo) and cytokinetic pre-abscission (Pre-Abs) stages in single a confocal section through the center of a cell doublet.

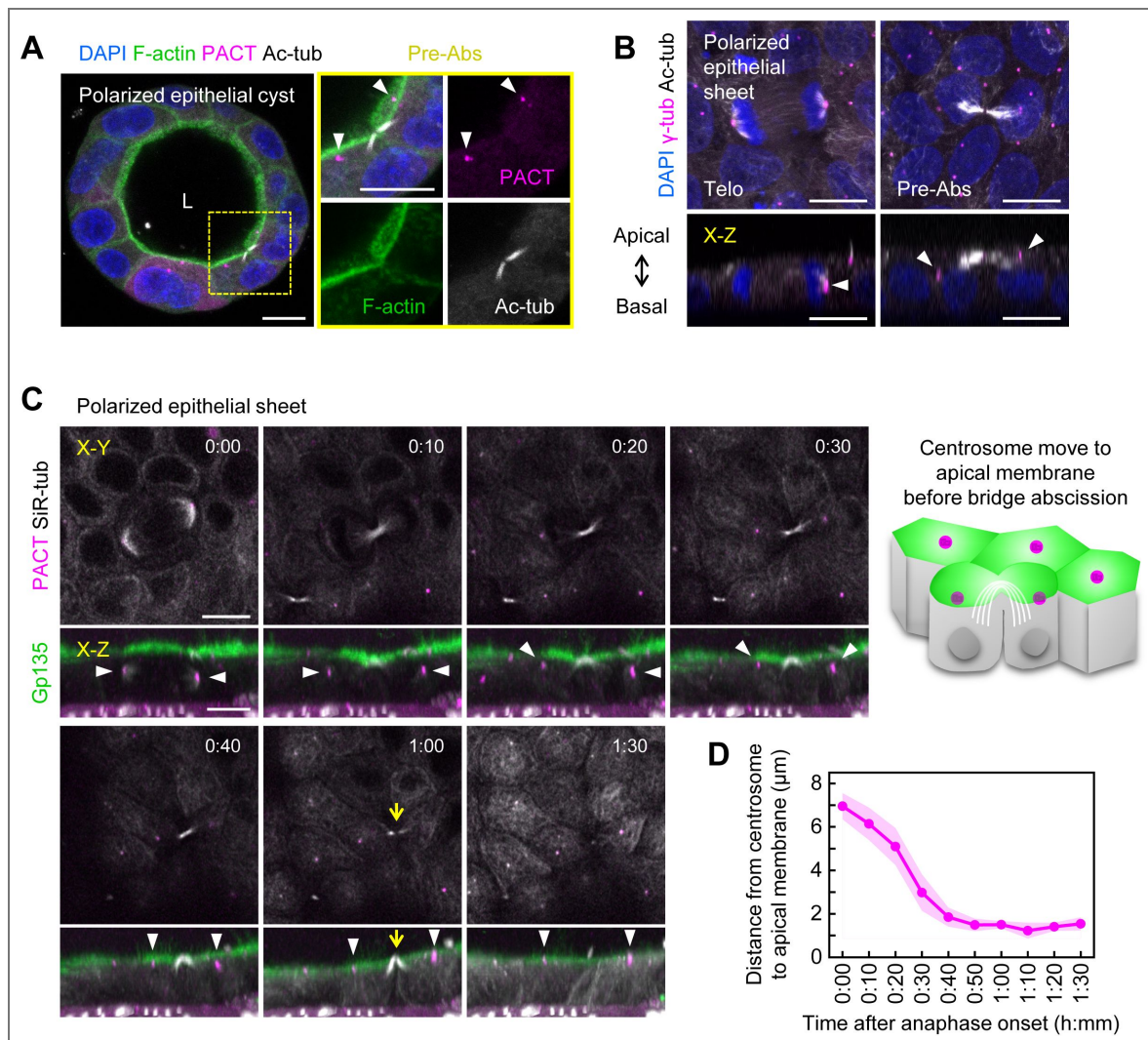
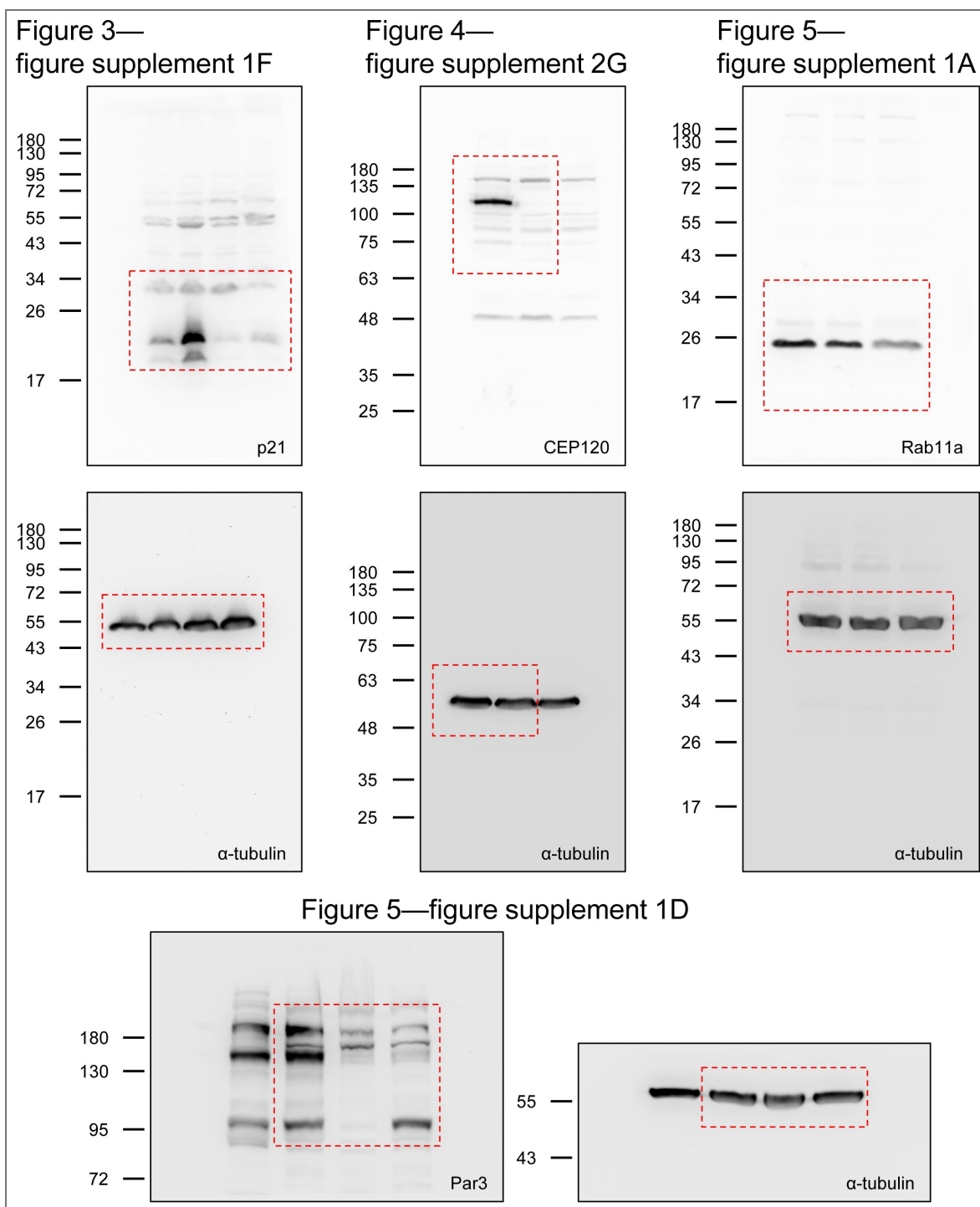


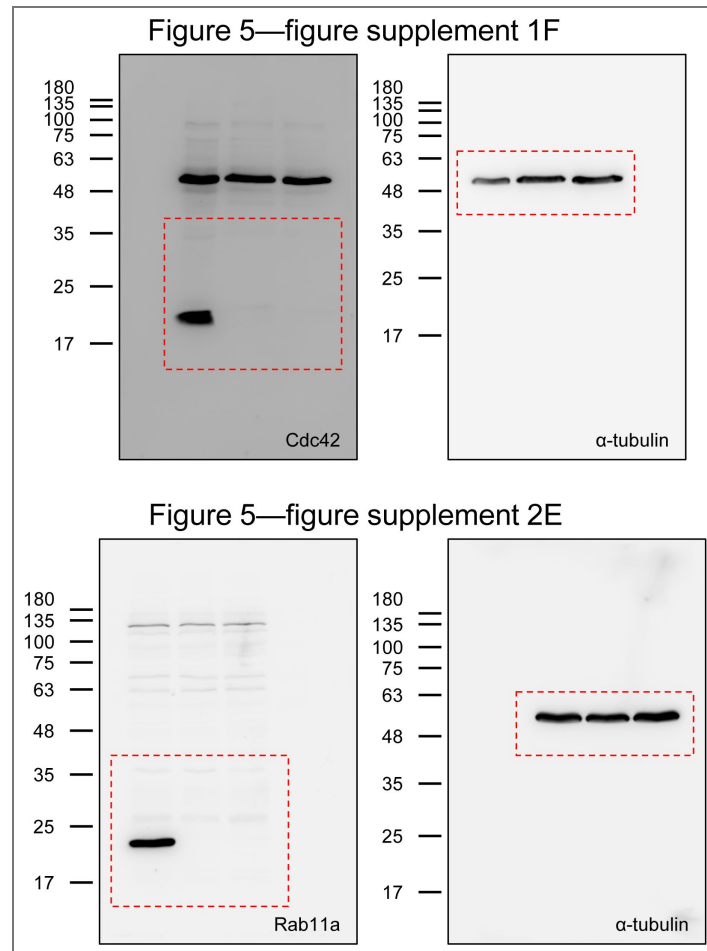
Figure 6—figure supplement 2. Centrosome position relative to the cytokinesis site in polarized epithelial sheets

(A–B) Polarized MDCK cyst and epithelial sheet with labeled markers: the centrosome markers PACT-mKO1 and γ -tubulin, acetyl-tubulin, F-actin, and DAPI. Insets show enlargements of the image in the boxed area. “L” indicates the lumen in (A). The z-projection of the top view and the X-Z cross-section of the side view (bottom) are shown in (B). Telo, telophase; Pre-Abs, pre-abscission. Arrowheads point to the centrosome position. Scale bar: 10 μm . (C) Time-lapse snapshots of MDCK cells expressing EGFP-Gp135 and PACT-mKO1 cultured on a Transwell insert to form polarized epithelial sheets. The microtubule probe SiR-tubulin (white) was applied before live cell imaging. The X-Y top view and the X-Z cross-section of the side view (bottom) are shown. Time stamps show hours and minutes, with 0:00 set to the first frame after anaphase onset. Arrowheads point to the centrosomes. The yellow arrow indicates the abscission of the cytokinetic bridge. Scale bar: 10 μm . A schematic depicts the centrosomes moving to the apical membrane before bridge abscission. (D) Change in distance from the centrosome to the apical membrane, as measured in each frame (data from three independent experiments, $n = 5$ cells). The graph presents the mean $\pm$ SD.



Source data 1. Uncropped Western blots are shown in this manuscript

(A) Western blots showing that p21 is present in MDCK WT and p53-KO cells with or without centrinone (CN) treatment are shown in Figure 3—figure supplement 1F. (B) Western blots showing the knockout of CEP120 from the p53-KO cell line are shown in Figure 4—figure supplement 2G. (C–D) Western blots show the depletion of Rab11a and Par3 in MDCK WT cells, as shown in Figure 5—figure supplement 1A and D.



Source data 2. Uncropped Western blots are shown in this manuscript

(A) Western blots showing the depletion of Cdc42 in MDCK WT cells are shown in [Figure 5—figure supplement 1F](#). (B) Western blots showing the knockout of Rab11a from MDCK WT cells are shown in [Figure 5—figure supplement 2E](#).

Acknowledgements

We thank the DNA Sequencing Core Facility (IBMS, AS-CFII-113-A12), Flow Cytometry Core Facility (IBMS, AS-CFII-111-212), and Light Microscopy Core Facility of the Institute of Biomedical Sciences, Academia Sinica, for their valuable assistance. We also thank Dr. Tzuu-Shuh Jou (NTU, Taiwan) for the Gp135 antibody and Dr. Kai Simons (MPI-CBG, Germany) for the EGFP-rabbit podocalyxin cDNA.

Additional information

Author contributions

Conceptualization: P.-K. Wang and T. K. Tang; Investigation: P.-K. Wang; Methodology: P.-K. Wang and K.-H. Lin; Resources: P.-K. Wang and K.-H. Lin; Writing – Original Draft: P.-K. Wang, K.-H. Lin and T. K. Tang; Writing – Review & Editing: P.-K. Wang, K.-H. Lin and T. K. Tang.

Funding

This work was supported by grants from Academia Sinica (AS-IA-109-L04) and the National Science and Technology Council, Taiwan (NSTC 112-2326-B001-010).

Funding

Funder	Grant reference number	Author
Academia Sinica (AS)	AS-IA-109-L04	Po-Kai Wang Tang K Tang
National Science and Technology Council (NSTC)	NSTC 112-2326-B001-010	Tang K Tang Po-Kai Wang

Author ORCID iDs

Keng-Hui Lin: <https://orcid.org/0000-0003-0638-7712>

Tang K Tang: <https://orcid.org/0000-0002-3660-0695>

Additional files

Figure 2A video. [Centrosome migration and apical membrane component trafficking during *de novo* epithelial polarization.](#) Centrosome migration and apical membrane component trafficking visualized by stable expression of PACT-mKO1 (magenta) and EGFP-Gp135 (green) in Matrigel-cultured MDCK cells. Nuclei were labeled with 500 nM SiR-DNA (blue). The cell was imaged for 1.5 hours (19 frames, 5 min intervals) during the cell division. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

Figure 2 - S1A video. [Centrosome migration and apical membrane component trafficking in adherent cells.](#) Centrosome migration and apical membrane component trafficking were visualized by stable expression of PACT-mKO1 (magenta) and EGFP-Gp135 (green) in MDCK cells cultured on a cover glass. Nuclei were labeled with 500 nM SiR-DNA (blue). During cell division, the cell was imaged for ~4 hours (20 frames, 15-minute intervals). Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

Figure 3A video 1. [Centrosome migration and apical membrane component trafficking in p53-KO cells.](#) Centrosome migration and apical membrane component trafficking visualized by stable expression of PACT-mKO1 (magenta) and EGFP-Gp135 (green) in Matrigel-cultured p53-KO cells. Nuclei were labeled with 500 nM SiR-DNA (blue). The cell was imaged for ~1.5 hours (21 frames, 5 min intervals) during the cell division. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

Figure 3A video 2.  Apical membrane component trafficking in centrosome-depleted cells. Apical membrane component trafficking is visualized by stable EGFP-Gp135 (green) expression in Matrigel-cultured p53-KO cells with centrinone (CN) treatment. Nuclei were labeled with 500 nM SiR-DNA (blue). The cell was imaged for ~2 hours (24 frames, 5 min intervals) during the cell division. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

Figure 6A video.  Centrosome migration and apical membrane component trafficking in aggregated cell doublets. Centrosome migration and apical membrane component trafficking visualized by stable expression of PACT-mKO1 (magenta) and EGFP-Gp135 (green) in aggregated MDCK cells. Nuclei were labeled with 500 nM SiR-DNA (cyan). The cell was imaged for 3 hours during the contact of two cells. Time stamps show hours and minutes, with 0:00 set at the beginning of live cell imaging. Scale bar: 10 μ m.

Figure 6G video.  Centrosome migration and apical membrane component trafficking in suspension-cultured cells. Centrosome migration and apical membrane component trafficking visualized by stable expression of PACT-mKO1 (magenta) and EGFP-Gp135 (green) in suspension-cultured MDCK cells. Nuclei were labeled with 500 nM SiR-DNA (cyan). Bright-field imaging showed cells suspended in 60 μ m diameter microwells. The cell was imaged for ~1.5 hours during the cell division. Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

Figure 6 - S2C video.  Centrosome migration in polarized epithelial sheets. Centrosome migration was visualized by stable expression of PACT-mKO1 (magenta) in MDCK epithelial sheets. The apical membrane and microtubule were labeled with EGFP-Gp135 (green) and 750 nM SiR-tubulin (white). During cell division, the cell was imaged for 2 hours (13 frames, 10 min intervals). Time stamps show hours and minutes, with 0:00 set at the first frame of anaphase onset. Scale bar: 10 μ m.

References

- Ahmed S. M., Macara I. G (2017) The Par3 polarity protein is an exocyst receptor essential for mammary cell survival. *Nat Commun* **8**:14867 <https://doi.org/10.1038/ncomms14867>
- Bisi S., Marchesi S., Rizvi A., Carra D., Beznoussenko G. V., Ferrara I., ..., Disanza A. (2020) IRSp53 controls plasma membrane shape and polarized transport at the nascent lumen in epithelial tubules. *Nat Commun* **11**:3516 <https://doi.org/10.1038/s41467-020-17091-x>
- Bornens M (2012) The centrosome in cells and organisms. *Science* **335**:422-426 <https://doi.org/10.1126/science.1209037>
- Bryant D. M., Datta A., Rodriguez-Fraticelli A. E., Peranen J., Martin-Belmonte F., Mostov K. E (2010) A molecular network for de novo generation of the apical surface and lumen. *Nat Cell Biol* **12**:1035-1045 <https://doi.org/10.1038/ncb2106>
- Bryant D. M., Roignot J., Datta A., Overeem A. W., Kim M., Yu W., ..., Mostov K. E. (2014) A molecular switch for the orientation of epithelial cell polarization. *Dev Cell* **31**:171-187 <https://doi.org/10.1016/j.devcel.2014.08.027>
- Buckley C. E., St Johnston D (2022) Apical-basal polarity and the control of epithelial form and function. *Nat Rev Mol Cell Biol* **23**:559-577 <https://doi.org/10.1038/s41580-022-00465-y>
- Burakov A. V., Nadezhdina E. S (2020) Centering and Shifting of Centrosomes in Cells. *Cells* **9** <https://doi.org/10.3390/cells9061351>
- Burute M., Prioux M., Blin G., Truchet S., Letort G., Tseng Q., ..., Thery M. (2017) Polarity Reversal by Centrosome Repositioning Primes Cell Scattering during Epithelial-to-Mesenchymal Transition. *Dev Cell* **40**:168-184 <https://doi.org/10.1016/j.devcel.2016.12.004>
- Chen X., Macara I. G (2005) Par-3 controls tight junction assembly through the Rac exchange factor Tiam1. *Nat Cell Biol* **7**:262-269 <https://doi.org/10.1038/ncb1226>

- Ching K., Wang J. T., Stearns T (2022) Long-range migration of centrioles to the apical surface of the olfactory epithelium. *eLife* **11** <https://doi.org/10.7554/eLife.74399>
- Chou S. Y., Hsu K. S., Otsu W., Hsu Y. C., Luo Y. C., Yeh C., ..., Sung C. H. (2016) CLIC4 regulates apical exocytosis and renal tube luminogenesis through retromer- and actin-mediated endocytic trafficking. *Nat Commun* **7**:10412 <https://doi.org/10.1038/ncomms10412>
- Comartin D., Gupta G. D., Fussner E., Coyaoud E., Hasegan M., Archinti M., ..., Pelletier L. (2013) CEP120 and SPICE1 cooperate with CPAP in centriole elongation. *Curr Biol* **23**:1360-1366 <https://doi.org/10.1016/j.cub.2013.06.002>
- Conduit P. T., Wainman A., Raff J. W (2015) Centrosome function and assembly in animal cells. *Nat Rev Mol Cell Biol* **16**:611-624 <https://doi.org/10.1038/nrm4062>
- de Anda F. C., Meletis K., Ge X., Rei D., Tsai L. H. (2010) Centrosome motility is essential for initial axon formation in the neocortex. *J Neurosci* **30**:10391-10406 <https://doi.org/10.1523/JNEUROSCI.0381-10.2010>
- de Anda F. C., Pollarolo G., Da Silva J. S., Camoletto P. G., Feiguin F., Dotti C. G. (2005) Centrosome localization determines neuronal polarity. *Nature* **436**:704-708 <https://doi.org/10.1038/nature03811>
- Feldman J. L., Priess J. R (2012) A role for the centrosome and PAR-3 in the hand-off of MTOC function during epithelial polarization. *Curr Biol* **22**:575-582 <https://doi.org/10.1016/j.cub.2012.02.044>
- Fong C. S., Mazo G., Das T., Goodman J., Kim M., O'Rourke B. P., ..., Tsou M. F. (2016) 53BP1 and USP28 mediate p53-dependent cell cycle arrest in response to centrosome loss and prolonged mitosis. *eLife* **5** <https://doi.org/10.7554/eLife.16270>
- Fong K. W., Choi Y. K., Rattner J. B., Qi R. Z (2008) CDK5RAP2 is a pericentriolar protein that functions in centrosomal attachment of the gamma-tubulin ring complex. *Mol Biol Cell* **19**:115-125 <https://doi.org/10.1091/mbc.e07-04-0371>
- Galvez-Santisteban M., Rodriguez-Fraticelli A. E., Bryant D. M., Vergarajauregui S., Yasuda T., Banon-Rodriguez I., ..., Martin-Belmonte F. (2012) Synaptotagmin-like proteins control the formation of a single apical membrane domain in epithelial cells. *Nat Cell Biol* **14**:838-849 <https://doi.org/10.1038/ncb2541>
- Gillingham A. K., Munro S (2000) The PACT domain, a conserved centrosomal targeting motif in the coiled-coil proteins AKAP450 and pericentrin. *EMBO Rep* **1**:524-529 <https://doi.org/10.1093/embo-reports/kvd105>
- Goldstein B., Macara I. G (2007) The PAR proteins: fundamental players in animal cell polarization. *Dev Cell* **13**:609-622 <https://doi.org/10.1016/j.devcel.2007.10.007>
- Gomes Pereira S., Dias Louro M. A., Bettencourt-Dias M. (2021) Biophysical and Quantitative Principles of Centrosome Biogenesis and Structure. *Annu Rev Cell Dev Biol* **37**:43-63 <https://doi.org/10.1146/annurev-cellbio-120219-051400>
- Hakanen J., Ruiz-Reig N., Tissir F (2019) Linking Cell Polarity to Cortical Development and Malformations. *Front Cell Neurosci* **13**:244 <https://doi.org/10.3389/fncel.2019.00244>
- Halaoui R., McCaffrey L (2015) Rewiring cell polarity signaling in cancer. *Oncogene* **34**:939-950 <https://doi.org/10.1038/onc.2014.59>
- Hannaford M. R., Liu R., Billington N., Swider Z. T., Galletta B. J., Fagerstrom C. J., ..., Rusan N. M. (2022) Pericentrin interacts with Kinesin-1 to drive centriole motility. *J Cell Biol* **221** <https://doi.org/10.1083/jcb.202112097>
- Hannaford M. R., Rusan N. M (2024) Positioning centrioles and centrosomes. *J Cell Biol* **223** <https://doi.org/10.1083/jcb.202311140>
- Hooikaas P. J., Damstra H. G., Gros O. J., van Riel W. E., Martin M., Smits Y. T., ..., Akhmanova A. (2020) Kinesin-4 KIF21B limits microtubule growth to allow rapid centrosome polarization in T cells. *eLife* **9** <https://doi.org/10.7554/eLife.62876>

- Horikoshi Y., Suzuki A., Yamanaka T., Sasaki K., Mizuno K., Sawada H., ..., Ohno S. (2009) Interaction between PAR-3 and the aPKC-PAR-6 complex is indispensable for apical domain development of epithelial cells. *J Cell Sci* **122**:1595-1606 <https://doi.org/10.1242/jcs.043174>
- Huang C. K., Paylaga G. J., Bupphathong S., Lin K. H. (2020) Spherical microwell arrays for studying single cells and microtissues in 3D confinement. *Biofabrication* **12**:025016 <https://doi.org/10.1088/1758-5090/ab6eda>
- Hung H. F., Hehnly H., Doxsey S. (2016) The Mother Centriole Appendage Protein Cenexin Modulates Lumen Formation through Spindle Orientation. *Curr Biol* **26**:793-801 <https://doi.org/10.1016/j.cub.2016.01.025>
- Iden S., Collard J. G. (2008) Crosstalk between small GTPases and polarity proteins in cell polarization. *Nat Rev Mol Cell Biol* **9**:846-859 <https://doi.org/10.1038/nrm2521>
- Jewett C. E., Prekeris R. (2018) Insane in the apical membrane: Trafficking events mediating apicobasal epithelial polarity during tube morphogenesis. *Traffic* <https://doi.org/10.1111/tra.12579>
- Jimenez A. J., Schaeffer A., De Pascalis C., Letort G., Vianay B., Bornens M., ..., Thery M. (2021) Actomyosin network geometry defines centrosome position. *Curr Biol* **31**:1206-1220.e1205. <https://doi.org/10.1016/j.cub.2021.01.002>
- Jonsdottir A. B., Dirks R. W., Vrolijk J., Ogmundsdottir H. M., Tanke H. J., Eyfjord J. E., Szuhai K. (2010) Centriole movements in mammalian epithelial cells during cytokinesis. *BMC Cell Biol* **11**:34 <https://doi.org/10.1186/1471-2121-11-34>
- Kapitein L. C., Peterman E. J., Kwok B. H., Kim J. H., Kapoor T. M., Schmidt C. F. (2005) The bipolar mitotic kinesin Eg5 moves on both microtubules that it crosslinks. *Nature* **435**:114-118 <https://doi.org/10.1038/nature03503>
- Klinkert K., Rocancourt M., Houdusse A., Echard A. (2016) Rab35 GTPase couples cell division with initiation of epithelial apico-basal polarity and lumen opening. *Nat Commun* **7**:11166 <https://doi.org/10.1038/ncomms11166>
- Krishnan N., Swoger M., Rathbun L. I., Fioramonti P. J., Freshour J., Bates M., ..., Hehnly H. (2022) Rab11 endosomes and Pericentrin coordinate centrosome movement during pre-abscission in vivo. *Life Sci Alliance* **5** <https://doi.org/10.26508/lsa.202201362>
- Laan L., Roth S., Dogterom M. (2012) End-on microtubule-dynein interactions and pulling-based positioning of microtubule organizing centers. *Cell Cycle* **11**:3750-3757 <https://doi.org/10.4161/cc.21753>
- Lambrus B. G., Daggubati V., Uetake Y., Scott P. M., Clutario K. M., Sluder G., Holland A. J. (2016) A USP28-53BP1-p53-p21 signaling axis arrests growth after centrosome loss or prolonged mitosis. *J Cell Biol* **214**:143-153 <https://doi.org/10.1083/jcb.201604054>
- Lee G. Y., Kenny P. A., Lee E. H., Bissell M. J. (2007) Three-dimensional culture models of normal and malignant breast epithelial cells. *Nat Methods* **4**:359-365 <https://doi.org/10.1038/nmeth1015>
- Li D., Mangan A., Cicchini L., Margolis B., Prekeris R. (2014) FIP5 phosphorylation during mitosis regulates apical trafficking and lumenogenesis. *EMBO Rep* **15**:428-437 <https://doi.org/10.1002/embr.201338128>
- Liang X., Weberling A., Hii C. Y., Zernicka-Goetz M., Buckley C. E. (2022) E-cadherin mediates apical membrane initiation site localisation during de novo polarisation of epithelial cavities. *EMBO J* **41**:e111021 <https://doi.org/10.15252/embj.2022111021>
- Lim H., Yu C. Y., Jou T. S. (2017) Galectin-8 regulates targeting of Gp135/podocalyxin and lumen formation at the apical surface of renal epithelial cells. *FASEB J* **31**:4917-4927 <https://doi.org/10.1096/fj.201601386R>
- Lin Y. N., Wu C. T., Lin Y. C., Hsu W. B., Tang C. J., Chang C. W., Tang T. K. (2013) CEP120 interacts with CPAP and positively regulates centriole elongation. *J Cell Biol* **202**:211-219 <https://doi.org/10.1083/jcb.201212060>

- Mangan A. J., Sietsema D. V., Li D., Moore J. K., Citi S., Prekeris R** (2016) Cingulin and actin mediate midbody-dependent apical lumen formation during polarization of epithelial cells. *Nat Commun* **7**:12426 <https://doi.org/10.1038/ncomms12426>
- Martin-Belmonte F., Gassama A., Datta A., Yu W., Rescher U., Gerke V., Mostov K** (2007) PTEN-mediated apical segregation of phosphoinositides controls epithelial morphogenesis through Cdc42. *Cell* **128**:383-397 <https://doi.org/10.1016/j.cell.2006.11.051>
- Martin M., Veloso A., Wu J., Katrukha E. A., Akhmanova A** (2018) Control of endothelial cell polarity and sprouting angiogenesis by non-centrosomal microtubules. *eLife* **7** <https://doi.org/10.7554/eLife.33864>
- Meder D., Shevchenko A., Simons K., Fullekrug J** (2005) Gp135/podocalyxin and NHERF-2 participate in the formation of a preapical domain during polarization of MDCK cells. *J Cell Biol* **168**:303-313 <https://doi.org/10.1083/jcb.200407072>
- Meiring J. C. M., Shneyer B. I., Akhmanova A** (2020) Generation and regulation of microtubule network asymmetry to drive cell polarity. *Curr Opin Cell Biol* **62**:86-95 <https://doi.org/10.1016/j.ceb.2019.10.004>
- Meitinger F., Anzola J. V., Kaulich M., Richardson A., Stender J. D., Benner C., ..., Oegema K.** (2016) 53BP1 and USP28 mediate p53 activation and G1 arrest after centrosome loss or extended mitotic duration. *J Cell Biol* **214**:155-166 <https://doi.org/10.1083/jcb.201604081>
- Mellman I., Nelson W. J** (2008) Coordinated protein sorting, targeting and distribution in polarized cells. *Nat Rev Mol Cell Biol* **9**:833-845 <https://doi.org/10.1038/nrm2525>
- Murdoch J. N., Henderson D. J., Doudney K., Gaston-Massuet C., Phillips H. M., Paternotte C., ..., Copp A. J.** (2003) Disruption of scribble (Scrib1) causes severe neural tube defects in the circletail mouse. *Hum Mol Genet* **12**:87-98 <https://doi.org/10.1093/hmg/ddg014>
- Ng K., Hirani N., Bland T., Borrego-Pinto J., Wagner S., Kreysing M., Goehring N. W** (2023) Cleavage furrow-directed cortical flows bias PAR polarization pathways to link cell polarity to cell division. *Curr Biol* **33**:4298-4311.e4296. <https://doi.org/10.1016/j.cub.2023.08.076>
- O'Brien L. E., Jou T. S., Pollack A. L., Zhang Q., Hansen S. H., Yurchenco P., Mostov K. E** (2001) Rac1 orientates epithelial apical polarity through effects on basolateral laminin assembly. *Nat Cell Biol* **3**:831-838 <https://doi.org/10.1038/ncb0901-831>
- Okumura M., Natsume T., Kanemaki M. T., Kiyomitsu T** (2018) Dynein-Dynactin-NuMA clusters generate cortical spindle-pulling forces as a multi-arm ensemble. *eLife* **7** <https://doi.org/10.7554/eLife.36559>
- Piel M., Nordberg J., Euteneuer U., Bornens M** (2001) Centrosome-dependent exit of cytokinesis in animal cells. *Science* **291**:1550-1553 <https://doi.org/10.1126/science.1057330>
- Pitaval A., Senger F., Letort G., Gidrol X., Guyon L., Sillibourne J., Thery M** (2017) Microtubule stabilization drives 3D centrosome migration to initiate primary ciliogenesis. *J Cell Biol* **216**:3713-3728 <https://doi.org/10.1083/jcb.201610039>
- Ran F. A., Hsu P. D., Wright J., Agarwala V., Scott D. A., Zhang F** (2013) Genome engineering using the CRISPR-Cas9 system. *Nat Protoc* **8**:2281-2308 <https://doi.org/10.1038/nprot.2013.143>
- Rathbun L. I., Colicino E. G., Manikas J., O'Connell J., Krishnan N., Reilly N. S., ..., Hehnlly H.** (2020) Cytokinetic bridge triggers de novo lumen formation in vivo. *Nat Commun* **11**:1269 <https://doi.org/10.1038/s41467-020-15002-8>
- Rodriguez-Boulan E., Macara I. G** (2014) Organization and execution of the epithelial polarity programme. *Nat Rev Mol Cell Biol* **15**:225-242 <https://doi.org/10.1038/nrm3775>
- Rodriguez-Fraticelli A. E., Auzan M., Alonso M. A., Bornens M., Martin-Belmonte F** (2012) Cell confinement controls centrosome positioning and lumen initiation during epithelial morphogenesis. *J Cell Biol* **198**:1011-1023 <https://doi.org/10.1083/jcb.201203075>

- Roland J. T., Bryant D. M., Datta A., Itzen A., Mostov K. E., Goldenring J. R (2011) Rab GTPase-Myo5B complexes control membrane recycling and epithelial polarization. *Proc Natl Acad Sci U S A* **108**:2789-2794 <https://doi.org/10.1073/pnas.1010754108>
- Roman-Fernandez A., Bryant D. M (2016) Complex Polarity: Building Multicellular Tissues Through Apical Membrane Traffic. *Traffic* **17**:1244-1261 <https://doi.org/10.1111/tra.12417>
- Roman-Fernandez A., Roignot J., Sandilands E., Nacke M., Mansour M. A., McGarry L., ..., Bryant D. M. (2018) The phospholipid PI(3,4)P(2) is an apical identity determinant. *Nat Commun* **9**:5041 <https://doi.org/10.1038/s41467-018-07464-8>
- Royer C., Lu X (2011) Epithelial cell polarity: a major gatekeeper against cancer?. *Cell Death Differ* **18**:1470-1477 <https://doi.org/10.1038/cdd.2011.60>
- Schluter M. A., Pfarr C. S., Pieczynski J., Whiteman E. L., Hurd T. W., Fan S., ..., Margolis B. (2009) Trafficking of Crumbs3 during cytokinesis is crucial for lumen formation. *Mol Biol Cell* **20**:4652-4663 <https://doi.org/10.1091/mbc.E09-02-0137>
- Schmoranz J., Fawcett J. P., Segura M., Tan S., Vallee R. B., Pawson T., Gundersen G. G (2009) Par3 and dynein associate to regulate local microtubule dynamics and centrosome orientation during migration. *Curr Biol* **19**:1065-1074 <https://doi.org/10.1016/j.cub.2009.05.065>
- Schneeberger K., Roth S., Nieuwenhuis E. E. S., Middendorp S (2018) Intestinal epithelial cell polarity defects in disease: lessons from microvillus inclusion disease. *Dis Model Mech* **11** <https://doi.org/10.1242/dmm.031088>
- St Johnston D., Ahringer J. (2010) Cell polarity in eggs and epithelia: parallels and diversity. *Cell* **141**:757-774 <https://doi.org/10.1016/j.cell.2010.05.011>
- Stiess M., Maghelli N., Kapitein L. C., Gomis-Ruth S., Wilsch-Brauninger M., Hoogenraad C. C., ..., Bradke F. (2010) Axon extension occurs independently of centrosomal microtubule nucleation. *Science* **327**:704-707 <https://doi.org/10.1126/science.1182179>
- Tang N., Marshall W. F (2012) Centrosome positioning in vertebrate development. *J Cell Sci* **125**:4951-4961 <https://doi.org/10.1242/jcs.038083>
- Tanos B. E., Yang H. J., Soni R., Wang W. J., Macaluso F. P., Asara J. M., Tsou M. F (2013) Centriole distal appendages promote membrane docking, leading to cilia initiation. *Genes Dev* **27**:163-168 <https://doi.org/10.1101/gad.207043.112>
- Tateishi K., Yamazaki Y., Nishida T., Watanabe S., Kunimoto K., Ishikawa H., Tsukita S (2013) Two appendages homologous between basal bodies and centrioles are formed using distinct Odf2 domains. *J Cell Biol* **203**:417-425 <https://doi.org/10.1083/jcb.201303071>
- Tsai J. J., Hsu W. B., Liu J. H., Chang C. W., Tang T. K (2019) CEP120 interacts with C2CD3 and Talpid3 and is required for centriole appendage assembly and ciliogenesis. *Sci Rep* **9**:6037 <https://doi.org/10.1038/s41598-019-42577-0>
- Vinopal S., Dupraz S., Alfadil E., Pietralla T., Bendre S., Stiess M., ..., Bradke F. (2023) Centrosomal microtubule nucleation regulates radial migration of projection neurons independently of polarization in the developing brain. *Neuron* **111**:1241-1263.e1216. <https://doi.org/10.1016/j.neuron.2023.01.020>
- Wakida N. M., Botvinick E. L., Lin J., Berns M. W (2010) An intact centrosome is required for the maintenance of polarization during directional cell migration. *PLoS One* **5**:e15462 <https://doi.org/10.1371/journal.pone.0015462>
- Wang T., Yanger K., Stanger B. Z., Cassio D., Bi E (2014) Cytokinesis defines a spatial landmark for hepatocyte polarization and apical lumen formation. *J Cell Sci* **127**:2483-2492 <https://doi.org/10.1242/jcs.139923>
- Watanabe S., Meitinger F., Shiao A. K., Oegema K., Desai A (2020) Centriole-independent mitotic spindle assembly relies on the PCNT-CDK5RAP2 pericentriolar matrix. *J Cell Biol* **219** <https://doi.org/10.1083/jcb.202006010>

- Wilson P. D (2011) Apico-basal polarity in polycystic kidney disease epithelia. *Biochim Biophys Acta* **1812**:1239-1248 <https://doi.org/10.1016/j.bbadis.2011.05.008>
- Wong Y. L., Anzola J. V., Davis R. L., Yoon M., Motamedi A., Kroll A., ..., Oegema K. (2015) Cell biology. Reversible centriole depletion with an inhibitor of Polo-like kinase 4. *Science* **348**:1155-1160 <https://doi.org/10.1126/science.aaa5111>
- Yamamoto S., Gaillard J., Vianay B., Guerin C., Orhant-Prioux M., Blanchoin L., Thery M (2022) Actin network architecture can ensure robust centering or sensitive decentering of the centrosome. *EMBO J* **41**:e111631 <https://doi.org/10.15252/embj.2022111631>
- Yi J., Wu X., Chung A. H., Chen J. K., Kapoor T. M., Hammer J. A (2013) Centrosome repositioning in T cells is biphasic and driven by microtubule end-on capture-shrinkage. *J Cell Biol* **202**:779-792 <https://doi.org/10.1083/jcb.201301004>
- Yu C. Y., Chen J. Y., Lin Y. Y., Shen K. F., Lin W. L., Chien C. L., ..., Jou T. S. (2007) A bipartite signal regulates the faithful delivery of apical domain marker podocalyxin/Gp135. *Mol Biol Cell* **18**:1710-1722 <https://doi.org/10.1091/mbc.e06-07-0629>
- Yu W., Datta A., Leroy P., O'Brien L. E., Mak G., Jou T. S., ..., Zegers M. M. (2005) Beta1-integrin orients epithelial polarity via Rac1 and laminin. *Mol Biol Cell* **16**:433-445 <https://doi.org/10.1091/mbc.e04-05-0435>
- Yu W., Shewan A. M., Brakeman P., Eastburn D. J., Datta A., Bryant D. M., ..., Mostov K. E. (2008) Involvement of RhoA, ROCK I and myosin II in inverted orientation of epithelial polarity. *EMBO Rep* **9**:923-929 <https://doi.org/10.1038/embor.2008.135>
- Zimmerman W. C., Sillibourne J., Rosa J., Doxsey S. J (2004) Mitosis-specific anchoring of gamma tubulin complexes by pericentrin controls spindle organization and mitotic entry. *Mol Biol Cell* **15**:3642-3657 <https://doi.org/10.1091/mbc.e03-11-0796>

Peer reviews

Reviewer #1 (Public review):

Summary:

Wang, Po-Kai et al., utilized the de novo polarization of MDCK cells cultured in Matrigel to assess the interdependence between polarity protein localization, centrosome positioning and apical membrane formation. They show that the inhibition of Plk4 with Centrione does not prevent apical membrane formation, but does result in its delay, a phenotype the authors attribute to the loss of centrosomes due to the inhibition of centriole duplication. However, the targeted mutagenesis of specific centrosome proteins implicated in the positioning of centrosomes in other cell types (CEP164, ODF2, PCNT and CEP120), as well as the use of dominant negative constructs to inhibit centrosomal microtubule nucleation did not affect centrosome positioning in 3D cultured MDCK cells. A screen of proteins previously implicated in MDCK polarization revealed that the polarity protein Par-3 was upstream of centrosome positioning, similar to other cell types.

Strengths:

The investigation into the temporal requirement and interdependence of previously proposed regulators of cell polarization and lumen formation is valuable. The authors have provided a detailed analysis of many of these components at defined stages of polarity establishment, and well demonstrate that centrosomes are not necessary for apical polarity formation, but are involved in the efficient establishment of the apical membrane.

Weaknesses:

Key questions remain regarding the structure of the intracellular cytoskeleton following depletion of centrosomes, centrosome proteins, or abrogation of centrosome microtubule nucleation. The authors strengthen their model that centrosomes are positioned independently of microtubule nucleation using dominant negative Cdk5RAP2 and NEDD-1 constructs, however, the structure of the intracellular microtubule network remains unresolved and will be an important avenue for future investigation.

<https://doi.org/10.7554/eLife.101088.2.sa2>

Reviewer #3 (Public review):

Here the Wang et al resubmit their manuscript describing the events in the establishment of polarity in MDCK cells cultured in vitro. As with the original version, the description is throughout and is important to the field to report as it establishes a hierarchy of events in polarization, placing Par3 upstream of centrosome positioning and apical membrane component trafficking. Unfortunately, in the revised version, the authors addressed almost none of my points. They did a cursory job of responding in the rebuttal letter but made little attempt to actually address what was being asked or to incorporate any of my suggestions into the manuscript. The particularly egregious examples are cited below:

Comments on revisions:

(1) My original main experimental concern was not addressed: I had originally asked what role microtubules play in the process of polarization (either centrosomal or non-centrosomal). An obvious model is that Gp135, Rab11, etc. are delivered to the AMIS on centrosomal microtubules. Centrosomes might be also be pulled to the AMIS via cortically derived microtubules as is the case in the *C. elegans* intestine where the centrosome moves apically on apical microtubules via dynein directed transport to the cortically anchored minus ends. The authors do not explore the role of microtubules in the revision, citing that it was not possible to observe the microtubules directly or to perform nocodazole experiments during polarization. Instead, the authors use a relatively new genetic tool to disrupt centrosomal microtubules. They appear to succeed in displacing centrosomal γ -tubulin using this tool, but without being able to observe microtubules, a remaining caveat of this experiment is that it is still unclear whether the authors have removed centrosomal microtubules. Compounding this issue is that this tool has never been used in MDCK cells. The authors conclude "we found that cells lacking centrosomal microtubules were still able to polarize and position the centrioles apically.", but they have not shown this, instead the data suggest this conclusion and the authors should acknowledge the caveat that they have no idea whether centrosomal microtubules are abolished. Similarly, the authors also state: "Additionally, although PCNT knockout cells show reduced microtubule nucleation ability, they still recruit a small amount of γ -tubulin". Where are the data that show that microtubule nucleation is reduced in these PCNT knock out cells?

(2) Many of my comments were addressed in the rebuttal, but not in the text.

The non-centrosomal GP135 in Figure 2 is not acknowledged or explained.

That the polarity index does not actually measure polarity, but nuclear-centrosome distance is not acknowledged or explained in the paper.

I still don't believe that the quantification in Figure 3D matches the images I am being shown in Figure 3A. In the centrinone treatment condition, there is certainly an enrichment of GP135 at the AMIS that is not detected in the quantification. The method described in the rebuttal might miss this enrichment if it is offset from line drawn between the centroid of the two nuclei.

Cell height changes in the centrosome depleted cysts are still referenced in the text ("the cell heights of the centrosome-depleted cysts are less uniform"), but no specific data or image is called out. Currently, Figure 3G is referenced, but that is a graph of GP135 intensity

In my original review, I called on the authors to comment on the striking similarity of the mechanisms they documented in MDCK cells to what has been shown in *in vivo* systems. The authors did not do this, instead restating in the rebuttal some features of what they found. But, the mechanisms shown here are remarkably similar to the polarization of primordia that generate tubular organs *in vivo*. Perhaps most striking is the similarity to the *C. elegans* intestine where Par3 localizes to the cortex at the site of an apical MTOC that pulls the centrosome to the apical surface via dynein (Feldman and Priess, 2012). Instead of discussing this similarity, the authors state: "Par3 is likely to regulate centrosome positioning through some intermediate molecules or mechanisms, but its specific mechanism is still unclear and requires further investigation." Given the acetylated tubulin signal emanating from the Par3 positive patch in Figure 5E and F, I suspect similar mechanisms to the *C. elegans* intestine are at play here. Such a parallel should be noted in the Discussion.

I had originally commented that "I find the results in Figure 6G puzzling. Why is ECM signaling required for Gp135 recruitment to the centrosome. Could the authors discuss what this means?" The authors responded that "The data in Figure 6G do not indicate that ECM signaling is required for the recruitment of Gp135 to the centrosome". In Figure 6G, the localization of GP135 to the centrosome appears significantly delayed compared to its localization to the centrosome in images where cells were cultured in Matrigel. Indeed, the authors argue that the centrosomal localization precedes and contributes to its localization to the AMIS. In the absence of ECM, GP135 localizes to the membrane before it localizes to the centrosome and its localization to the centrosome appears significantly reduced. Thus, my original and current interpretation is that ECM signaling is somehow required for the centrosomal targeting of GP135. One could make a competition argument, i.e. that the cortex in the absence of ECM is somehow a more desirable place to localize than the centrosome, but this experiment also argues that the centrosome does not need to be a source of this material in order for it to end up on the cortex.

(3) There needs to be precision in the language used in many places:

I don't understand this line in the abstract: "When cultured in Matrigel, *de novo* polarization of a single epithelial cell is often coupled with mitosis." If a cell has divided, it is no longer a single cell.

The authors state in the Introduction "Because of its strong ability to nucleate microtubules, the centrosome functions as the primary microtubule organizing center", but then state ""In polarized epithelial cells, the centrosome is localized at the apical region during interphase, which contributes to the construction of an asymmetric microtubule network conducive to polarized vesicle trafficking". In the latter statement, I assume the authors are describing the well-characterized apical microtubule network in epithelial cells that is non-centrosomal. Thus, the latter sentence is at odds with the former.

The authors continually refer to Par3 as a tight junction protein. "Par3, which controls tight junction assembly to partition the apical surface from the basolateral surface". To my knowledge, PARD3 is an apical protein with similar localization to *C. elegans* PAR-3 and *Drosophila* Bazooka. PARD3B is a junctional protein. I assume that the antibody that the authors are using is to PARD3 and not PARD3B? Can the authors please clarify this in the text.

<https://doi.org/10.7554/eLife.101088.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Wang, Po-Kai, et al., utilized the *de novo* polarization of MDCK cells cultured in Matrigel to assess the interdependence between polarity protein localization, centrosome positioning, and apical membrane formation. They show that the inhibition of Plk4 with Centrinone does not prevent apical membrane formation, but does result in its delay, a phenotype the authors attribute to the loss of centrosomes due to the inhibition of centriole duplication. However, the targeted mutagenesis of specific centrosome proteins implicated in the positioning of centrosomes in other cell types (CEP164, ODF2, PCNT, and CEP120) did not affect centrosome positioning in 3D cultured MDCK cells. A screen of proteins previously implicated in MDCK polarization revealed that the polarity protein Par-3 was upstream of centrosome positioning, similar to other cell types.

Strengths:

The investigation into the temporal requirement and interdependence of previously proposed regulators of cell polarization and lumen formation is valuable to the community. Wang et al., have provided a detailed analysis of many of these components at defined stages of polarity establishment. Furthermore, the generation of PCNT, p53, ODF2, Cep120, and Cep164 knockout MDCK cell lines is likely valuable to the community.

Weaknesses:

Additional quantifications would highly improve this manuscript, for example it is unclear whether the centrosome perturbation affects gamma tubulin levels and therefore microtubule nucleation, it is also not clear how they affect the localization of the trafficking machinery/polarity proteins. For example, in Figure 4, the authors measure the intensity of Gp134 at the apical membrane initiation site following cytokinesis, but there is no measure of Gp134 at the centrosome prior to this.

We thank the reviewer for this important suggestion. Previous studies have shown that genes encoding appendage proteins and CEP120 do not regulate γ -tubulin recruitment to centrosomes (Betleja, Nanjundappa, Cheng, & Mahjoub, 2018; Vasquez-Limeta & Loncarek, 2021). Although the loss of PCNT reduces γ -tubulin levels, this reduction is partially compensated by AKAP450. Even in the case of PCNT/AKAP450 double knockouts, low levels of γ -tubulin remain at the centrosome (Gavilan et al., 2018), suggesting that it is difficult to completely eliminate γ -tubulin by perturbing centrosomal genes alone.

To directly address this question, in the revised manuscript (Page 8, Paragraph 4; Figure 4—figure supplement 3), we employed a recently reported method to block γ -tubulin recruitment by co-expressing two constructs: the centrosome-targeting carboxy-terminal domain (C-CTD) of CDK5RAP2 and the γ -tubulin-binding domain of NEDD1 (N-gTBD). This approach effectively depleted γ -tubulin and abolished microtubule nucleation at the centrosome (Vinopal et al., 2023). Interestingly, despite the reduced efficiency of apical vesicle trafficking, these cells were still able to establish polarity, with centrioles positioned apically. These results suggest that microtubule nucleation at the centrosomes (centrosomal microtubules) facilitates—but is not essential for—polarity establishment.

Regarding Figure 4, we assume the reviewer was referring to Gp135 rather than Gp134. In the revised manuscript (Page 8, Paragraph 2; Figure 4I), we observed a slight decrease in Gp135 intensity near PCNT-KO centrosomes at the pre-Abs stage. However, its localization at the AMIS following cytokinesis remained unaffected. These results suggest that the loss of PCNT has a limited impact on Gp135 localization.

Reviewer #2 (Public review):

Summary:

The authors decoupled several players that are thought to contribute to the establishment of epithelial polarity and determined their causal relationship. This provides a new picture of the respective roles of junctional proteins (Par3), the centrosome, and endomembrane compartments (Cdc42, Rab11, Gp135) from upstream to downstream.

Their conclusions are based on live imaging of all players during the early steps of polarity establishment and on the knock-down of their expression in the simplest ever model of epithelial polarity: a cell doublet surrounded by ECM.

The position of the centrosome is often taken as a readout for the orientation of the cell polarity axis. There is a long-standing debate about the actual role of the centrosome in the establishment of this polarity axis. Here, using a minimal model of epithelial polarization, a doublet of daughters MDCK cultured in Matrigel, the authors made several key observations that bring new light to our understanding of a mechanism that has been studied for many years without being fully explained:

(1) They showed that centriole can reach their polarized position without most of their microtubule-anchoring structures. These observations challenge the standard model according to which centrosomes are moved by the production and transmission of forces along microtubules.

(2) However) they showed that epithelial polarity can be established in the absence of a centriole.

(3) (Somehow more expectedly) they also showed that epithelial polarity can't be established in the absence of Par3.

(4) They found that most other polarity players that are transported through the cytoplasm in lipid vesicles, and finally fused to the basal or apical pole of epithelial cells, are moved along an axis which is defined by the position of centrosome and orientation of microtubules.

(5) Surprisingly, two non-daughter cells that were brought in contact (for 6h) could partially polarize by recruiting a few Par3 molecules but not the other polarity markers.

(6) Even more surprisingly, in the absence of ECM, Par 3 and centrosomes could move to their proper position close to the intercellular junction after cytokinesis but other polarity markers (at least GP135) localized to the opposite, non-adhesive, side. So the polarity of the centrosome-microtubule network could be dissociated from the localisation of GP135 (which was believed to be transported along this network).

Strengths:

(1) The simplicity and reproducibility of the system allow a very quantitative description of cell polarity and protein localisation.

(2) The experiments are quite straightforward, well-executed, and properly analyzed.

(3) *The writing is clear and conclusions are convincing.*

Weaknesses:

(1) *The simplicity of the system may not capture some of the mechanisms involved in the establishment of cell polarity in more physiological conditions (fluid flow, electrical potential, ion gradients,...).*

We agree that certain mechanisms may not be captured by this simplified system. However, the model enables us to observe intrinsic cellular responses, minimize external environmental variables, and gain new insights into how epithelial cells position their centrosomes and establish polarity.

(2) *The absence of centriole in centrinone-treated cells might not prevent the coalescence of centrosomal protein in a kind of MTOC which might still orient microtubules and intracellular traffic. How are microtubules organized in the absence of centriole? If they still form a radial array, the absence of a centriole at the center of it somehow does not conflict with classical views in the field.*

Previous studies have shown that in the absence of centrioles, centrosomal proteins can relocate to alternative microtubule-organizing centers (MTOCs), such as the Golgi apparatus (Gavilan et al., 2018). Furthermore, centriole loss leads to increased nucleation of non-centrosomal microtubules (Martin, Veloso, Wu, Katrukha, & Akhmanova, 2018). However, these microtubules typically do not form the classical radial array or a distinct star-like organization.

While this non-centrosomal microtubule network can still support polarity establishment, it does so less efficiently—similar to what is observed in p53-deficient cells undergoing centriole-independent mitosis (Meitinger et al., 2016). Thus, although the absence of centrioles does not completely prevent microtubule-based organization or polarity establishment, it impairs their spatial coordination and reduces overall efficiency compared to a centriole-centered microtubule-organizing center (MTOC).

(3) *The mechanism is still far from clear and this study shines some light on our lack of understanding. Basic and key questions remain:*

(a) *How is the centrosome moved toward the Par3-rich pole? This is particularly difficult to answer if the mechanism does not imply the anchoring of MTs to the centriole or PCM.*

Previous studies have shown that Par3 interacts with dynein, potentially anchoring it at the cell cortex (Schmoranz et al., 2009). This interaction enables dynein, a minus-end directed motor, to exert pulling forces on microtubules, thereby promoting centrosome movement toward the Par3-enriched pole.

In our experiments (Figure 4), we attempted to disrupt centrosomal microtubule nucleation by knocking out multiple genes involved in centrosome structure and function, including ODF2 and PCNT. Under these perturbations, γ -tubulin still remained detectable at the centrosome, and we were unable to completely eliminate centrosomal microtubules.

To address this question more directly, we employed a strategy to deplete γ -tubulin from centrosomes by co-expressing the centrosome-targeting C-terminal domain (C-CTD) of CDK5RAP2 and the γ -tubulin-binding domain of NEDD1 (N-gTBD). As shown in the new data of the revised manuscript (Page 8, Paragraph 4; Figure 4—figure supplement 3), this approach effectively depleted γ -tubulin from centrosomes, thereby abolishing microtubule nucleation at the centrosome.

Surprisingly, even under these conditions, centrioles remained apically positioned (Page 8, Paragraph 4; Figure 4—figure supplement 3), indicating that centrosomal microtubules are not essential for centrosome movement during polarization.

Given these findings, we agree that the precise mechanism by which the Par3-enriched cortex attracts or guides centrosome movement remains unclear. Although dynein–Par3 interactions may contribute, further studies are needed to elucidate how centrosome repositioning occurs in the absence of microtubule-based pulling forces from the centrosome itself.

(b) What happens during cytokinesis that organises Par3 and intercellular junction in a way that can't be achieved by simply bringing two cells together? In larger epithelia cells have neighbours that are not daughters, still, they can form tight junctions with Par3 which participates in the establishment of cell polarity as much as those that are closer to the cytokinetic bridge (as judged by the overall cell symmetry). Is the protocol of cell aggregation fully capturing the interaction mechanism of non-daughter cells?

We speculate that a key difference between cytokinesis and simple cell-cell contact lies in the presence or absence of actomyosin contractility during the process of cell division. Specifically, contraction of the cytokinetic ring generates mechanical forces between the two daughter cells, which are absent when two non-daughter cells are simply brought together. While adjacent epithelial cells can indeed form tight junctions and recruit Par3, the lack of shared cortical tension and contractile actin networks between non-daughter cells may lead to differences in how polarity is initiated. This mechanical input during cytokinesis may serve as an organizing signal for centrosome positioning. This idea is supported by recent work showing that the actin cytoskeleton can influence centrosome positioning (Jimenez et al., 2021), suggesting that contractile actin structures formed during cytokinesis may contribute to spatial organization in a manner that cannot be replicated by simple aggregation.

In our experiments, we simply captured two cells that were in contact within Matrigel. We cannot say for sure that it captures all the interaction mechanisms of non-daughter cells, but it does provide a contrast to daughter cells produced by cytokinesis.

Reviewer #3 (Public review):

Here, Wang et al. aim to clarify the role of the centrosome and conserved polarity regulators in apical membrane formation during the polarization of MDCK cells cultured in 3D. Through well-presented and rigorous studies, the authors focused on the emergence of polarity as a single MDCK cell divided in 3D culture to form a two-cell cyst with a nascent lumen. Focusing on these very initial stages, rather than in later large cyst formation as in most studies, is a real strength of this study. The authors found that conserved polarity regulators Gp135/podocalyxin, Crb3, Cdc42, and the recycling endosome component Rab11a all localize to the centrosome before localizing to the apical membrane initiation site (AMIS) following cytokinesis. This protein relocalization was concomitant with a repositioning of centrosomes towards the AMIS. In contrast, Par3, aPKC, and the junctional components E-cadherin and ZO1 localize directly to the AMIS without first localizing to the centrosome. Based on the timing of the localization of these proteins, these observational studies suggested that Par3 is upstream of centrosome repositioning towards the AMIS and that the centrosome might be required for delivery of apical/luminal proteins to the AMIS.

To test this hypothesis, the authors generated numerous new cell lines and/or employed pharmacological inhibitors to determine the hierarchy of localization among these components. They found that removal of the centrosome via centrinone treatment severely delayed and weakened the delivery of Gp135 to the AMIS and single lumen

formation, although normal lumenogenesis was apparently rescued with time. This effect was not due to the presence of CEP164, ODF2, CEP120, or Pericentrin. Par3 depletion perturbed the repositioning of the centrosome towards the AMIS and the relocalization of the Gp135 and Rab11 to the AMIS, causing these proteins to get stuck at the centrosome. Finally, the authors culture the MDCK cells in several ways (forced aggregation and ECM depleted) to try and further uncouple localization of the pertinent components, finding that Par3 can localize to the cell-cell interface in the absence of cell division. Par3 localized to the edge of the cell-cell contacts in the absence of ECM and this localization was not sufficient to orient the centrosomes to this site, indicating the importance of other factors in centrosome recruitment.

Together, these data suggest a model where Par3 positions the centrosome at the AMIS and is required for the efficient transfer of more downstream polarity determinants (Gp135 and Rab11) to the apical membrane from the centrosome. The authors present solid and compelling data and are well-positioned to directly test this model with their existing system and tools. In particular, one obvious mechanism here is that centrosome-based microtubules help to efficiently direct the transport of molecules required to reinforce polarity and/or promote lumenogenesis. This model is not really explored by the authors except by Pericentrin and subdistal appendage depletion and the authors do not test whether these perturbations affect centrosomal microtubules. Exploring the role of microtubules in this process could considerably add to the mechanisms presented here. In its current state, this paper is a careful observation of the events of MCDK polarization and will fill a knowledge gap in this field. However, the mechanism could be significantly bolstered with existing tools, thereby elevating our understanding of how polarity emerges in this system.

We agree that further exploration of microtubule dynamics could strengthen the mechanistic framework of our study. In our initial experiments, we disrupted centrosome function through genetic perturbations (e.g., knockout of PCNT, CEP120, CEP164, and ODF2). However, consistent with previous reports (Gavilan et al., 2018; Tateishi et al., 2013), we found that single-gene deletions did not completely eliminate centrosomal microtubules. Furthermore, imaging microtubule organization in 3D culture presents technical challenges. Due to the increased density of microtubules during cell rounding, we were unable to obtain clear microtubule filament structures—either using α -tubulin staining in fixed cells or SiR-tubulin labeling in live cells. Instead, the signal appeared diffusely distributed throughout the cytosol.

To overcome this, we employed a recently reported approach by co-expressing the centrosome-targeting carboxy-terminal domain (C-CTD) of CDK5RAP2 and the γ -tubulin-binding domain (gTBD) of NEDD1 to completely deplete γ -tubulin and abolish centrosomal microtubule nucleation (Vinopal et al., 2023). In our new data presented in the revised manuscript (Page 8, Paragraph 4; Figure 4—figure supplement 3), we found that cells lacking centrosomal microtubules were still able to polarize and position the centrioles apically. However, the efficiency of polarized transport of Gp135 vesicles to the apical membrane was reduced. These findings suggest that centrosomal microtubules are not essential for polarity establishment but may contribute to efficient apical transport.

Reference

- Betleja, E., Nanjundappa, R., Cheng, T., & Mahjoub, M. R. (2018). A novel Cep120-dependent mechanism inhibits centriole maturation in quiescent cells. *Elife*, 7. doi:10.7554/eLife.35439
- Gavilan, M. P., Gandolfo, P., Balestra, F. R., Arias, F., Bornens, M., & Rios, R. M. (2018). The dual role of the centrosome in organizing the microtubule network in interphase. *EMBO Rep*, 19(11). doi:10.15252/embr.201845942

- Jimenez, A. J., Schaeffer, A., De Pascalis, C., Letort, G., Vianay, B., Bornens, M., . . . Thery, M. (2021). Acto-myosin network geometry defines centrosome position. *Curr Biol*, 31(6), 1206-1220 e1205. doi:10.1016/j.cub.2021.01.002
- Martin, M., Veloso, A., Wu, J., Katrukha, E. A., & Akhmanova, A. (2018). Control of endothelial cell polarity and sprouting angiogenesis by non-centrosomal microtubules. *Elife*, 7. doi:10.7554/eLife.33864
- Meitinger, F., Anzola, J. V., Kaulich, M., Richardson, A., Stender, J. D., Benner, C., . . . Oegema, K. (2016). 53BP1 and USP28 mediate p53 activation and G1 arrest after centrosome loss or extended mitotic duration. *J Cell Biol*, 214(2), 155-166. doi:10.1083/jcb.201604081
- Schmoranzner, J., Fawcett, J. P., Segura, M., Tan, S., Vallee, R. B., Pawson, T., & Gundersen, G. G. (2009). Par3 and dynein associate to regulate local microtubule dynamics and centrosome orientation during migration. *Curr Biol*, 19(13), 1065-1074. doi:10.1016/j.cub.2009.05.065
- Tateishi, K., Yamazaki, Y., Nishida, T., Watanabe, S., Kunitomo, K., Ishikawa, H., & Tsukita, S. (2013). Two appendages homologous between basal bodies and centrioles are formed using distinct Odf2 domains. *J Cell Biol*, 203(3), 417-425. doi:10.1083/jcb.201303071
- Vasquez-Limeta, A., & Loncarek, J. (2021). Human centrosome organization and function in interphase and mitosis. *Semin Cell Dev Biol*, 117, 30-41. doi:10.1016/j.semcdb.2021.03.020
- Vinopal, S., Dupraz, S., Alfadil, E., Pietralla, T., Bendre, S., Stiess, M., . . . Bradke, F. (2023). Centrosomal microtubule nucleation regulates radial migration of projection neurons independently of polarization in the developing brain. *Neuron*, 111(8), 1241-1263 e1216. doi:10.1016/j.neuron.2023.01.020.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Figures:

(1) Figure 3 B+C - Although in comparison to Figure 2 it appears the p53 mutation does not affect θ N-C, or Lo-c. the figure would benefit from direct comparison to control cells.

We appreciate your suggestion to improve the clarity of the figure. In response, we have revised Figure 3B+C to include control cell data, allowing for clearer side-by-side comparisons in the updated figures.

(2) Figure 3D - Clarify if both were normalized to time point 0:00 of the p53 KO. The image used appears that Gp135 intensity increases substantially between 0:00 and 0:15 in the figure, but the graph suggests that the intensity is the same if not slightly lower.

Figure 3D – The data were normalized to the respective 0:00 time point for each condition. Because the intensity profile was measured along a line connecting the two nuclei, Gp135 signal could only be detected if it appeared along this line. However, the images shown are maximum-intensity projections, meaning that Gp135 signals from peripheral regions are projected onto the center of the image. This may create the appearance of increased intensity at certain time points (e.g., Figure 3A, p53-KO + CN, 0:00–0:15).

(3) Figure 4A: The diagram does not accurately represent the effect of the mutations, for example, PCNT mutation likely doesn't completely disrupt PCM (given gamma-tubulin is still visible in the staining), but instead results in its disorganization, Cep164 also wouldn't be expected to completely ablate distal appendages.

Thank you for your comment. We have modified the figure in the revised manuscript (Figure 4A) to more clearly depict the defective DAs.

(4) Figure 4 + Supplements: A more in-depth characterization of the mutations would help address the previous comment and strengthen the manuscript. Especially as these components have previously been implicated in centrosome transport.

Thank you for your valuable suggestion. As noted in previous studies, CEP164 is essential for distal appendage function and basal body docking, with its loss resulting in blocked ciliogenesis (Tanos et al., 2013); CEP120 is required for centriole elongation and distal appendage formation, and its loss also results in blocked ciliogenesis (Comartin et al., 2013; Lin et al., 2013; Tsai, Hsu, Liu, Chang, & Tang, 2019); ODF2 functions upstream in the formation of subdistal appendages, and its loss eliminates these structures and impairs microtubule anchoring (Tateishi et al., 2013); and PCNT functions as a PCM scaffold, necessary for the recruitment of PCM components and for microtubule nucleation at the centrosome (Fong, Choi, Rattner, & Qi, 2008; Zimmerman, Sillibourne, Rosa, & Doxsey, 2004).

Given that the phenotypes of these mutants have been well characterized in the literature. Here, we further focus on their roles in centrosome migration and polarized vesicle trafficking within the specific context of our study.

(5) Figure 4: It would be interesting to measure the Gp135 intensity at the centrosomes, given that the model proposes it is trafficked from the centrosomes to the AMIS.

Thank you for your suggestion. We have included measurements of Gp135 intensity at the centrosomes during the Pre-Abs stage in the revised figure (Figure 4I). Our data show no significant differences in Gp135 intensity between wild-type (WT) and CEP164-, ODF2-, or CEP120-knockout (KO) cell lines. However, a slight decrease in Gp135 intensity was observed in PCNT-KO cells.

(6) Figure 6F shows that in suspension culture polarity is reversed, however, in Figure 6G gp135 still localizes to the cytokinetic furrow prior to polarity reversal. Given this paper demonstrates Par-3 is upstream of centrosome positioning, it would be important to have temporal data of how Par-3 localizes prior to the ring observed in 6F.

Thank you for your comment. We have included a temporal analysis of Par3 localization using fixed-cell staining in the revised figure (Figure 6—figure supplement 1D). This analysis shows that Par3 also localizes to the cytokinesis site during the Pre-Abs stage, prior to ring formation observed during the Post-CK stage (Figure 6F). Interestingly, during the Pre-Abs stage, the centrosomes also migrate toward the center of the cell doublets in suspension culture, and Gp135 surrounding the centrosomes is also recruited to a region near the center (Figure 6—figure supplement 1E). These data suggest that Par3 also is initially recruited to the cytokinesis site before polarity reversal, potentially promoting centrosome migration. The main difference from Matrigel culture is the peripheral localization of Par3 and Gp135 in suspension, which is likely due to the lack of external ECM signaling.

Results:

(1) Page 7 Paragraph 1 - consistently use AMIS (Apical membrane initiation site) rather than "the apical site".

Thank you for your helpful comment. We have revised the manuscript (Page 7, Paragraph 1) and will now use "AMIS" (Apical Membrane Initiation Site) instead of "the apical site" throughout the text.

(2) Page 7 Paragraph 4 - A single sentence explaining why the p53 background had to be used for the Cep120 deletion would be beneficial. Did the cell line have a reduced

centrosome number? Does this effect apical membrane initiation similar to centrinone?

We have revised the text (Page 7, Paragraph 4) to clarify that we were unable to generate a CEP120 KO line in p53-WT cells for unknown reasons. CEP120-KO cells have a normal number of centrosome, but their centrioles are shorter. Because this KO line still contains centrioles, the effect is different from centrinone treatment, which results in a complete loss of centrioles.

(3) Page 10 paragraph 4 - This paragraph is confusing to read. I understand that in the cysts and epithelial sheet the cytokinetic furrow is apical, therefore a movement towards the AMIS could be due to its coincidence with the furrow. However, the phrasing "....we found that centrosomes move towards the apical membrane initiation site direction before bridge abscission. Taken together these findings indicate the position is strongly associated with the site of cytokinesis but not with the apical membrane" is confusing to the reader.

We have revised the manuscript (Page 11, paragraph 4) to change the AMIS as the center of the cell doublet. During de novo epithelial polarization, the apical membrane has not yet formed at the Pre-Abs stage. However, at the Pre-Abs stage, the centrosome has already migrated toward the site of cytokinesis, suggesting that centrosome positioning is correlated with the site of cell division. A similar phenomenon occurs in fully polarized epithelial cysts and sheets, where the centrosomes also migrate before bridge abscission. Thus, we propose that the position of the centrosome is closely associated with the site of cytokinesis and is independent of apical membrane formation.

Discussion

(1) Page 11, Paragraph 2 - citations needed when discussing previous studies.

Thank you for your suggestion. We have included the necessary references to the discussion of the previous studies in the revised manuscript (Page 12, Paragraph 2).

(2) Page 12, Paragraph 2 - This section of the discussion would be strengthened by discussing the role of the actomyosin network in defining centrosome position (Jimenez et al., 2021). It seems plausible that the differences observed in the different conditions could be due to altered actomyosin architecture. Especially where the cells haven't undergone cytokinesis.

We appreciate the suggestion of a role for the actomyosin network in determining centrosome positioning. Recent studies have indeed highlighted the role of the actomyosin network in regulating centrosome centering and off-centering (Jimenez et al., 2021). During the pre-abscission stage of cell division, the actomyosin network undergoes significant dynamic changes, with the contractile ring forming at the center and actin levels decreasing at the cell periphery. In contrast, under aggregated cell conditions—meaning cells that have not undergone division—the actomyosin network does not exhibit such dynamic changes. The loss of actomyosin remodeling may therefore influence whether the centrosome moves. Thus, alterations in actomyosin architecture may contribute to the differences observed under various conditions, particularly when cells have not yet completed cytokinesis. We have revised Paragraph 2 on Page 13 to briefly mention the referenced study and to propose that the actomyosin network may influence centrosome positioning, contributing to our observed results. This addition strengthens the discussion and clarifies our findings.

(3) Page 12 paragraph 3 - Given that centrosome translocation during cytokinesis in MDCK cells (this study) appears to be similar to that observed in HeLa cells and the zebrafish Kupffers vesicle (Krishnan et al., 2022) it would be interesting to discuss why Rab11a and PCNT may not be essential to centrosome positioning in MDCK cells.

Thank you for your insightful comment. We agree that it is interesting that centrosome translocation during cytokinesis in MDCK cells (as observed in our study) is similar to that observed in HeLa cells and zebrafish Kupffer's vesicle (Krishnan et al., 2022). However, there are notable differences between these systems that may help explain why Rab11a and PCNT are not essential for centrosome positioning in MDCK cells.

Our study used 3D culture of MDCK cells, while the reference study examined adherent culture of HeLa cells. In the adherent culture, cells attached to the culture surface form large actin stress fibers on their basal side, which weakens the actin networks in the apical and intercellular regions. In contrast, the 3D culture system used in our study better preserves cell polarity and the integrity of the actin network, which might contribute to centrosome positioning independent of Rab11a and PCNT. Differences in culture conditions and actin network architecture may explain why Rab11a and PCNT are not required for centrosome positioning in MDCK cells.

Furthermore, the referenced study focused on Rab11a and PCNT in zebrafish embryos at 3.3–5 hours post-fertilization (hpf), a time point before the formation of the Kupffer's vesicle. At this stage, the cells they examined may not yet have become epithelial cells, which may also influence the requirement of Rab11a and PCNT for centrosome positioning. We hypothesize that during the pre-abscission stage, centrosome migration toward the cytokinetic bridge occurs primarily in epithelial cells, and that the polarity and centrosome positioning mechanisms in these cells may differ from those in other cell types, such as zebrafish embryos.

Furthermore, data from Krishnan et al. (2022) suggest that cytokinesis failure in *pcnt*^{+/-} heterozygous embryos and Rab11a functional-blocked embryos may be due to the presence of supernumerary centrosomes. Consistent with this, our data show that blocking cytokinesis inhibits centrosome movement in MDCK cells. However, in our MDCK cell lines with PCNT or Rab11a knockdown, we did not observe significant cytokinesis failure, and centrosome migration proceeded normally.

Reviewer #2 (Recommendations for the authors):

Suggestions for experiments:

(1) A description of the organization of microtubules in the absence of centriole, or in the absence of ECM would be interesting to understand how polarity markers end up where you observed them. This easy experiment may significantly improve our understanding of this system.

Previous studies have shown that in the absence of centrioles, microtubule organization undergoes significant changes. Specifically, the number of non-centrosomal microtubules increases, and these microtubules are not radially arranged, leading to the absence of focused microtubule organizing centers in centriolar-deficient cells (Martin, Veloso, Wu, Katrukha, & Akhmanova, 2018). This disorganized microtubule network reduces the efficiency of vesicle transport during de novo epithelial polarization at the mitotic preabscission stage.

In contrast, the organization of microtubules under ECM-free conditions remains less well characterized. Here, we show that while the ECM plays a critical role in establishing the direction of epithelial polarity, it does not influence the positioning of the centrosome, the microtubule-organizing center (MTOC).

(2) Would it be possible to knock down ODF2 and pericentrin to completely disconnect the centrosome from microtubules?

ODF2 is the base of subdistal appendages. When ODF2 is knocked out, it affects the recruitment of all downstream proteins to the subdistal appendages (Mazo, Soplop, Wang, Uryu, & Tsou, 2016). One study has shown that ODF2 knockout cells almost completely lost subdistal appendage structures and significantly reduced the microtubule asters surrounding the centrioles (Tateishi et al., 2013). However, although pericentrin (PCNT) is the main scaffold of the pericentriolar matrix (PCM) of centrosomes, the microtubule organization ability of centrosomes can be compensated by AKAP450, a paralog of PCNT, after PCNT knockout. A previous study has even shown that in cells with a double knockout of PCNT and AKAP450, γ -tubulin can still be recruited to the centrosomes, and centrosomes can still nucleate microtubules (Gavilan et al., 2018). This suggests that there are other proteins or pathways that promote microtubule nucleation on centrosomes. We are unsure whether the triple knockout of ODF2, PCNT, and AKAP450 can completely disconnect the centrosome from microtubules. However, a recent study reported a simpler approach involving the expression of dominant-negative fragments of the γ -tubulin-binding protein NEDD1 and the activator CDK5RAP2 at the centrosome (Vinopal et al., 2023). In our revised manuscript (Page 8, Paragraph 4; Figure 4—figure supplement 3), we applied this strategy, which resulted in the depletion of nearly all γ -tubulin from the centrosome. This indicates a strong suppression of centrosomal microtubule nucleation and an effective disconnection of the centrosome from the microtubule network.

(3) The study does not distinguish the role of cytokinesis from the role of tight junctions, which form only after cytokinesis and not simply by bringing cells into contact. Would it be feasible and interesting to study the polarization after cytokinesis in cells that could not form tight junctions (due to the absence of Ecad or ZO1 for example)?

Studying cell polarization after cytokinesis in cells unable to form tight junctions is a promising area of research.

Recent studies have shown that mouse embryonic stem cells (mESCs) cultured in Matrigel can form ZO-1-labelled tight junctions at the midpoint of cell–cell contact even in the absence of cell division. However, in the absence of E-cadherin, ZO-1 localization is significantly impaired. Interestingly, despite the loss of E-cadherin, the Golgi apparatus and centrosomes remain oriented toward the cell–cell interface (Liang, Weberling, Hii, Zernicka-Goetz, & Buckley, 2022). These findings suggest that cell polarity can be maintained independently of tight junction formation, highlighting the potential value of studying cell polarization that lack tight junctions.

Furthermore, while studies have explored the effects of knocking down tight junction components such as JAM-A and Cingulin on lumen formation in MDCK 3D cultures (Mangan et al., 2016; Tuncay et al., 2015), the role of ZO-1 in this context remains underexplored. Cingulin knockdown has been shown to disrupt endosome targeting and the formation of the AMIS, while both JAM-A and Cingulin knockdown result in actin accumulation at multiple points, leading to the formation of multi-lumen structures rather than a reversal of polarity. However, previous research has not specifically investigated centrosome positioning in JAM-A and Cingulin knockdown cells, an area that could provide valuable insights into how polarity is maintained in the absence of tight junctions.

Writing details:

(1) The migration of the centrosome in the absence of appendages or PCM is proposed to be ensured by compensatory mechanisms ensuring the robustness of microtubule anchoring to the centrosome. It could also be envisaged that the centrosome motion does not require this anchoring and that other yet unknown moving mechanisms, based on an actin network for example, might exist.

Thank you for your valuable comments. We agree that there may indeed be some unexpected mechanisms that allow centrosomes to move independently of microtubule anchoring to the centrosome, such as mechanisms based on actin filaments or noncentrosomal microtubules; these mechanisms are worth further investigation.

In response to your suggestion, in the Paragraph 5 of the discussion section, we further clarified that while a microtubule anchoring mechanism might be one explanation, other mechanisms could also influence centrosome movement in the absence of appendages or PCM. Additionally, we revised the Paragraph 4 regarding the possibility of actin network-driven centrosome movement and emphasized the importance of future research for a deeper understanding of these processes.

(2) The actual conclusion of the study of Martin et al (eLife 2018) is not simply that centrosome is not involved in cell polarization but that it hinders cell polarization!

Thank you for your valuable feedback. We agree with the findings of Martin et al. (eLife 2018) that centrosome is not irrelevant to cell polarity, but rather they inhibit cell polarization. Therefore, we have revised the manuscript (Page 2, Paragraph 2) to more accurately reflect this viewpoint.

(3) This study recalls some conclusions of the study by Burute et al (Dev Cell 2017), in particular the role of Par3 in driving centrosome toward the intercellular junction of daughter cells after cytokinesis. It would be welcome to comment on the results of this study in light of their work.

Thank you for your valuable feedback. The study by Burute et al. (Dev Cell, 2017) showed that in micropattern-cultures of MCF10A cells, the cells exhibit polarity and localize their centrosomes towards the intercellular junction, while downregulation of Par3 gene expression disrupts this centrosome positioning. This result is similar to our findings in 3D cultured MDCK cells and consistent with previous studies in *C. elegans* intestinal cells and migrating NIH 3T3 cells (Feldman & Priess, 2012; Schmoranz et al., 2009), indicating that Par3 indeed influences centrosome positioning in different cellular systems. However, Par3 does not directly localize to the centrosome; rather, it localizes to the cell cortex or cell-cell junctions. Therefore, Par3 likely regulates centrosome positioning through other intermediary molecules or mechanisms, but the specific mechanism remains unclear and requires further investigation.

(4) Could the term apico-basal be used in the absence of a basement membrane to form a basal pole?

We understand that using the term "apico-basal" in the absence of a basement membrane might raise some questions. Traditionally, the apico-basal axis refers to the polarity of epithelial cells, where the apical surface faces the lumen or external environment, and the basal surface is oriented toward the basement membrane. However, in the absence of a basement membrane, such as in certain in vitro systems or under specific experimental conditions, polarity along a similar axis can still be observed. In such cases, the term "apico-basal" can still be used to describe the polarity between the apical domain and the region where it contacts the substrate or adjacent cells.

(5) The absence of centrosome movement to the intercellular bridge in spread cells in culture is not so surprising considering the work of Lafaurie-Janvore et al (Science 2018) about the role of cell spreading in the regulation of bridge tension and abscission delay.

Thank you for your valuable comment. Indeed, previous studies have shown that in some cell types, the centrosome does move toward the intercellular bridge in spread cells (Krishnan et al., 2022; Piel, Nordberg, Euteneuer, & Bornens, 2001), but other studies have suggested that

this movement may not be significant and it may not occur in universally observed across all cell types (Jonsdottir et al., 2010). In our study, we aim to demonstrate that this phenomenon is more pronounced in 3D culture systems compared to 2D spread cell culture systems. Previous studies and our work have observed that centrosome migration occurs during the pre-abscission stage, but whether this migration is directly related to cytokinetic bridge tension or the time of abscission remains an open question. Further research is needed to explore the potential relationship between centrosome positioning, cytokinetic bridge tension, and the timing of abscission.

(6) GP135 (podocalyxin) has been proposed to have anti-adhesive/lubricant properties (hence its pro-invasive effect). Could it be possible that once localized at the cell surface it is systematically moved away from regions that are anchored to either the ECM or adjacent cells? So its localization away from the centrosome in an ECM-free experiment would not be a consequence of defective targeting but relocalization after reaching the plasma membrane?

Thank you for your valuable comment. We agree that GP135 may indeed move directly across the cell surface, away from the region where it interacts with the ECM or adjacent cells. This re-localization could be due to its anti-adhesive or lubricating properties, which may facilitate its displacement from these adhesive sites. To validate this, it is necessary to employ higher-resolution real-time imaging system to observe the dynamic behavior of GP135 on the cell surface.

However, this does not contradict our main conclusion. Under suspension culture conditions without ECM, the centrosome positioning in cell doublets is indeed decoupled from apical membrane orientation. This suggests that the localization of the centrosome and the apical membrane is regulated by different mechanisms. Specifically, the GP135 protein tends to accumulate away from areas of contact with the ECM or adjacent cells, possibly through movement within the cell membrane or by recycling endosome transport. In contrast, centrosome positioning is closely related to the cytokinesis site. Our study clearly elucidates the differences between these two polarity properties.

Reviewer #3 (Recommendations for the authors):

Major:

(1) To me, a clear implication of these studies is that Gp135, Rab11, etc. are delivered to the AMIS on centrosomal microtubules. The authors do not explore this model except to say that depletion of SD appendage or pericentrin has no effect on the protein relocalization to the AMIS. However, the authors do not observe microtubule association with the centrosome in these KO conditions. This analysis is imperative to interpret existing results since these are new KO conditions in this cell/culture system and parallel pathways (e.g. CDK5RAP2) are known to contribute to microtubule association with the centrosome. An ability to comment on the mechanism by which the centrosome contributes to the efficiency of polarization would greatly enhance the paper.

Microtubule requirement could also be tested in numerous additional ways requiring varying degrees of new experiments:

(a) faster live cell imaging at abscission to see if the deposition of those components appears to traffic on MTs;

(b) live cell imaging with microtubules (e.g. SPY-tubulin) and/or EB1 to determine the origin and polarity of microtubules at the pertinent stages;

For (a) and (b), because the cells were cultured in Matrigel, they tended to be round up, with a dense internal structure that made observation difficult. In contrast, under adherent

culture conditions, the cells were flattened with a more dispersed internal structures, making them easier to observe. We had previously used SPY-tubulin to label microtubules for live cell imaging; however, due to the dense microtubule structure in 3D culture, the image contrast was reduced, and we could not clearly observe the microtubule network within the cells.

(c) acute nocodazole treatment at abscission to determine the effect on protein localization.

Regarding the method of using nocodazole to study microtubule requirements at the abscission stage, we believe that nocodazole treatment may lead to cytokinesis failure. Cell division failure results in the formation of binucleated cells, which are unable to establish cell polarity. Furthermore, nocodazole treatment cannot distinguish between centrosomal and non-centrosomal microtubules, making it unsuitable for studying the specific role of centrosomal microtubules in this process.

In our new data (Figure 4-figure supplementary 3) presented in the revised manuscript, we employed a recently reported method by co-expressing of the centrosome-targeting carboxy-terminal domain (C-CTD) of CDK5RAP2 and the γ -tubulin-binding domain (gTBD) of NEDD1 to completely deplete γ -tubulin and abolish centrosomal microtubule nucleation (Vinopal et al., 2023). We found that cells lacking centrosomal microtubules were still able to polarize and position the centrioles apically. However, the efficiency of polarized transport of Gp135 vesicles to the apical membrane was reduced. These findings suggest that centrosomal microtubules are not essential for polarity establishment but may contribute to facilitate efficient apical transport.

(2) Similar to the expanded analysis of the role of microtubules in this system, it would be excellent if the author could expand on the role of Par3 and the centrosome, although this reviewer recognizes that the authors have already done substantial work. For example, what are the consequences of Gp135 and/or Rab11 getting stuck at the centrosome? Do the authors have any later images to determine when and if these components ever leave the centrosome? Existing literature focuses on the more downstream consequence of Par3 removal on single-lumen formation.

Similarly, could the authors expand on the description of polarity disruption following centrinone treatment? It is clear that Gp135 recruitment is disrupted, but how and when do things get fixed and what else is disrupted at the very earliest stages of AMIS formation? The authors have an excellent opportunity to really expand on what is known about the requirements for these conserved components.

Regarding the use of centrinone in treatment, we speculate that Gp135 can still accumulate at the AMIS over time, although the efficiency of its recruitment may be reduced.

Furthermore, under similar conditions, other apical membrane components (such as the Crumbs3 protein) may exhibit similar characteristics to Gp135 protein.

(3) Perhaps satisfying both of the above asks, could the authors do a faster time-lapse at the relevant time points, i.e. as proteins are being recruited to the AMIS (time points between 1Aiv and v)? This type of imaging again might help shed light on the mechanism.

We believe the above questions are very important and may require further experimental verification in the future.

Minor:

(1) What is the green patch of Gp135 in Figure 2A that does not colocalize with the centrosome? Is this another source of Gp135 that is being delivered to the AMIS? This

type of patch is also visible in Figure 3A 15 and 30-minute panels.

During mitosis, membrane-composed organelles such as the Golgi apparatus are typically dispersed throughout the cytoplasm. However, during the pre-abscission stage, these organelles begin to reassemble and cluster around the centrosome. Furthermore, they also accumulate in the region between the nucleus and the cytokinetic bridge, corresponding to the “patch” mentioned in Figure 2A.

Live cell imaging results showed that this Gp135 patch initially appears in a region not associated with the centrosome. Subsequently, they were either directly transported to the AMIS or fused with the centrosome-associated Gp135 and transported together. Notably, this patch was only observed when Gp135 was overexpressed in cells. No such distinct protein patches were observed when staining endogenous Gp135 protein (Figure 1A), suggesting that overexpression of Gp135 protein may lead to a localized increase in its concentration in that region.

(2) I am confused by the "polarity index" quantification as this appears to just be a nucleus centrosome distance measurement and wouldn't, for example, distinguish if the centrosomes separated from the nucleus but were on the basal side of the cell.

The position of the centrosome within the cell (i.e., its distance from the nucleus) can indeed serve as an indicator of cell polarity (Burute et al., 2017). We acknowledge that this quantitative method does not directly capture the specific direction in which the centrosome deviates from the cell center. To address this limitation, we have incorporated information about the angle between the nucleus and the centrosome, which allows for a more accurate description of changes in cell polarity (Rodriguez-Fraticelli, Auzan, Alonso, Bornens, & Martin-Belmonte, 2012).

(3) How is GP135 "at AMIS" measured? Is an arbitrary line drawn? This is important later when comparing to centrinone treatment in Figure 3D where the quantification does not seem to accurately capture the enrichment of Gp135 that is seen in the images.

To measure the expression level of Gp135 in the "AMIS" region of the cell, we first connected the centers of the two cell nuclei in three-dimensional space to form a straight line. Then, we used the Gp135 expression intensity at the midpoint of this line as the representative value for the AMIS region. This method is based on the assumption that the AMIS region is most likely located between the centers of the two cell nuclei. Therefore, this quantitative method provides a standardized assessment tool for comparing Gp135 expression levels under different conditions.

(4) The authors reference cell height (p.7) but no data for this measurement are shown

Thank you for the comment. Although we did not perform quantitative measurements, the differences in cell height are clearly visible in Figure 3E (p53-KO + CN), which visually illustrates this phenomenon.

(5) Can the authors comment on the seeming reduction of Par3 in p53 KO cells?

We did not observe a reduction of Par3 in p53-KO cells in our experiments.

(6) Can the authors make sense of the E-cad localization: Figure 5, Supplement 2.

Our study revealed that E-cadherin begins to accumulate at the cell-cell contact sites during the pre-abscission stage. Its appearance is similar to that of ZO-1, which also appears near the cell division site during this phase. Therefore, the behavior of E-cadherin contrasts sharply with that of Gp135, further highlighting the unique trafficking mechanisms of apical membrane proteins during this process.

(7) I find the results in Figure 6G puzzling. Why is ECM signaling required for Gp135 recruitment to the centrosome. Could the authors discuss what this means?

We appreciate the reviewer's valuable comments and thank you for the opportunity to clarify this point. The data in Figure 6G do not indicate that ECM signaling is required for the recruitment of Gp135 to the centrosome. Rather, our findings suggest that even in the absence of ECM, the centrosomes can migrate to a polarized position similar to that in Matrigel culture. This suggests that centrosome migration and the orientation of the nucleus-centrosome axis may be independent of ECM signaling and are primarily driven by cytokinesis alone.

Regarding the localization of Gp135, previous studies have shown that ECM signaling through integrin promotes endocytosis, which is crucial for the internalization of Gp135 from the cell membrane and its subsequent transport to the AMIS (Buckley & St Johnston, 2022). Our study found that, prior to its accumulation at the AMIS, Gp135 transiently localizes around the centrosome. In the absence of ECM, due to reduced endocytosis, Gp135 primarily remains on the cell membrane and does not undergo intracellular trafficking.

(8) The authors end the Discussion stating that these studies may have implication for in vivo settings, yet do not discuss the striking similarities to the *C. elegans* and *Drosophila* intestine or the findings from any other more observational studies of tubular epithelial systems in vivo (e.g. mouse kidney polarization, zebrafish neuroepithelium, etc.). These models should be discussed.

Thank you for your valuable comment. Indeed, all types of epithelial tissues or tubular epithelial systems in vivo share some common features during cell division, which have been well-documented across various species.

These features include: during interphase, the centrosome is located at the apical surface of the cells; after the cell enters mitosis, the centrosome moves to the lateral side of the cell to regulate spindle orientation; and during cytokinesis, the cleavage furrow ingresses asymmetrically from the basal to the apical side, with the cytokinetic bridge positioned at the apical surface. Our study using MDCK 3D culture and transwell culture systems successfully mimicked these key features, demonstrating that these in vitro models are of significant value for studying cell polarization dynamics.

Based on our observations, we speculate that the centrosome may return to the apical surface after anaphase, just before bridge abscission. This is consistent with our findings from studies using MDCK 3D cultures and transwell systems, which showed that the centrosome relocates prior to the final stages of cytokinesis.

Additionally, we propose that de novo polarization of the kidney tubule in vivo may not solely depend on the aggregation and mesenchymal-epithelial transition (MET) of the metanephric mesenchyme. It may also be related to the cell division process, which triggers centrosome migration and polarized vesicle trafficking. These processes likely contribute to enhancing cell polarization, as we observed in our in vitro models.

We hope this will further clarify the potential implications of our findings for in vivo model studies, as well as and their broader impact on the field of tubular epithelial cell polarization research.

(9) There are several grammatical issues/typos throughout the paper. A careful readthrough is required. For example:

this sentence makes no sense "that the centrosome acts as a hub of apical recycling endosomes and centrosome migration during cytokinetic pre-abscission before apical

| *membrane components are targeted to the AMIS"*

We carefully reviewed the paper and made necessary revisions to address the issues raised. In particular, we revised certain sentences to improve clarity and readability (Page 5, Paragraph 3).

| (10) *P.8: have been previously reported [to be] involved in MDCK...*

We appreciate the reviewer's valuable suggestions. We have revised the sentence accordingly (Page 9, Paragraph 2).

| (11) *This sentence seems misplaced: "Cultured conditions influence cellular polarization preferences."*

The sentence itself is fine, but to improve the coherence and clarity of the paragraph, we adjusted the paragraph structure and added some transitional phrases (Page 13, Paragraph 1).

| (12) *"Play a downstream role in Par3 recruitment" doesn't make sense, this should just be downstream of Par3 recruitment.*

Thank you for your suggestion. We have revised the wording accordingly, changing it to "downstream of Par3 recruitment" (Page 10, Paragraph 2).

Reference

Buckley, C. E., & St Johnston, D. (2022). Apical-basal polarity and the control of epithelial form and function. *Nat Rev Mol Cell Biol*, 23(8), 559-577. doi:10.1038/s41580-022-00465-y

Burute, M., Prioux, M., Blin, G., Truchet, S., Letort, G., Tseng, Q., . . . Thery, M. (2017). Polarity Reversal by Centrosome Repositioning Primes Cell Scattering during Epithelial-to-Mesenchymal Transition. *Dev Cell*, 40(2), 168-184. doi:10.1016/j.devcel.2016.12.004

Comartin, D., Gupta, G. D., Fussner, E., Coyaoud, E., Hasegan, M., Archinti, M., . . . Pelletier, L. (2013). CEP120 and SPICE1 cooperate with CPAP in centriole elongation. *Curr Biol*, 23(14), 1360-1366.

doi:10.1016/j.cub.2013.06.002

Feldman, J. L., & Priess, J. R. (2012). A role for the centrosome and PAR-3 in the hand-off of MTOC function during epithelial polarization. *Curr Biol*, 22(7), 575-582. doi:10.1016/j.cub.2012.02.044

Fong, K. W., Choi, Y. K., Rattner, J. B., & Qi, R. Z. (2008). CDK5RAP2 is a pericentriolar protein that functions in centrosomal attachment of the gamma-tubulin ring complex. *Mol Biol Cell*, 19(1), 115-125. doi:10.1091/mbc.e07-04-0371

Gavilan, M. P., Gandolfo, P., Balestra, F. R., Arias, F., Bornens, M., & Rios, R. M. (2018). The dual role of the centrosome in organizing the microtubule network in interphase. *EMBO Rep*, 19(11). doi:10.15252/embr.201845942

Jimenez, A. J., Schaeffer, A., De Pascalis, C., Letort, G., Vianay, B., Bornens, M., . . . Thery, M. (2021). Acto-myosin network geometry defines centrosome position. *Curr Biol*, 31(6), 1206-1220 e1205. doi:10.1016/j.cub.2021.01.002

Jonsdottir, A. B., Dirks, R. W., Vrolijk, J., Ogmundsdottir, H. M., Tanke, H. J., Eyfjord, J. E., & Szuhai, K. (2010). Centriole movements in mammalian epithelial cells during cytokinesis. *BMC Cell Biol*, 11, 34. doi:10.1186/1471-2121-11-34

- Krishnan, N., Swoger, M., Rathbun, L. I., Fioramonti, P. J., Freshour, J., Bates, M., . . . Hehnly, H. (2022). Rab11 endosomes and Pericentrin coordinate centrosome movement during preabscission in vivo. *Life Sci Alliance*, 5(7). doi:10.26508/lsa.202201362
- Liang, X., Weberling, A., Hii, C. Y., Zernicka-Goetz, M., & Buckley, C. E. (2022). E-cadherin mediates apical membrane initiation site localisation during de novo polarisation of epithelial cavities. *EMBO J*, 41(24), e111021. doi:10.15252/emboj.2022111021
- Lin, Y. N., Wu, C. T., Lin, Y. C., Hsu, W. B., Tang, C. J., Chang, C. W., & Tang, T. K. (2013). CEP120 interacts with CPAP and positively regulates centriole elongation. *J Cell Biol*, 202(2), 211219. doi:10.1083/jcb.201212060
- Mangan, A. J., Sietsema, D. V., Li, D., Moore, J. K., Citi, S., & Prekeris, R. (2016). Cingulin and actin mediate midbody-dependent apical lumen formation during polarization of epithelial cells. *Nat Commun*, 7, 12426. doi:10.1038/ncomms12426
- Martin, M., Veloso, A., Wu, J., Katrukha, E. A., & Akhmanova, A. (2018). Control of endothelial cell polarity and sprouting angiogenesis by non-centrosomal microtubules. *Elife*, 7. doi:10.7554/eLife.33864
- Mazo, G., Soplop, N., Wang, W. J., Uryu, K., & Tsou, M. F. (2016). Spatial Control of Primary Ciliogenesis by Subdistal Appendages Alters Sensation-Associated Properties of Cilia. *Dev Cell*, 39(4), 424-437. doi:10.1016/j.devcel.2016.10.006
- Piel, M., Nordberg, J., Euteneuer, U., & Bornens, M. (2001). Centrosome-dependent exit of cytokinesis in animal cells. *Science*, 291(5508), 1550-1553. doi:10.1126/science.1057330
- Rodriguez-Fraticelli, A. E., Auzan, M., Alonso, M. A., Bornens, M., & Martin-Belmonte, F. (2012). Cell confinement controls centrosome positioning and lumen initiation during epithelial morphogenesis. *J Cell Biol*, 198(6), 1011-1023. doi:10.1083/jcb.201203075
- Schmoranzner, J., Fawcett, J. P., Segura, M., Tan, S., Vallee, R. B., Pawson, T., & Gundersen, G. G. (2009). Par3 and dynein associate to regulate local microtubule dynamics and centrosome orientation during migration. *Curr Biol*, 19(13), 1065-1074. doi:10.1016/j.cub.2009.05.065
- Tanos, B. E., Yang, H. J., Soni, R., Wang, W. J., Macaluso, F. P., Asara, J. M., & Tsou, M. F. (2013). Centriole distal appendages promote membrane docking, leading to cilia initiation. *Genes Dev*, 27(2), 163-168. doi:10.1101/gad.207043.112
- Tateishi, K., Yamazaki, Y., Nishida, T., Watanabe, S., Kunitomo, K., Ishikawa, H., & Tsukita, S. (2013). Two appendages homologous between basal bodies and centrioles are formed using distinct Odf2 domains. *J Cell Biol*, 203(3), 417-425. doi:10.1083/jcb.201303071
- Tsai, J. J., Hsu, W. B., Liu, J. H., Chang, C. W., & Tang, T. K. (2019). CEP120 interacts with C2CD3 and Talpid3 and is required for centriole appendage assembly and ciliogenesis. *Sci Rep*, 9(1), 6037. doi:10.1038/s41598-019-42577-0
- Tuncay, H., Brinkmann, B. F., Steinbacher, T., Schurmann, A., Gerke, V., Iden, S., & Ebnet, K. (2015). JAM-A regulates cortical dynein localization through Cdc42 to control planar spindle orientation during mitosis. *Nat Commun*, 6, 8128. doi:10.1038/ncomms9128
- Vinopal, S., Dupraz, S., Alfadil, E., Pietralla, T., Bendre, S., Stiess, M., . . . Bradke, F. (2023). Centrosomal microtubule nucleation regulates radial migration of projection neurons independently of polarization in the developing brain. *Neuron*, 111(8), 1241-1263 e1216. doi:10.1016/j.neuron.2023.01.020
- Zimmerman, W. C., Sillibourne, J., Rosa, J., & Doxsey, S. J. (2004). Mitosis-specific anchoring of gamma tubulin complexes by pericentrin controls spindle organization and mitotic entry.

Mol Biol Cell, 15(8), 3642-3657. doi:10.1091/mbc.e03-11-0796.

<https://doi.org/10.7554/eLife.101088.2.sa0>